SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Ramapuram, Chennai-600089

FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

Academic Year (ODD 2024 – 2025)

21CSC302J –COMPUTER NETWORK

RAMAPURAM
SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

SEMESTER- V

Class : B.Tech
Department : Computer Science and Engineering
Year : III

Course Code	21CSC302J	Course Name	COMPUTER NETWORKS	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*School of Computing*	**Data Book / Codes / Standards**		*Nil*	

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*Define the layered network architecture*
CLR-2:	*Produce knowledge in IP addressing*
CLR-3:	*Identify suitable routing algorithms based on geographical location of the devices*
CLR-4:	*Apply the concept of Error detection to identify the errors in data.*
CLR-5:	*Exploring reliable and unreliable protocols*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	**Program Outcomes (PO)**												**Program Specific outcomes**		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*Apply the knowledge of communication*	*3*	-	-	-	*3*	-	-	-	-	-	-	-	*1*	-	-
CO-2:	*Construct the network using addressing schemes*	*3*	-	-	*2*	-	-	-	-	-	-	-	-	*1*	-	-
CO-3:	*Design and implement the various Routing Protocols*	*3*	-	-	*2*	*3*	-	-	-	-	-	-	-	*1*	-	-
CO-4:	*Identify and correct the errors in transmission*	*3*	-	-	-	-	-	-	-	-	-	-	-	*1*	-	-
CO-5:	*Analyze the services provided by Transport and Application layers*	*3*	-	-	-	-	-	-	-	-	-	-	-	*1*	-	-

Unit-1 - Introduction to Networks ***15 Hour***

Network Types: LAN, MAN, PAN, WAN - Network Topology : BUS, STAR, RING, MESH, HYBRID - Switching : Circuit Switching, Packet Switching - OSI Layered Architecture - TCP/IP Model - Physical Layer Overview - Latency, Bandwidth, Delay - Guided Media : Twisted pair, Coaxial cable, Fiber optic cable - Unguided Media : Radio waves, Microwaves, Infrared.
Lab 1: Introduction to Packet Tracer, Peer to Peer communication, study of cables and its colour codes
Lab 2: Implementation of Network Topologies
Lab 3: Router Configuration (Creating Passwords, Configuring Interfaces)

Unit-2 - Addressing ***15 Hour***

IPV4 Addressing - Address space - Classful addressing - Subnet mask - FLSM - Classless Addressing - VLSM – NAT – Super netting - Network Devices: Hub, Repeaters, Switch, Bridge, Router
Lab 4: IP addressing and Sub netting (VLSM)
Lab 5: Static and Default Routing
Lab 6: NAT Configuration

Unit-3 - Routing ***15 Hour***

Forwarding of IP Packets — Static and Default Routing — Unicast Routing Algorithms: Distance Vector Routing, Link State Routing, Path Vector Routing — Protocols: RIP V1, RIP V2, OSPF, BGP, EIGRP — Multicasting Basics — IPV6 Addressing Basics
Lab 7: Implementation of RIP version 1
Lab 8: Implementation of RIP version 2
Lab 9: Implementation of Single Area OSPF

Unit-4 – Medium Access Control	***15 Hour***
ALOHA ,CSMA/CD, CSMA/CA, Ethernet, Token Ring - Flow Control :Stop and Wait, Sliding Window - Error Control: Stop and Wait ARQ, Sliding Window ARQ - Error Detection : Parity Check, Checksum, CRC - Error Correction: Hamming codes - Data-Link Layer Protocols : HDLC, PPP. *Lab 10: Implementation of Multi Area OSPF* *Lab 11: PPP Configuration* *Lab 12: HDLC Configuration*	
Unit-5 – Transport and Application Layer Protocols	***15 Hour***
Port Numbers — User Datagram Protocol — Transmission Control Protocol — WWW and HTTP — FTP — Email –Telnet – DNS. *Lab 13: Implementation of BGP* *Lab 14: Implementation of EIGRP* *Lab 15: Telnet Configuration*	

Learning Resources	1. *Behrouz A. Forouzan, "Data Communication and Networking",5th ed.,2010* 2. *Bhushan Trivedi, " Data Communication and Networks" 2016*	3. *William Stallings, " Data and Computer Communications",9th ed.,2010* 4. *Todd Lammle, "CCNA Study Guide",7th ed.,2011*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	*Remember*	*20%*	-	-	*20%*	*20%*	-
Level 2	*Understand*	*40%*	-	-	*40%*	*40%*	-
Level 3	*Apply*	*40%*	-	-	*40%*	*40%*	-
Level 4	*Analyze*	-	-	-	-	-	-
Level 5	*Evaluate*	-	-	-	-	-	-
Level 6	*Create*	-	-	-	-	-	-
	Total	*100 %*		*100 %*		*100 %*	

Course Designers

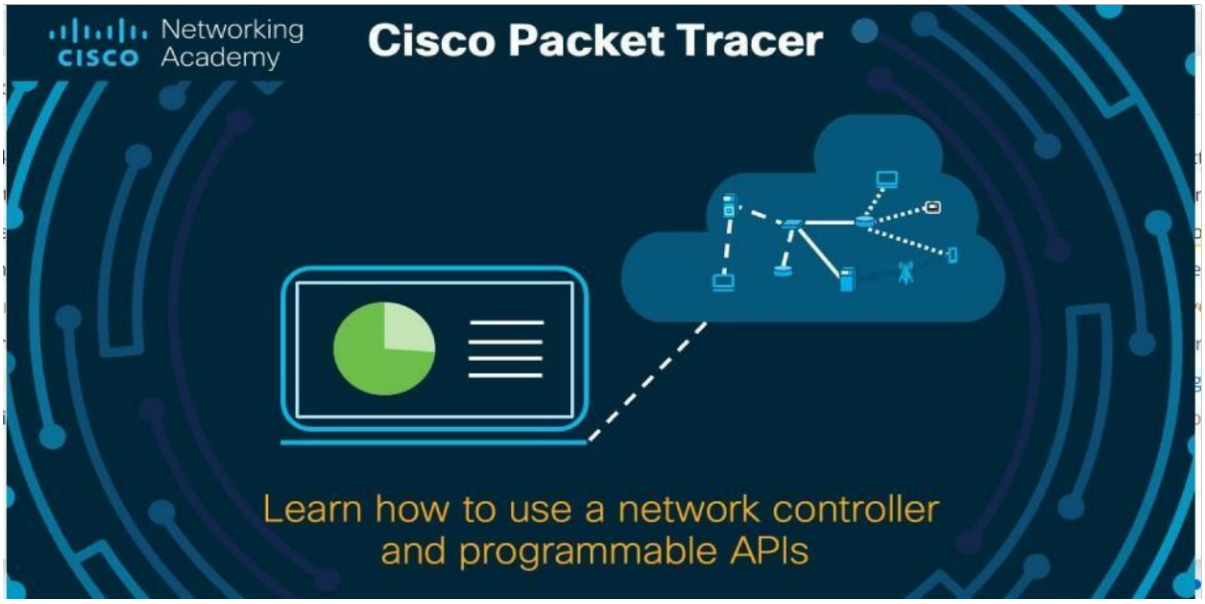
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. *Mr. Srinivasan Varadharajan, Senior Principal Software Engineer, Manhattan Associates, Atlanta, United States*	1. *Dr. I.Joe Louis Paul, Associate Professor, SSN College of Engineering, TamilNadu*	1. *Dr. S. Metilda Florence, SRMIST*

Session 2 Periods	Exercise	CLO	Page No
Lab 1	1.a - Introduction to Packet Tracer		
	1.b - Networking Commands (Windows/ Unix)		
	1.c – Study of Cables, Network Commands and devices		
Lab 2	2. a – Peer to Peer Network Implementation		
	2. b – Implementation of Network Topologies		
Lab 3	Demonstration of Router Configuration (Creating Password, Configuring interfaces)		
Lab 4	IP Addressing and Subnetting (VLSM)		
Lab 5	Configuration of Static & Default Routing		
Lab 6	NAT Configuration		
Lab 7	Implementation of RIP version 1		
	Implementation of RIP version 2		
Lab 8	Implementation of single area OSPF		
Lab 9	Implementation of multi-area OSPF		
Lab 10	PPP & HDLC Configuration		
Lab 11	Implementation of BGP		
Lab 12	Implementation of EIGRP		
Lab 13	TELNET Configuration		
Case Study Implementation			

EXERCISE 1

1. A) INTRODUCTION TO PACKET TRACER

Cisco Packet Tracer is a free application that enables you to practice network configuration and troubleshooting on your desktop or laptop computer. It enables you to mimic networks without having physical access to the underlying hardware. Along with networking, you may improve your Internet of Things (IoT) and cybersecurity skills through education and practice. You have the option of creating a network from scratch, using a pre-built sample network, or completing lab projects. While Packet Tracer is not a substitute for practising on physical routers, switches, firewalls, and servers, it does offer a number of advantages.

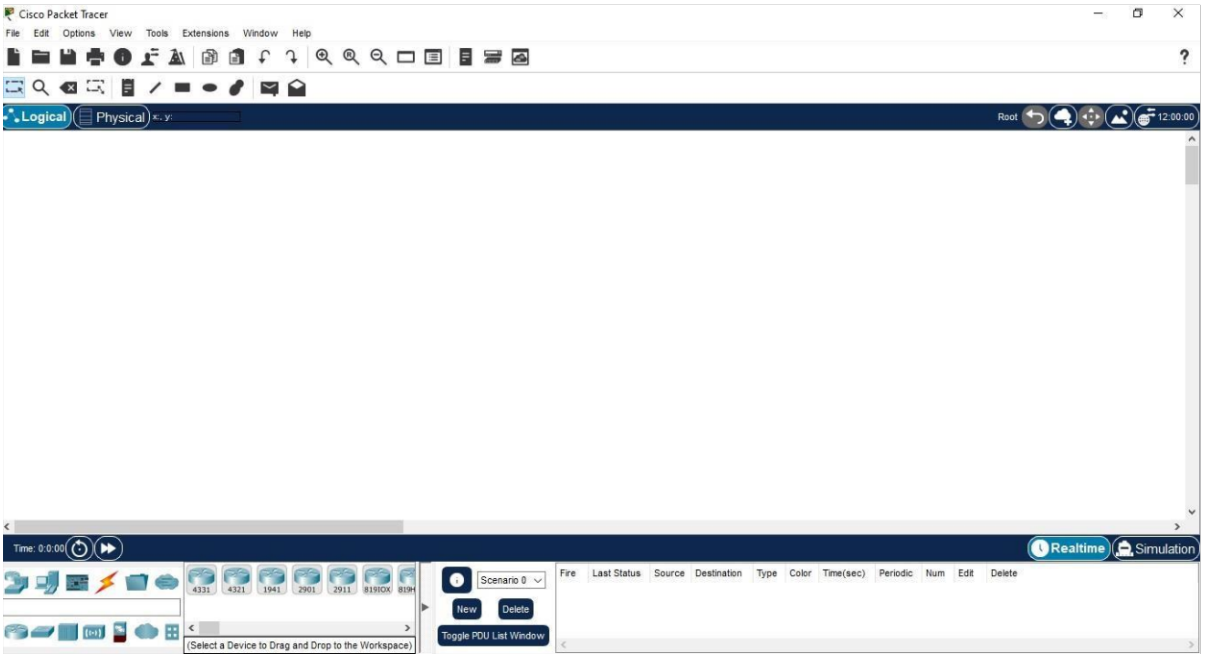


1.a What are the Benefits of Using Packet Tracer?

Imagine being able to peer inside a small business network or the internet. Have you ever wished to create an Internet of Things system that would notify you through the phone if there was an issue in your home environment? Welcome to Cisco Packet Tracer, the simulation environment that may assist you in doing all of these tasks and more. It is intended to familiarize you with the Cisco Packet Tracer network simulation and visualization tool.

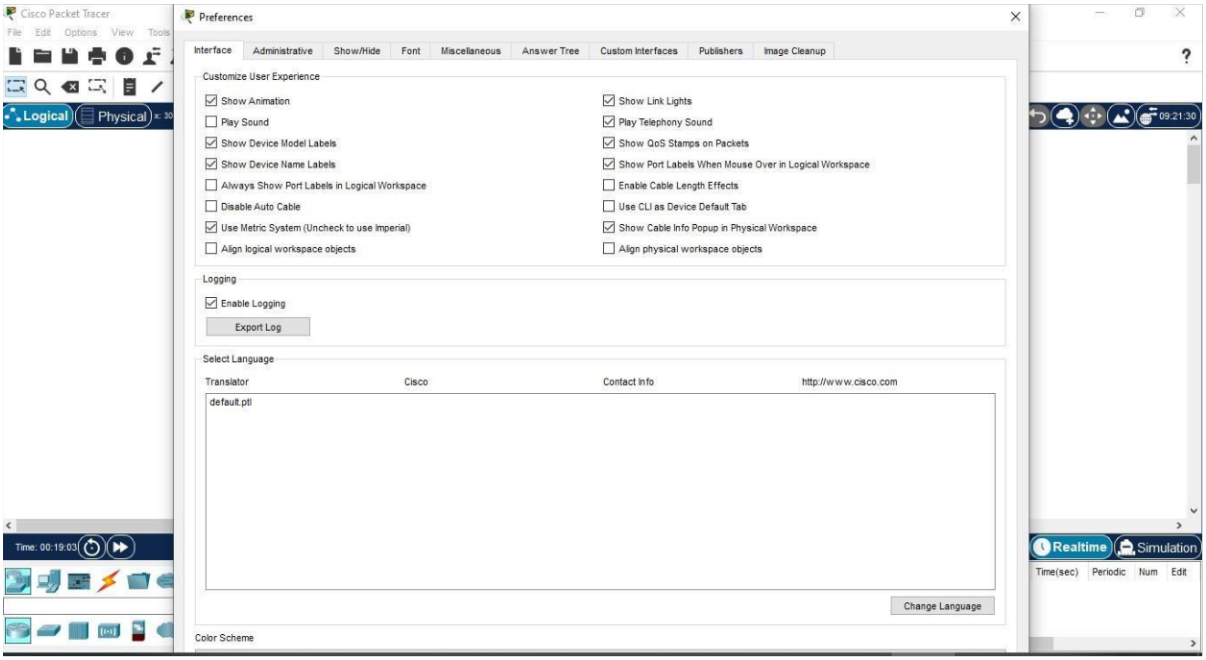
In Packet Tracer, you will design your own network (PT). Additionally, you will learn about the many sorts of PT files.

1.b Packet Tracer UI:



Packet Tracer is a tool that allows you to simulate real networks. It provides three main menus that you can use for the following:

- Add devices and connect them via cables or wireless.
- Select, delete, inspect, label, and group components within your network.
- Manage your network.



The network management menu lets you do the following:

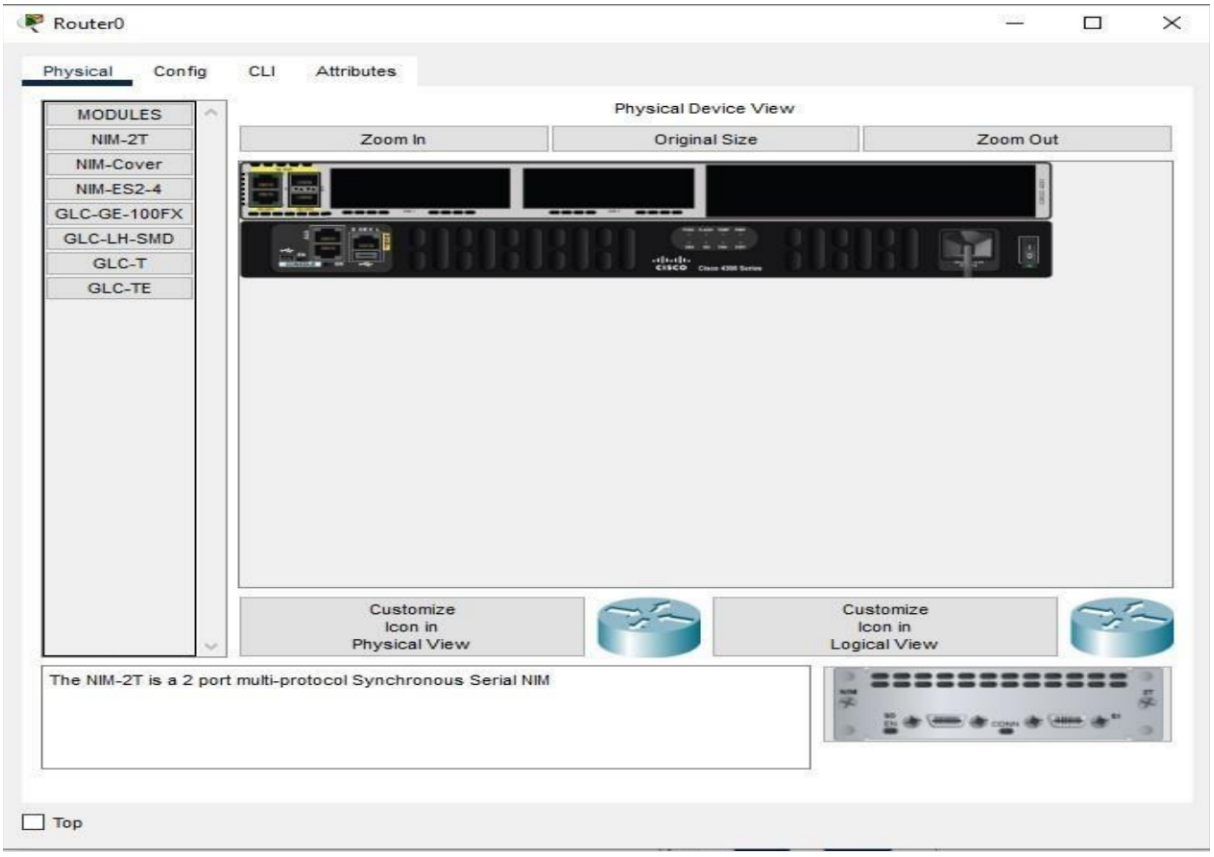
- Open an existing/sample network.
- Save your current network.
- Modify your user profile or your preferences.

Packet Tracer also provides a variety of tabs for device configuration including the following:

- Physical
- Config
- CLI
- Desktop
- Services

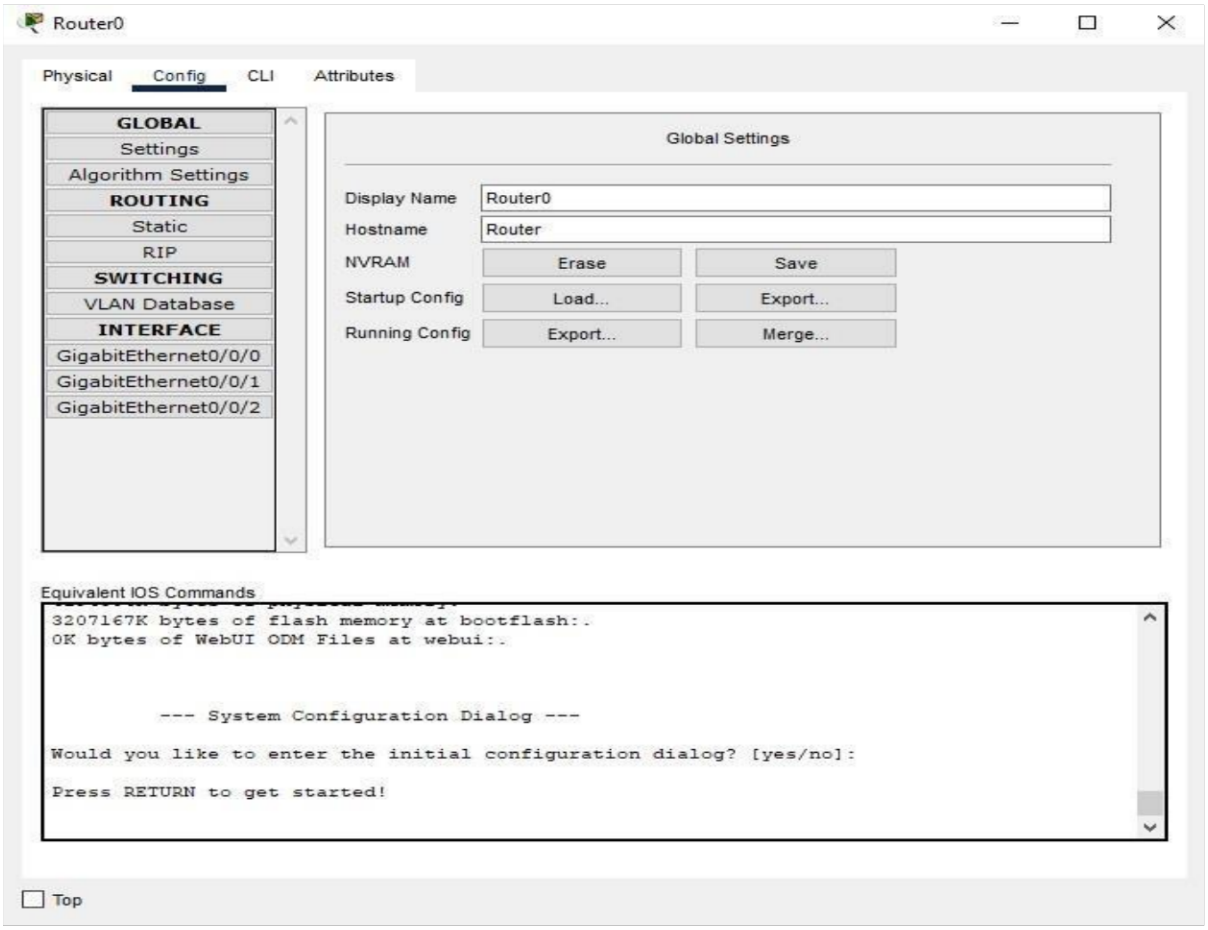
The tabs that are shown depend on the device you are currently configuring.

Physical Tab



The Physical tab provides an interface for interacting with the device including powering it on or off or installing different modules, such as a wireless network interface card (NIC).

Config Tab

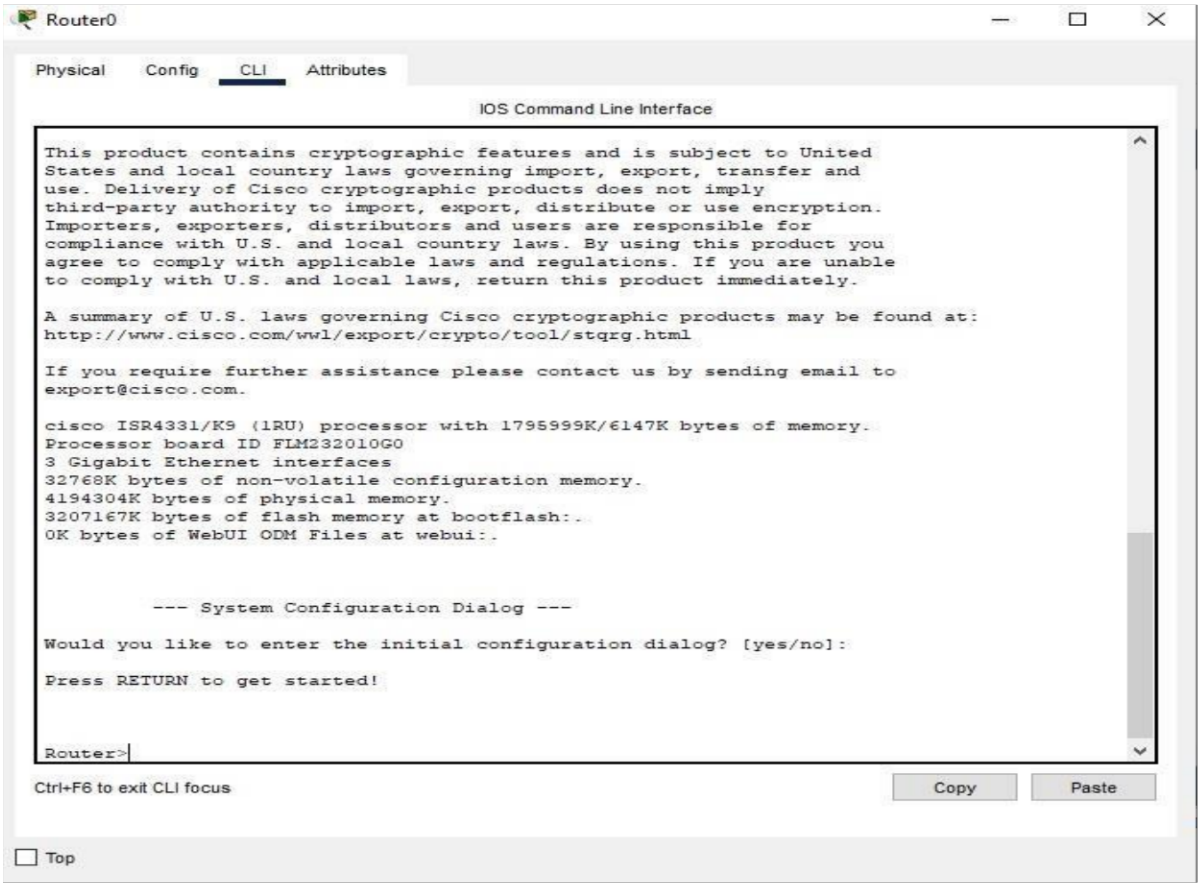


For intermediate devices such as routers and switches, there are two ways to access device configurations. Configurations can be accessed via a Config tab, which is a Graphical User Interface (GUI). Configurations can also be accessed using a command line interface (CLI).

The Config tab does not simulate the functionality of a device. This tab is unique to Packet Tracer. If you don't know how to use the command line interface, this tab provides a way to use a Packet Tracer-only GUI to configure basic settings. As settings are changed in the GUI, the equivalent CLI commands appear in the Equivalent IOS Commands window. This helps you to learn the CLI commands and the Cisco Internetwork Operating System (IOS) while you are using the Config tab.

For example, in the figure, the user has configured MyRouter as the name of the device. The Equivalent IOS Commands window shows the IOS command that achieves the same results in the CLI. In addition, device configuration files can be saved, loaded, erased, and exported here.

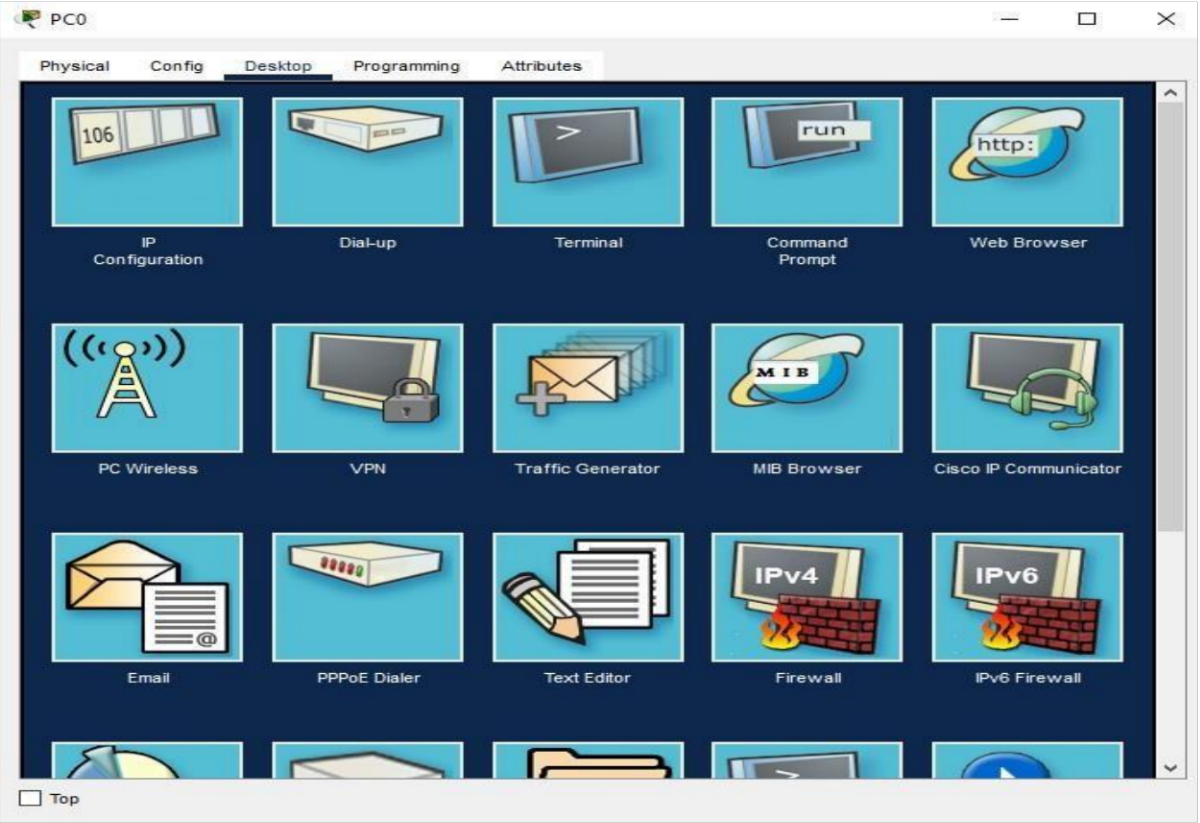
CLI Tab



The CLI tab provides access to the command line interface of a Cisco device. Using the CLI tab requires knowledge of device configuration with IOS. Here, you can practice configuring Cisco devices at the command line. CLI configuration is a necessary skill for more advanced networking implementations.

Note: Any commands that were entered from the Config tab are also shown in the CLI tab.

Desktop Tab



For some end devices, such as PCs and laptops, Packet Tracer provides a desktop interface that gives you access to IP configuration, wireless configuration, a command prompt, a web browser, and other applications.

1.B NETWORKING COMMANDS

VNStat

It is one of the most complete network commands. It works on all Linux and BSD systems, and allows us to monitor network traffic from the console.

- Installation is simple and fairly quick, allowing monitoring of all network interfaces.
- With VNStat we can collect all traffic needed from any configured interface.
- One of the big differences between VNStat and other tools is that VNStat collects kernel data instead of the interface itself, which means a lighter execution for the system.
- It will not require administrator permissions to run.
- It has the ability to store gathered information so your information never goes missing, even if the system crashes or reboots itself.
- You can set Vnstat to listen to traffic, daily or by billing period, as well as many other options.
- It stands out for its flexibility when configuring the reading of traffic.
- Finally, it is possible to set Vnstat output to generate console graphics and even customize them with colours.

Ping (Unix/Windows)

Ping dates from the 70s and is known for being one of the most basic network commands. However, it is not as simple as we believe and has many more uses than those we already know. It is based on the ICMP protocol and is used to determine:

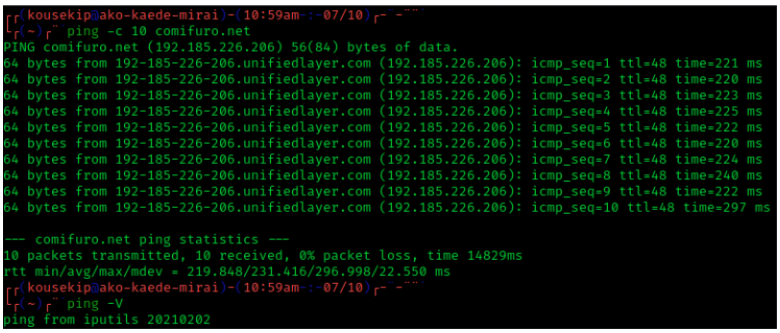
- If there is connectivity between your machine and another machine on the network.
- It's used to measure the "speed" or latency time.

It is a command that exists on all operating systems that support TCP/IP, and it is a basic command that you should know.

Ping is known for having dozens of parameters and the one that we find more useful is the one responsible for monitoring “the number of packages to send.” There are networks that undo the first package, so it is essential to send at least three so we can check that at least one has arrived without being discarded. For this, we use the -c parameter.

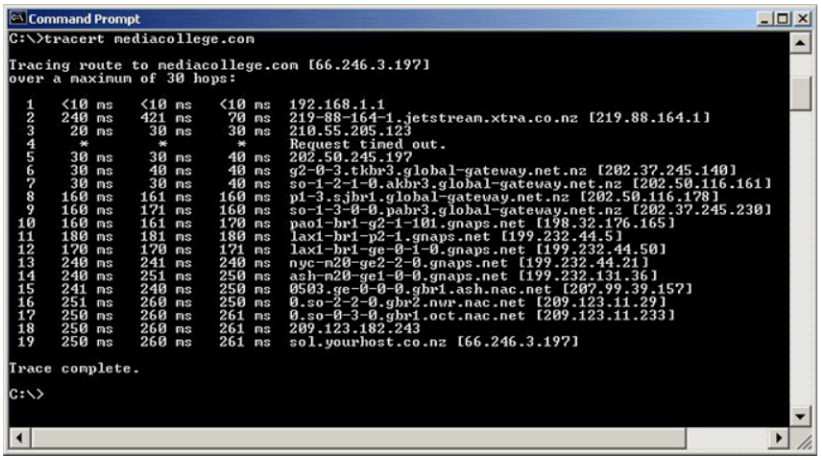
The same technique can be used to determine the loss percentage of packages in our network, sending ten packages and seeing if any gets lost. The number of packages that usually get lost in the network will surprise you. (This tool is included in Pandora FMS)

Execution: Ping name/System IP



Traceroute (Unix/Windows)

The main objective of this tool is to know the traveling path of a package through our network. This network command will tell us where the package is going through (machines, switches, routers) and check that our network is working properly. If you encounter any problems, it will allow us to have a rough idea about where the fault lies.



Execution:

traceroute –n (on Unix / Linux)

tracert –d (on Windows)

Arp (Unix/Windows)

This network command is used to change and view the ARP table, which contains the mappings between the IP address and the MAC address. It only sees the connections in our local area network segment (LAN), so it could be called “low level”. However, it’s used to discover what machines are directly connected to our host or what machines we are connected to. It is a diagnostic tool, and sometimes it can be interesting to monitor it in order to discard ARP Poisoning attacks, which are one of the most common forms of phishing attacks in local networks.

Execution: arp –a

Curl and wget (Unix/ Windows)

These are essential commands to do HTTP, HTTPS or FTP requests to remote servers. It allows you to download files or whole web pages, even recursively (it literally allows us to make a “copy” of a website, including images). It supports cookies and allows you to send POST requests, in addition to “simulate a” user agent, use a http proxy or even a SOCKS4/5 proxy.

One of the most common utilities in integration with Pandora FMS, is to verify the contents of a specific web page. Because wget / curl allows us to download the entire contents of a web, it is easy to compare the MD5 of that content with a value previously verified. If it changes, it means that the Web has been altered.

Netstat (Unix/Windows)

Network command identifies all TCP connections and UDP open on a machine. Besides this, it allows us to know the following information:

- Routing tables to meet our network interfaces and its outputs.
- Ethernet statistics that show sent and received packages and possible errors.
- To know the id of the process that is being used by the connection.
- Netstat is another basic command as Ping that meets many elementary functions.

Whois (Unix/ Windows)

This network command is used to query data domains: to find out who owns the domain, when that domain expires, to view the configured logs, contact details, etc. Its use is

highly recommended to contact the administrators of the domains or when incidents of migration of services such as mail and web happen.

To use 'whois' on Windows you need to download the software from this url: https://technet.microsoft.com/en-us/sysinternals/whois.aspx

SSH (Unix/Linux/Windows)

Command to run terminals on remote machines safely. SSH allows any user to run a console just by registering and entering his credentials. So you can run the commands you want as if you were in local.

More details you need to know about SSH:

Putty is recommended when using SSH in Windows. You can find it here: http://www.putty.org/

- To enable a remote computer to connect to our server via SSH, an SSH server must be installed and set up as FreeSSHd.
- SSH also allows to obtain an interactive remote Shell, execute remote commands and copy files in both directions.
- SSH is the natural replacement of classic tools like Telnet or FTP, and has become a basic tool in the administration of systems over the years. It is extremely powerful despite its complex combinations of symmetric encryption and authentication schemes, and verification, and it is the target of continuous attacks.

TCPDump (Unix/Linux/Windows)

It is one of the "basic" tools of network commands, and when used right, goes on to become a great ally for network administrators, system administrators or programmers.

TCPDump is an advanced command used to inspect traffic from different interfaces of a machine so you can get the exchanged packages. You can dump output to file so then you can analyse it with more powerful sniffers and graphical interfaces such as Wireshark. For Windows, you must use WinDump.

Ngrep (Unix/Linux/Windows)

- The grep command power is taken to the network.

- It is a TCPDump with a substring text filter in real time.
- It has a very powerful filtering system for regular expressions and it is typically used to process files generated by tcpdump, wireshark, etc.
- It is a communication package filter over HTTP, SMTP, FTP, DNS and other protocols.

NMAP (Unix/Windows)

NMAP is considered the father of the general network scanners. Although today there are more reliable tools for some tasks (like Fping), NMAP is a very versatile tool for scanning networks. It is used to determine which hosts are alive in a network and to do different ways of scanning.

Netcat (Windows/Unix)

NetCat, or NC, is the network command most versatile that exists nowadays and one of the lightest. However, its use requires some imagination. Only if you've played with scripting, you will understand the subtlety of its name: NetCat. It is a tool designed to be used as a destination of a redirect (one pipe or |). It is used to send or receive information about a connection. For example, a WEB request to service would be something as simple as:

echo -e "GET http://pandorafms.com HTTP/1.0\n\n" | nc pandorafms.com 80

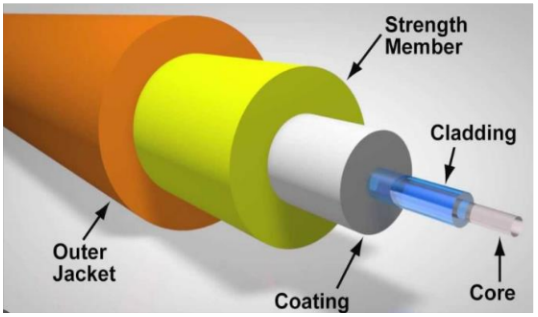
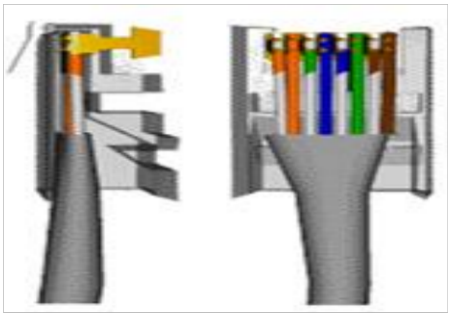
Lsof (Unix/Windows)

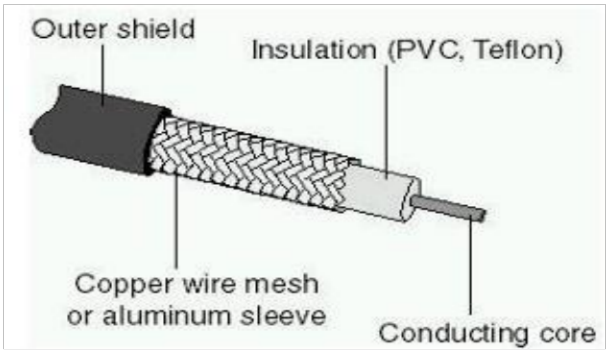
The 'lsof' command is not only used as a network tool, but also is used to identify which files have an open process. In Unix environments, a file can be a network connection, so that is used to know which ports have an open particular running process, something extremely useful in specific cases.

IPtraf (Linux)

Special command to obtain traffic statistics. It has a ncurses interface (text) to analyze real-time traffic passing through an interface. It allows you to work at low-level and to see what pairs of connections are established on each machine, and to see in detail the traffic connection of every pair, all in real-time.

1.C – Study of Cables & it's Color codes

Term	Description	Cross Sectional Diagram
BNC	British Naval Connector is the connector used with coaxial cables.	
RJ – 11	A registered jack 11 is a telephone connector used on modern telephone lines.	
RJ – 45	A registered jack 45 is an eight-wire connector used to connect computers to category 5 unshielded twisted pair (UTP) cables in a network.	
Fiber Optic Cable	Fiber optic cable uses light to transmit information across a network. The core of the cable is made of glass, which is protected by a layer of gel or plastic. A plastic cover surrounds the entire cable.	
UTP	Unshielded Twisted-Pair Cable is network cable that consists of up to 4 pairs of wires. Each pair is twisted around each other at a different rate and the entire cable is encased in a protective plastic covering. The twisting of the wires in cables is to help prevent EMI (Electro- Magnetic Interference).	

Coaxial Cable	Thin coaxial cable is often referred to as ThinNet. It consists of a copper wire surrounded first by a layer of plastic, then a layer of metal mesh and a final layer of protective plastic. It is used for peer-to-peer networking	

NETWORK DEVICES:

1. **Repeater:** Functioning at Physical Layer. A**repeater** is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports, so cannot be use to connect for more than two devices.

2. **Hub:** An **Ethernet hub**, **active hub**, **network hub**, **repeater hub**, **hub** or **concentrator** is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

3. **Switch:** A**network switch** or **switching hub** is a computer networking device that connects network segments. The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.

4. **Bridge:** A **network bridge** connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term *bridge* formally means a device that behaves according to the IEEE 802.1D standard. A bridge and switch are very much alike; a switch being a bridge with numerous ports. *Switch* or *Layer 2 switch* is often used interchangeably with *bridge* .Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.

5. **Router:** A **router** is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large collection of interconnected networks, the routers exchange information about target system addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

6. **Gate Way:** In a communications network, a network node equipped for interfacing with another network that uses different protocols.

• A gateway may contain devices such as protocol translators, impedance matching devices, rate converters, fault isolators, or signal translators as necessary to provide system interoperability. It also requires the establishment of mutually acceptable administrative procedures between both networks.

• A protocol translation/mapping gateway interconnects networks with different network protocol technologies by performing the required protocol conversions.

EXERCISE 2: CABLING – STRAIGHT THROUGH AND CROSS-OVER CABLING

Ethernet cable:

An Ethernet cable is a network cable used for high-speed wired network connections between two devices. This network cable is made of four-pair cable, which is consists of twisted pair conductors. It is used for data transmission at both ends of the cable, which is called RJ45 connector.

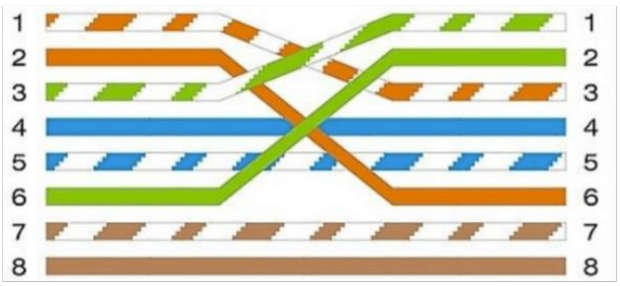
The Ethernet cables are categorized as Cat 5, Cat 5e, Cat 6, and UTP cable. Cat 5 cable can support a 10/100 Mbps Ethernet network while Cat 5e and Cat 6 cable to support Ethernet network running at 10/100/1000 Mbps.

Straight Through Cable:

Straight Through Cable

Straight-through cable is a type of CAT5 with RJ-45 connectors at each end, and each has the same pin out. It is in accordance with either the T568A or T568B standards. It uses the same color code throughout the LAN for consistency. This type of twisted-pair cable is used in LAN to connect a computer or a network hub such as a router. It is one of the most common types of network cable.

Crossover Cable:



Crossover Cable

A Crossover cable is a type of CAT 5 where one end isT568A configuration and the other end as T568BConfiguration. In this type of cable connection, Pin 1 is crossed with Pin 3, and Pin 2 is crossed with Pin 6.

Crossover cable is used to connect two or more computing devices. The internal wiring of crossover cables reverses the transmission and receive signals. It is widely used to connect two devices of the same type: e.g., two computers or two switches to each other.

In regard to physical appearance, Crossover Ethernet cables are very much similar to regular Ethernet cables. Still, they are different with regard to the order with which the wires are arranged. This type of Ethernet cable is made to connect to network devices of the same kind over Ethernet directly. Crossover cables are mostly used to connect two hosts directly.

Devices Connectivity:

DEVICES	HUB	SWITCH	ROUTER	PC
HUB	**CO**	**CO**	**ST**	**ST**
SWITCH	**CO**	**CO**	**ST**	**ST**
ROUTER	**ST**	**ST**	**CO**	**CO**
PC	**ST**	**ST**	**CO**	**CO**

EXERCISE 2.A DEMONSTRATION OF P2P NETWORK

Objective:

To demonstrate the Copper Cross-over cabling by designing a Peer to Peer Network

Components:

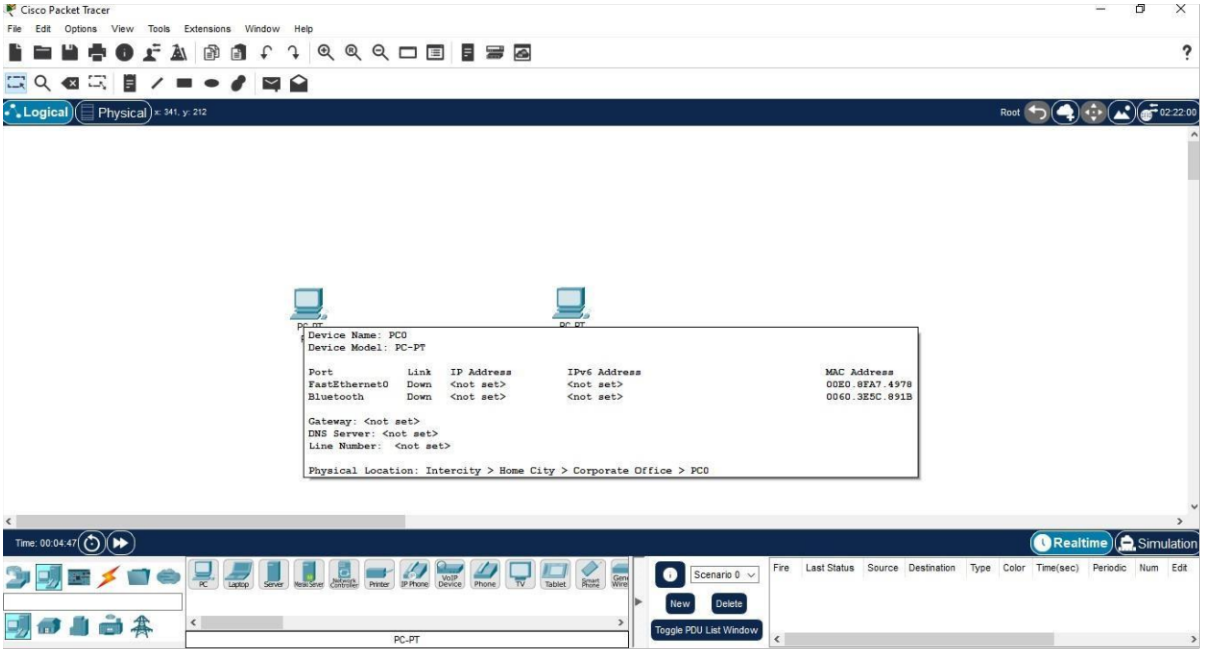
Devices	Required Nos
PCs	2
Copper Cross – Over Cable	1

Addressing Table:

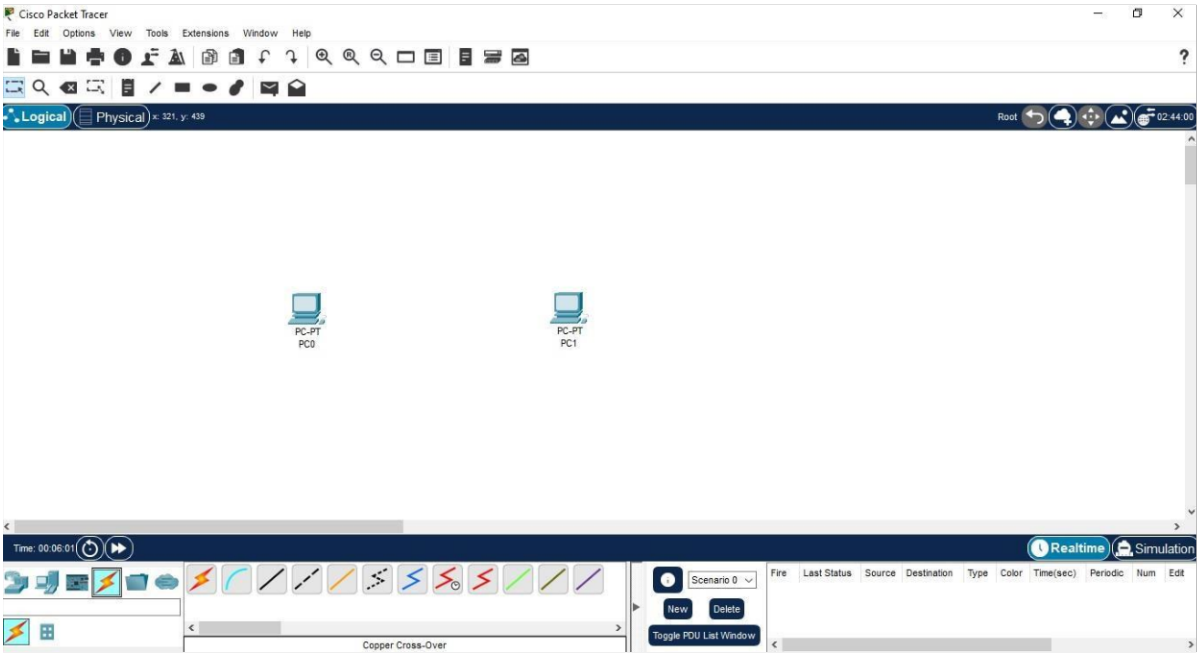
Device	Interface	IP Address	Subnet Mask
PC0	Fa0/0	192.168.10.1	255.255.255.0
PC1	Fa0/0	192.168.10.2	255.255.255.0

Procedure:

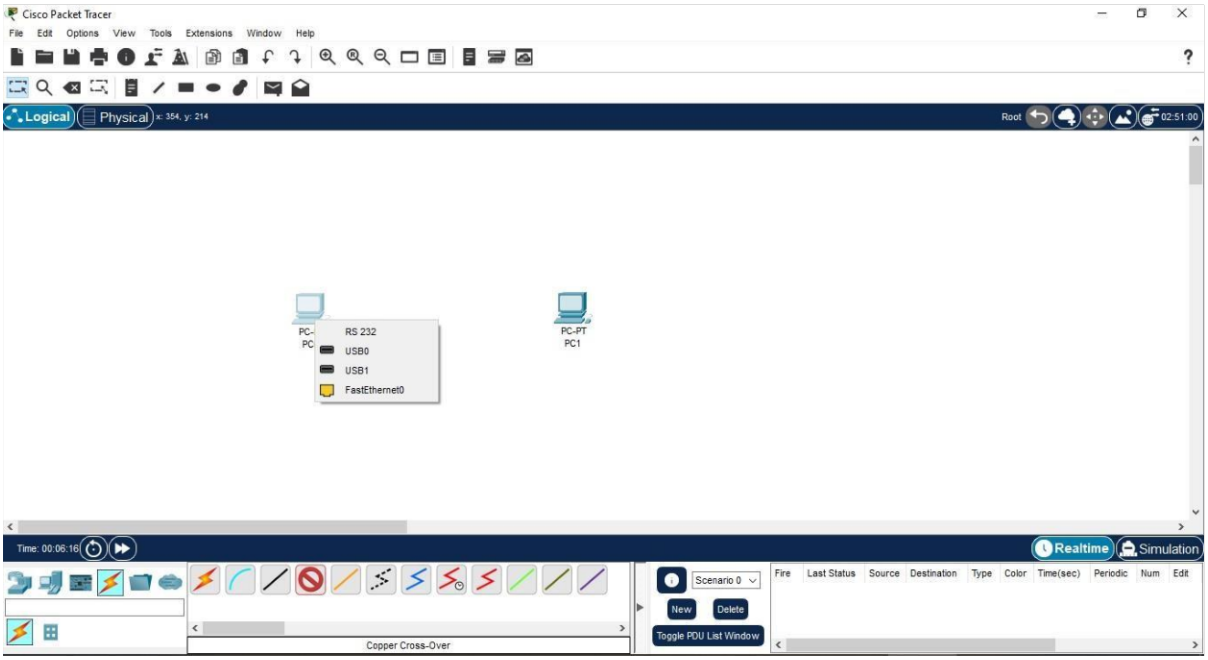
Step 1: Drag 2 PCs in the console area. Each PC will have interfaces as shown in the figure.



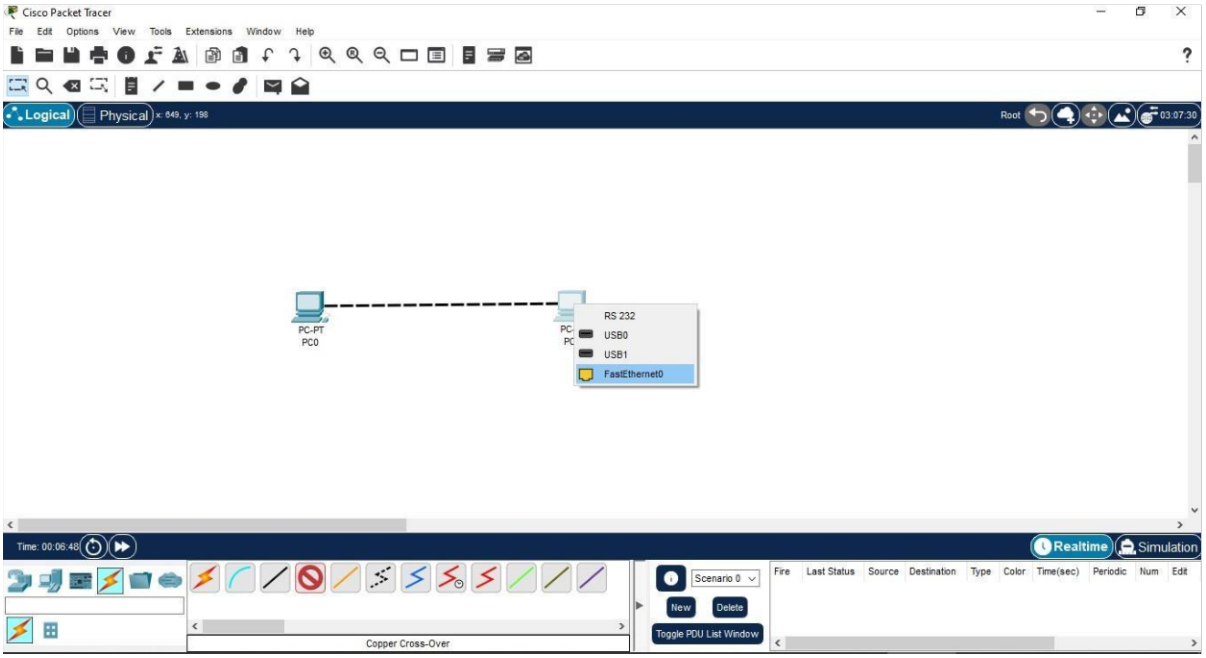
Step 2: Select Connectivity & Copper cross-over cable.



Step 3: Click on PC0 to get the interface options. Select Fa0/0

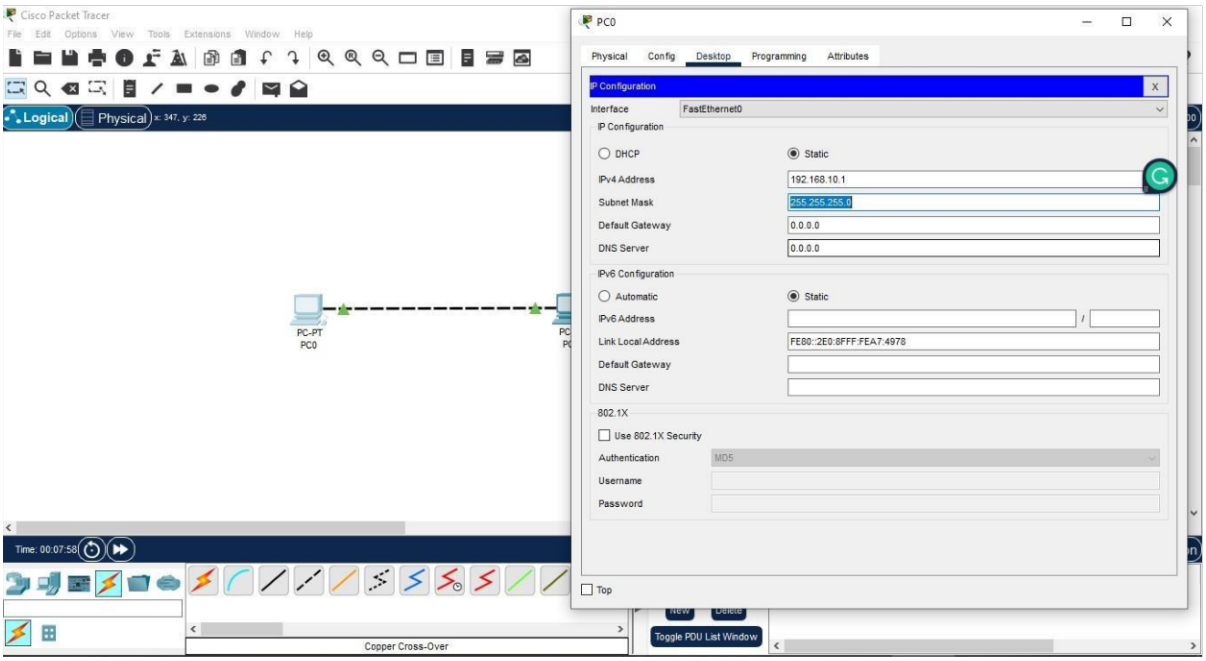


Step 4: Click on PC1 to get the interface options and select Fa0/0.

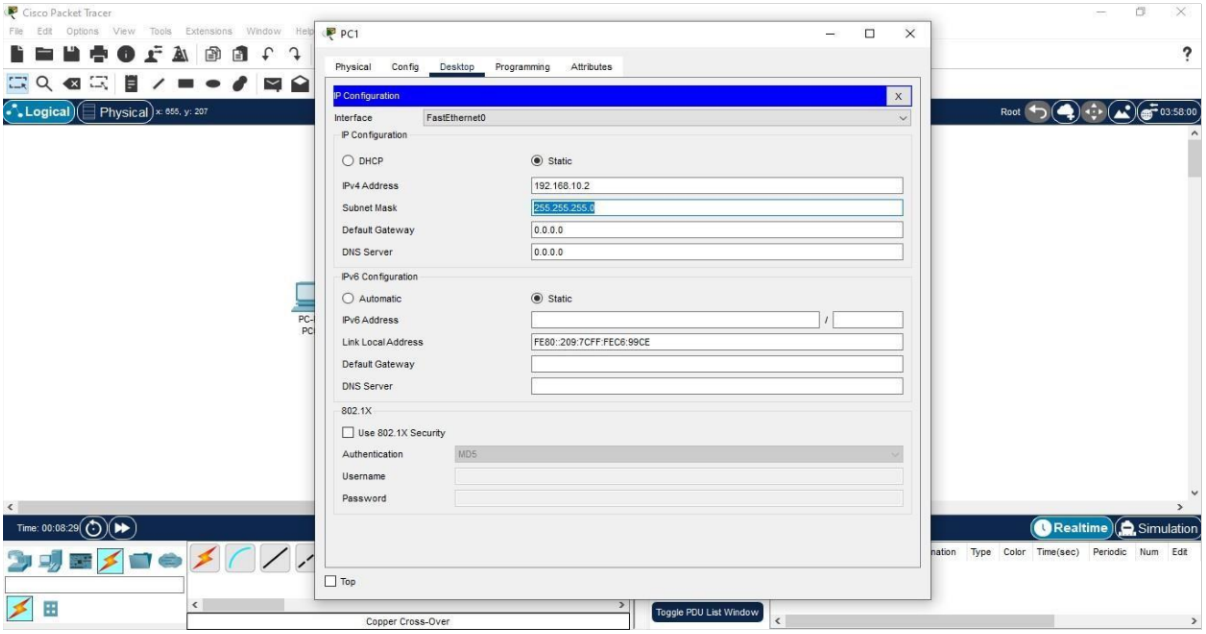


Step 5: Now the PCs are physically connected. To establish logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on IP Configuration icon.
- Configure as in the following figure

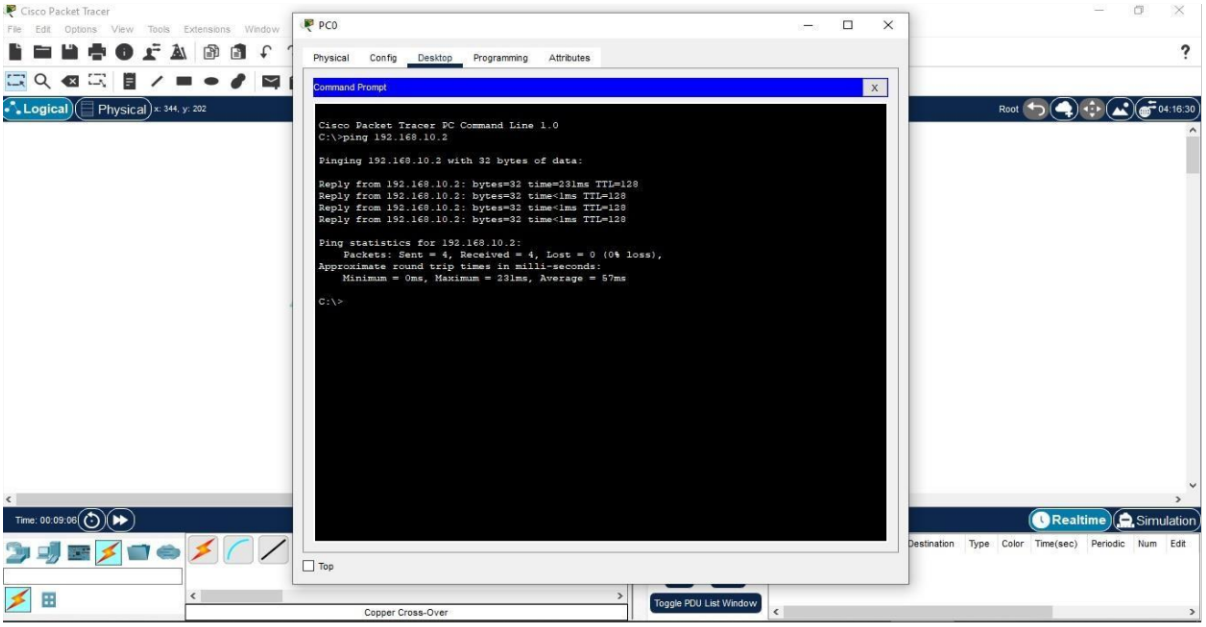


Step 6: Configure IP address for PC1 with the same procedure.



Step 7: Now both the PCs are physically and logically connected. To check the logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.2 to fetch the output as follows



EXERCISE 2.B - DEMONSTRATION OF SIMPLE LAN & TOPOLOGIES

Objective: To demonstrate the straight through cabling by designing a Local Area Network

Components:

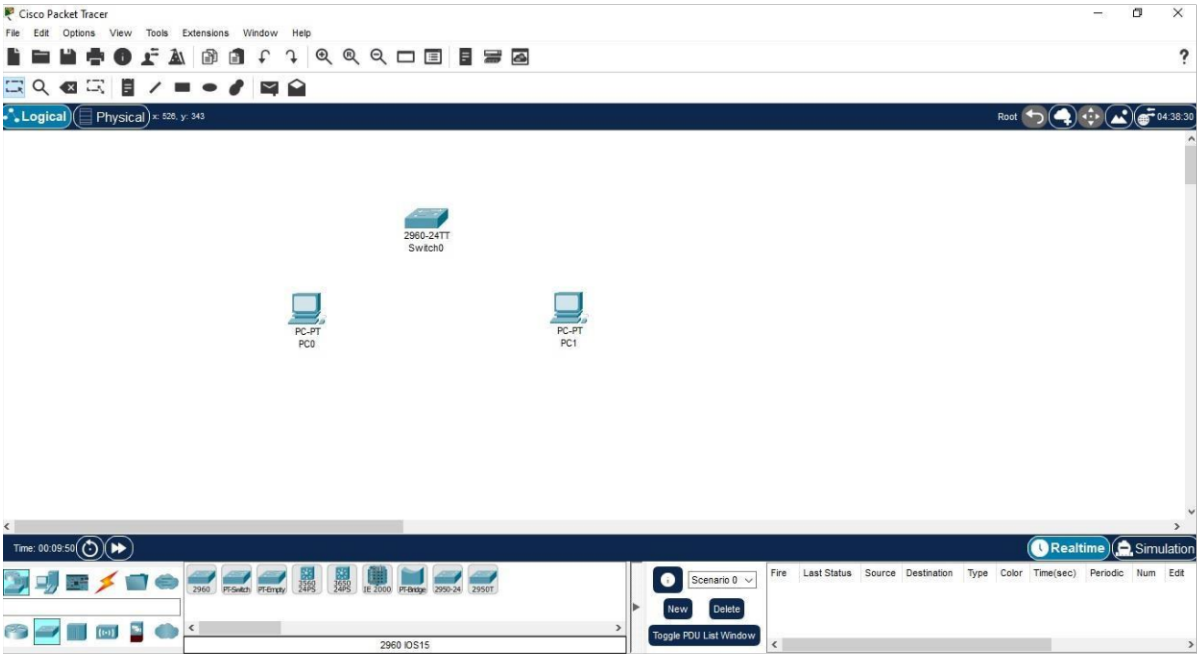
Devices	Required Nos
PCs	2
Copper Straight – Through Cables	2
Switch	1

Addressing Table:

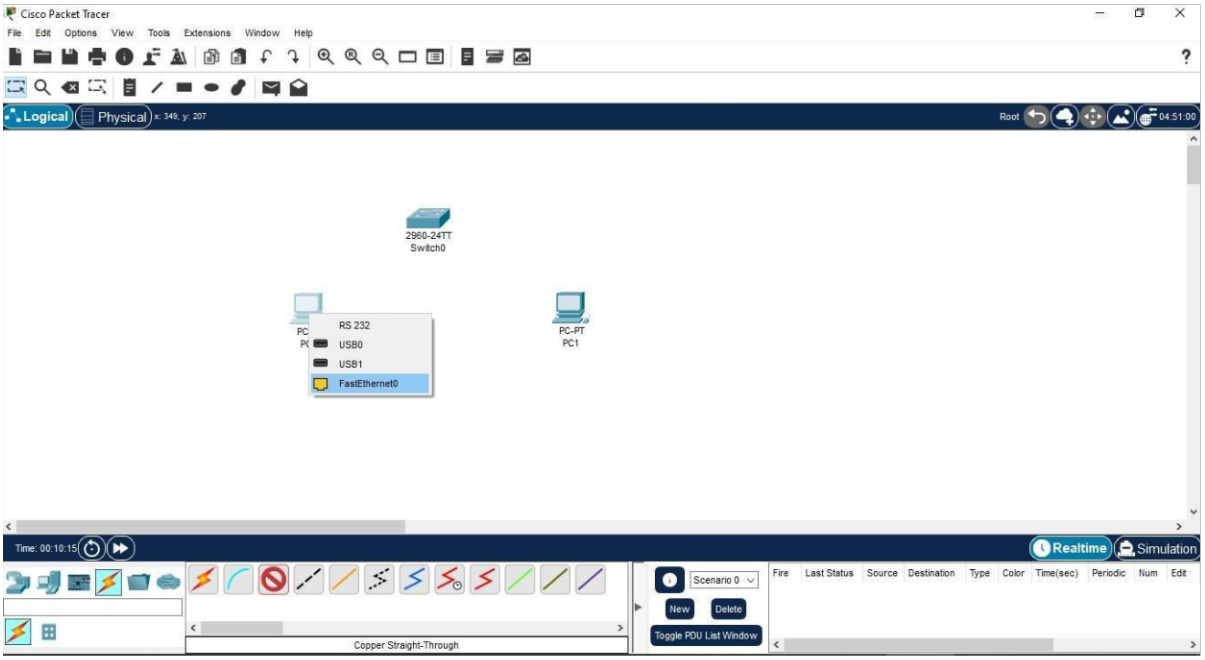
Device	Interface	IP Address	Subnet Mask
PC0	Fa0/0	192.168.10.1	255.255.255.0
PC1	Fa0/0	192.168.10.2	255.255.255.0

Procedure:

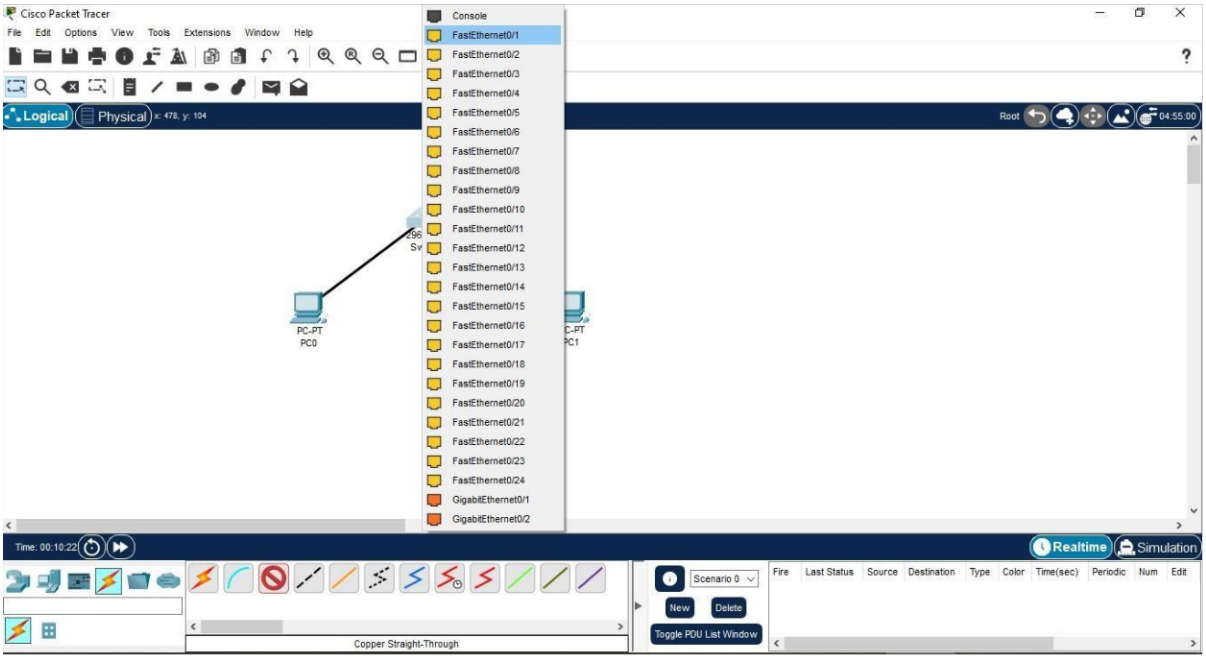
Step 1: Drag 2 PCs and a switch in the console area. Each PC will have interfaces as shown in the figure.



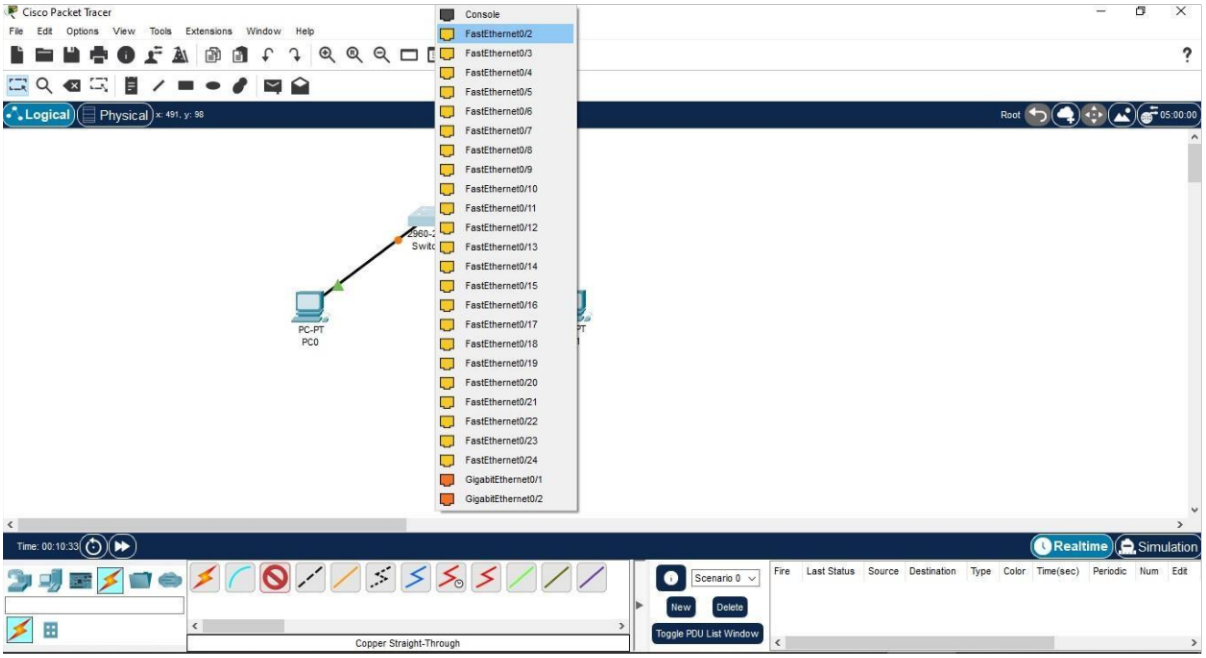
Step 2: Select Connectivity & Copper Straight-Through cable. Click on PC0 to get the interface options. Select Fa0/0



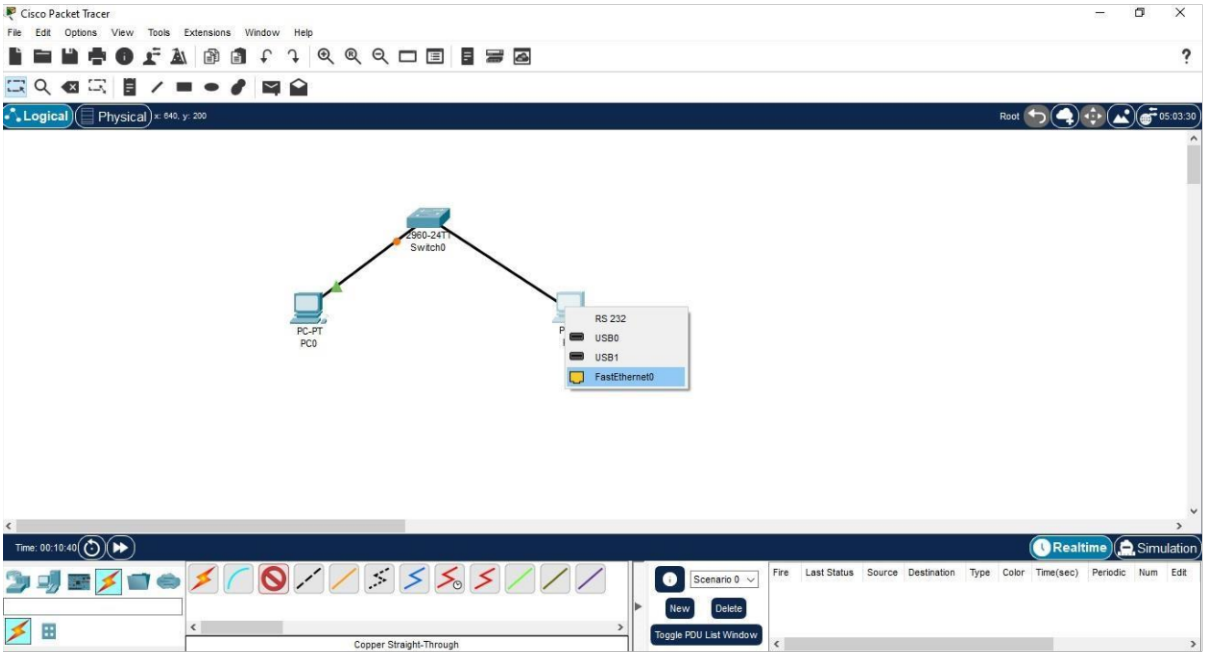
Step 3: Click on Switch to get the interface options and select Fa0/0.



Step 4: Now PC0 and Switch are physically connected. Again select copper straight-through cable and again click on Switch to get the interface options and select Fa0/1.



Step 5: Click on PC1 to get the interface options and select Fa0/0.

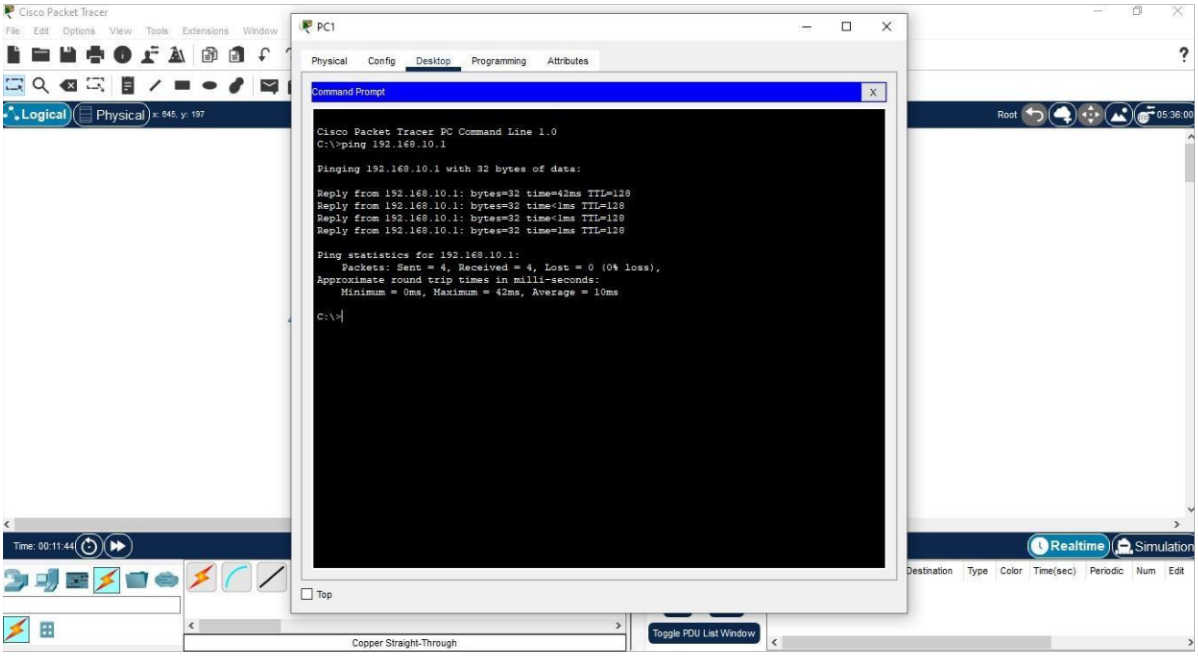


Step 6: Now the PCs are physically connected through switch. To establish logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on IP Configuration icon.
- Configure the ip address 192.168.10.1 and subnet mask 255.255.255.0
- Repeat the same procedure for PC1 and configure with the ip address 192.168.10.2 and subnet mask 255.255.255.0

Step 7: Now both the PCs are physically and logically connected. To check the logical connectivity,

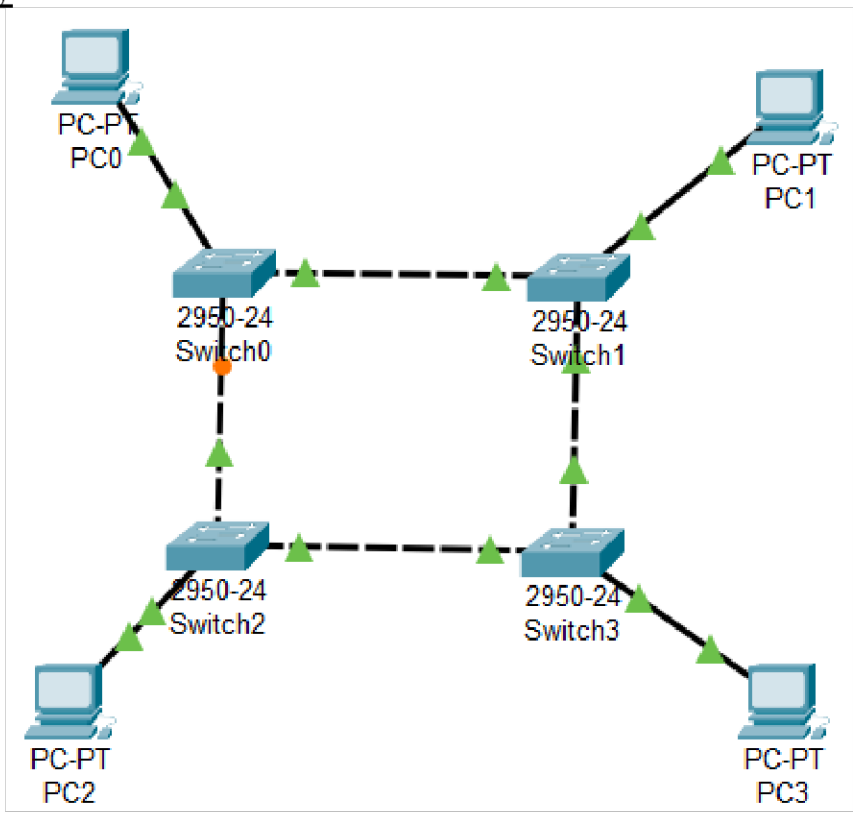
- Click on PC1.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.1 to fetch the output as follows



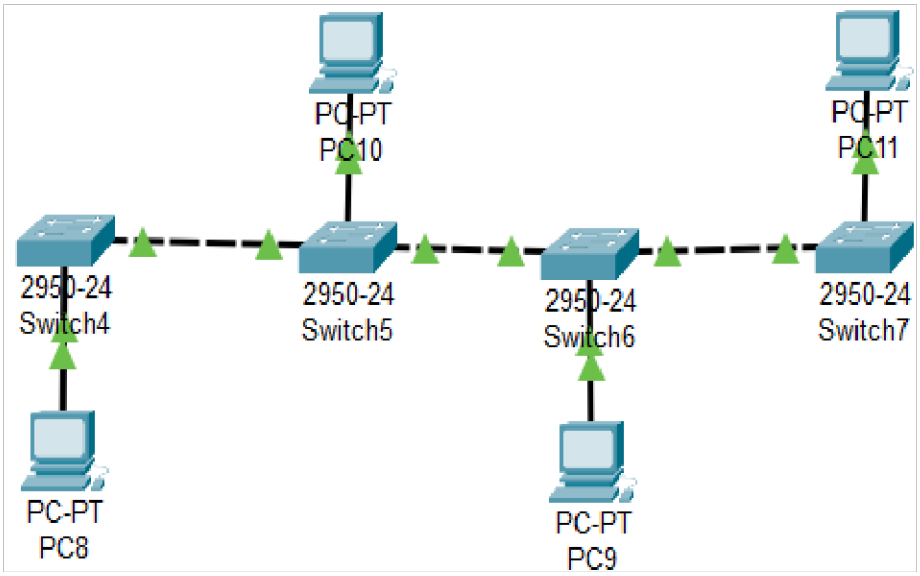
Now you have to implement the different types of Network Topologies as given below.

Note: Remember you need to assign IP address for all the PC's.

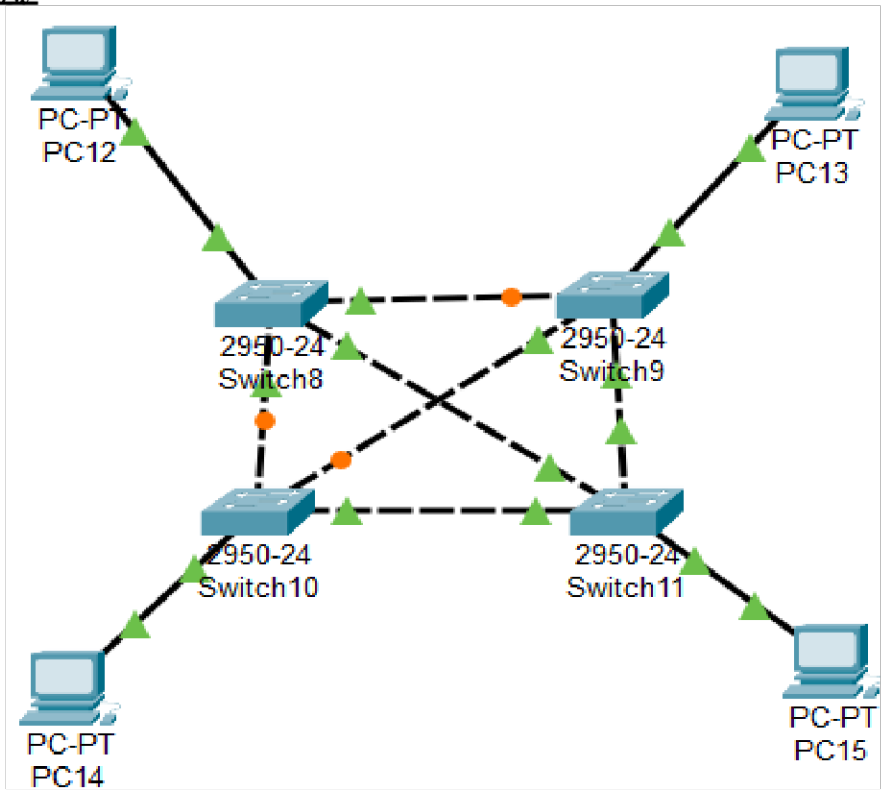
Ring Topology



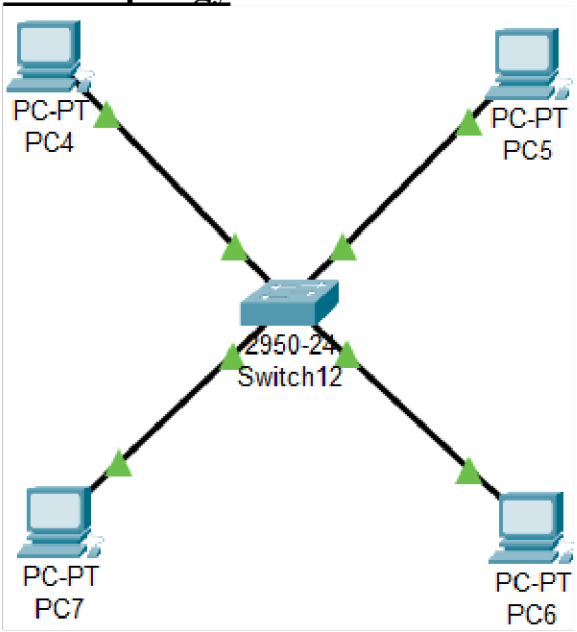
Bus Topology

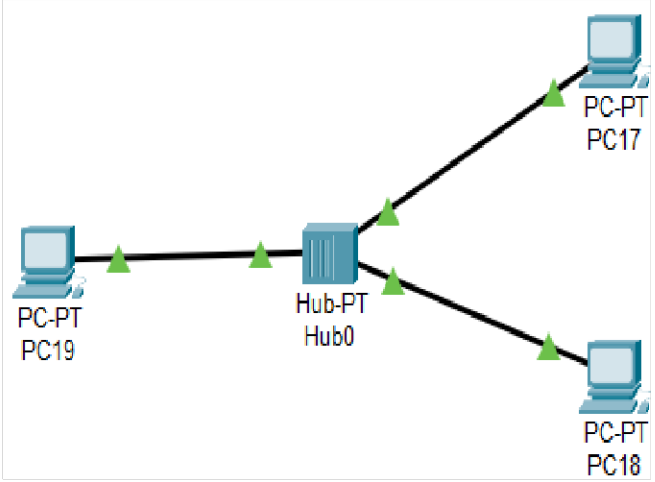


Mesh Topology

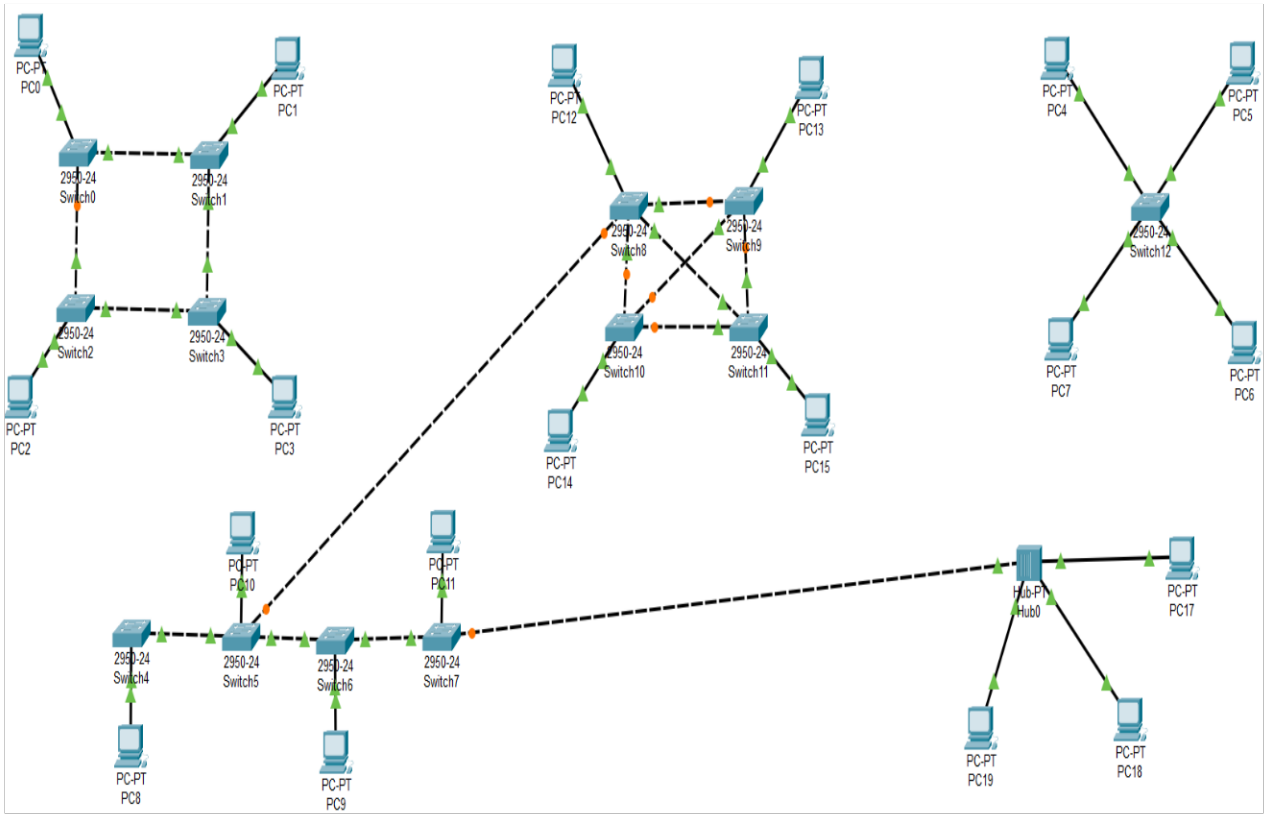


Star Topology





Hybrid Topology



Result:

Thus, the demonstration of P2P and Network topologies is done successfully.

EXERCISE 3: ROUTER CONFIGURATION THROUGH A CONSOLE

Objective:

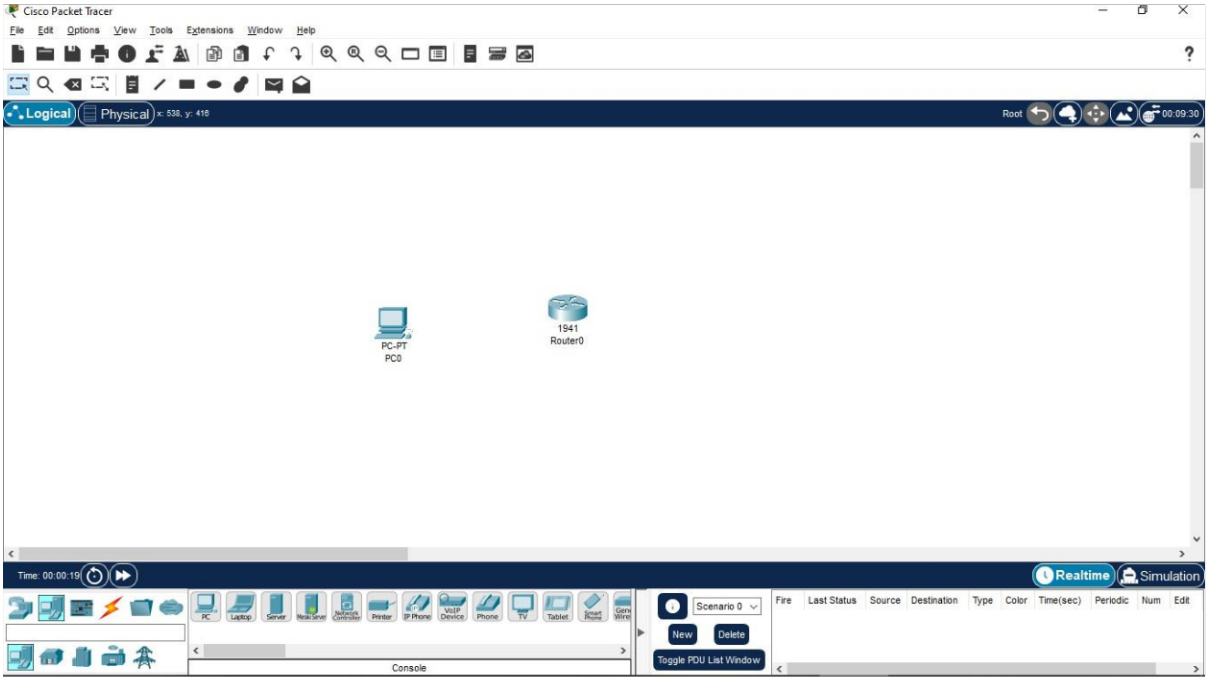
To demonstrate the configuration of Router with authentication through a console

Components:

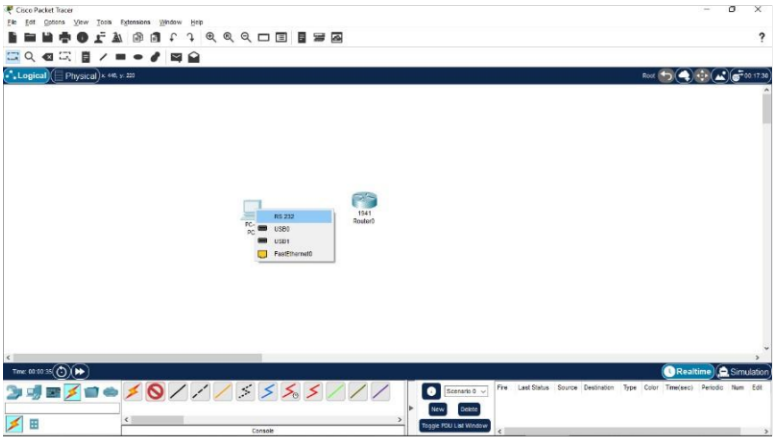
Devices	Required Nos
PCs	1
Console Cables	1
Router 1941	1

Procedure:

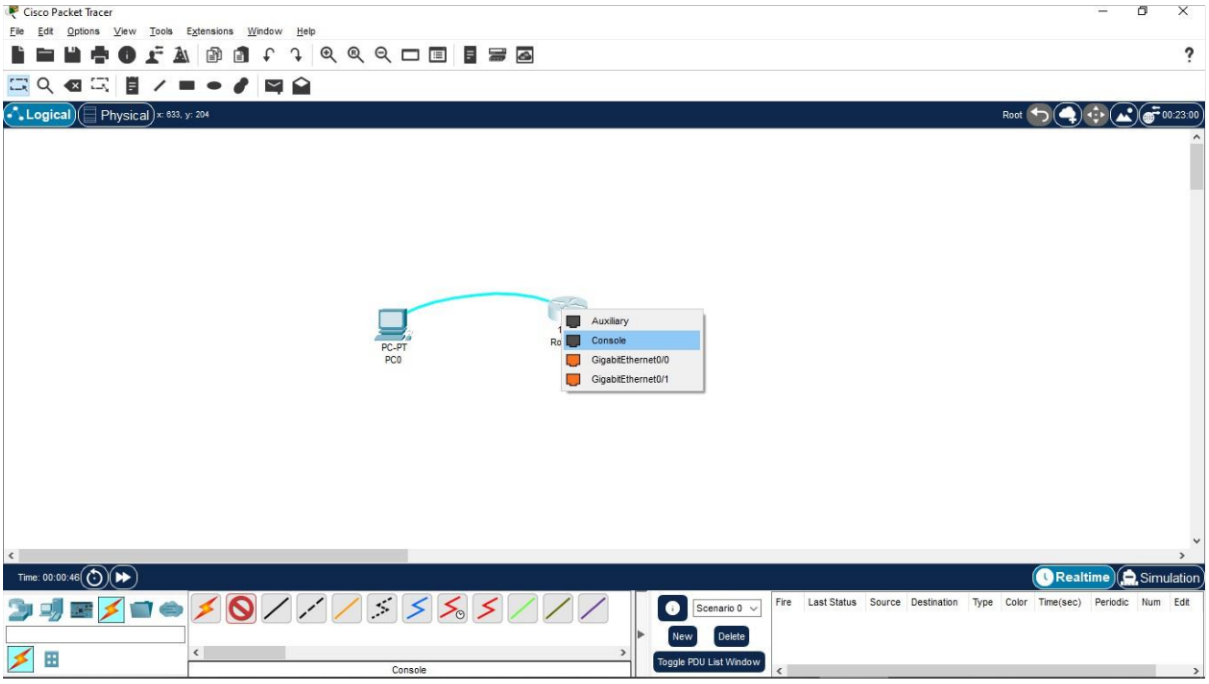
Step 1: Drag 1 PC and 1 number of 1941 Router in the console area



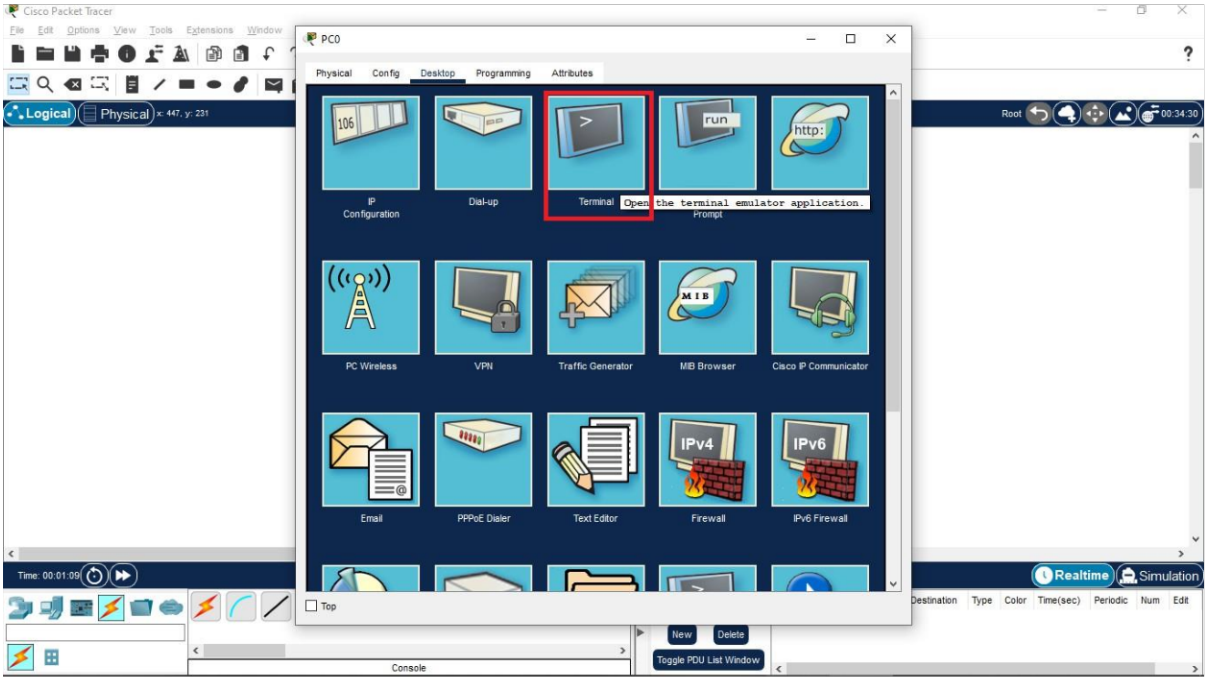
Step 2: Click on Connectivity and select Console cable. Click on PC and select RS232 interface.



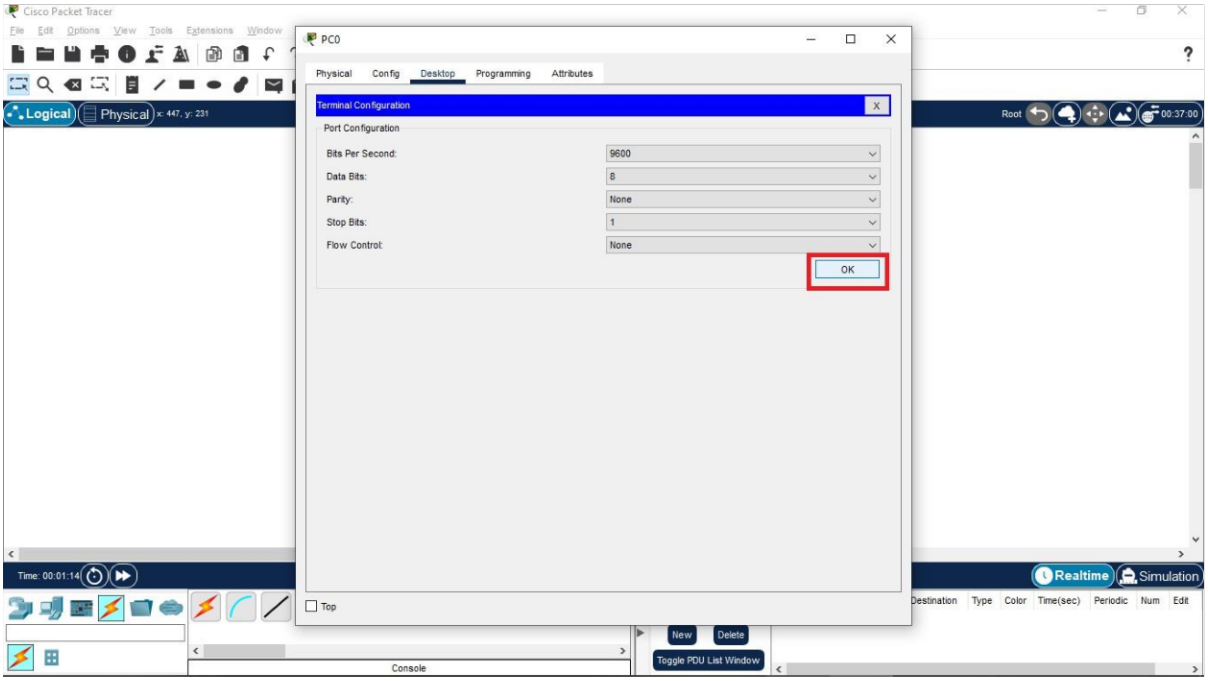
Step 3: Now click on Router and select Console interface to connect PC and Router with Console cable



Step 4: Now click on PC0 and go to Desktop => Terminal



Step 5: Click on OK for default Terminal parameters.

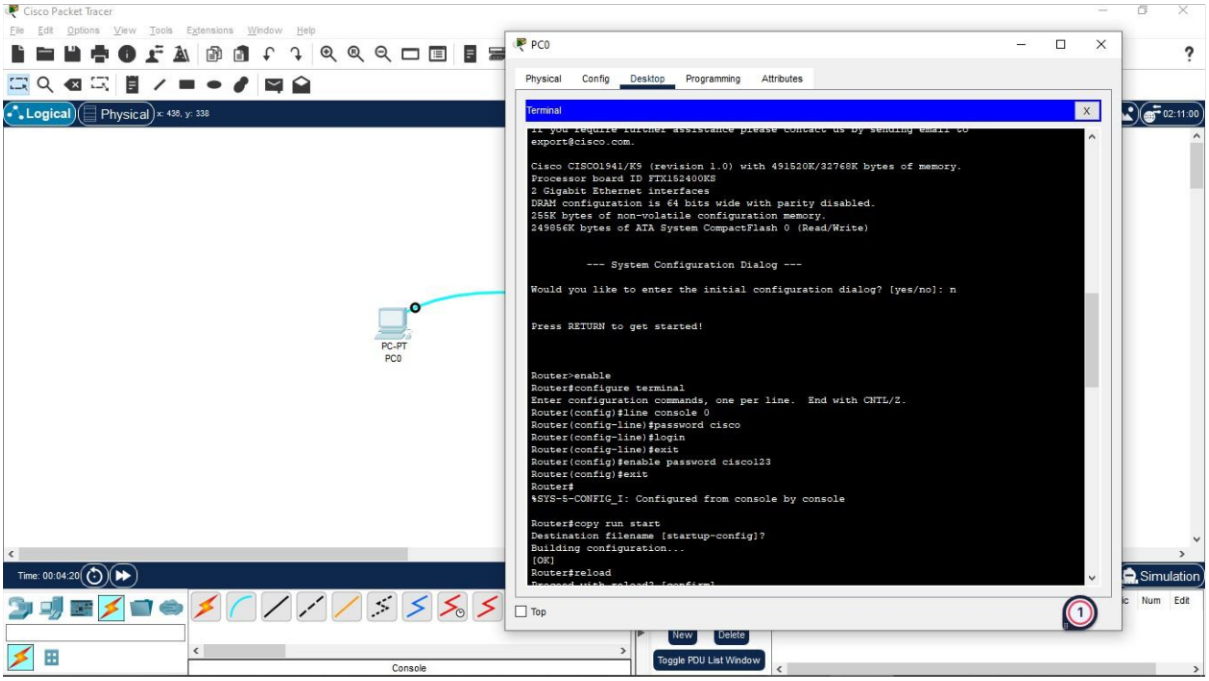


Step 6:

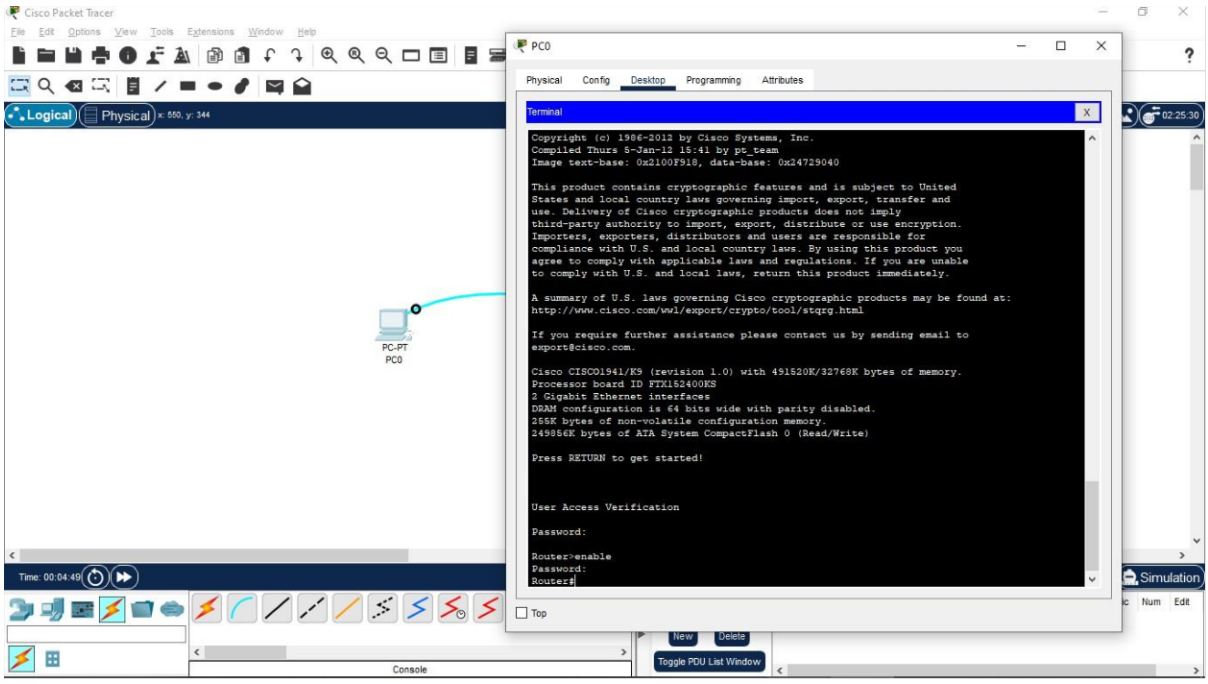
- Enter "no" when prompted, and type the following to configure "line console password" as "cisco" and privilege mode password as "cisco123"
- The "copy run start" command copies the running configuration parameters to start-up configuration parameters such that when you reload or reboot the router the configured parameters will not be erased from the Router memory RAM.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable password cisco123
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#copy run start
Destination filename [startup-config]?
```

```
Building configuration...
[OK]
Router#reload
Proceed with reload? [confirm]
```



Step 7: After reloading enter “cisco” as the password, when prompted to remote access and the password “cisco123” when you type “enable” to use Privilege mode access of Router from PC.



Configure Router Interfaces

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# description Link to LAN 1
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ip address 192.168.11.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# description Link to LAN 2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:acad:3::225/64
R1(config-if)# description Link to R2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
```

Result:

Thus, the initial configuration of a router is done successfully.

EXERCISE 4 - A: CONFIGURATION OF IP ADDRESS IN ROUTER

Objective: To demonstrate the configuration of IP Address in router

Pre-requisite: IP Address, Range of IP Address and Classes of IP Address

Components:

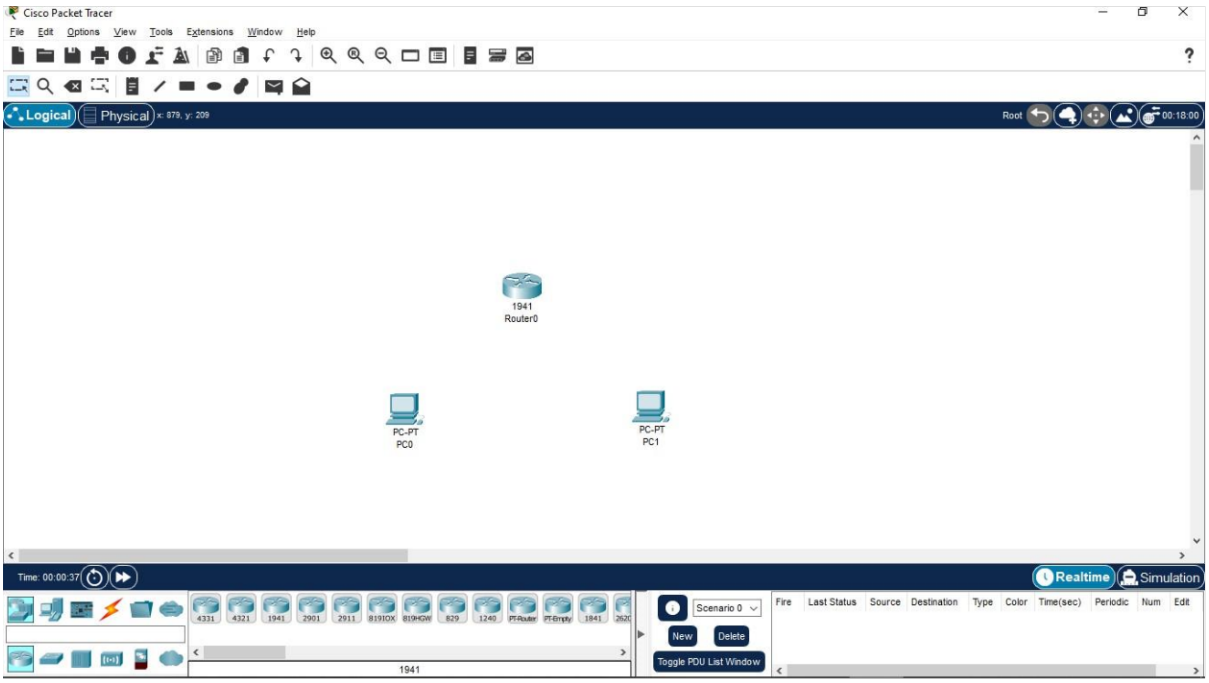
Devices	Required Nos
PCs	2
Copper cross-over Cables	2
Router	1

Addressing Table:

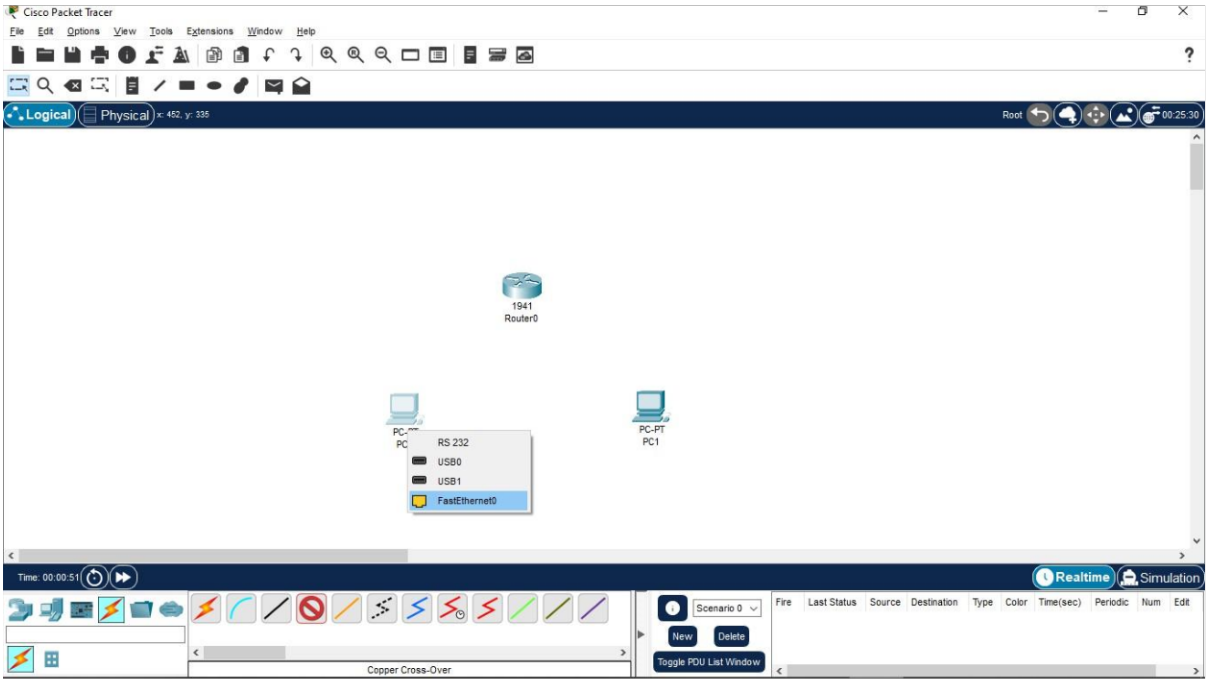
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/0	192.168.11.2	255.255.255.0	192.168.11.1
Router0	Gigabit 0/0	192.168.10.1	255.255.255.0	-
Router0	Gigabit 0/1	192.168.11.1	255.255.255.0	-

Procedure:

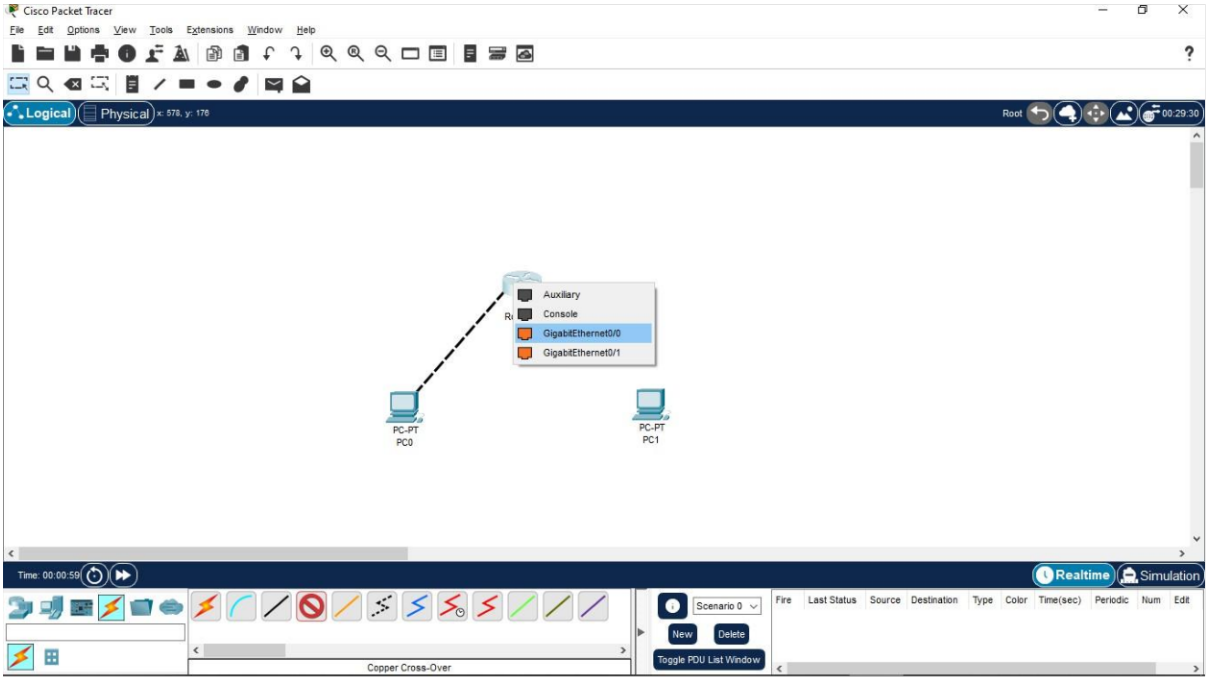
Step 1: Drag 2 PCs and a router in the console area.



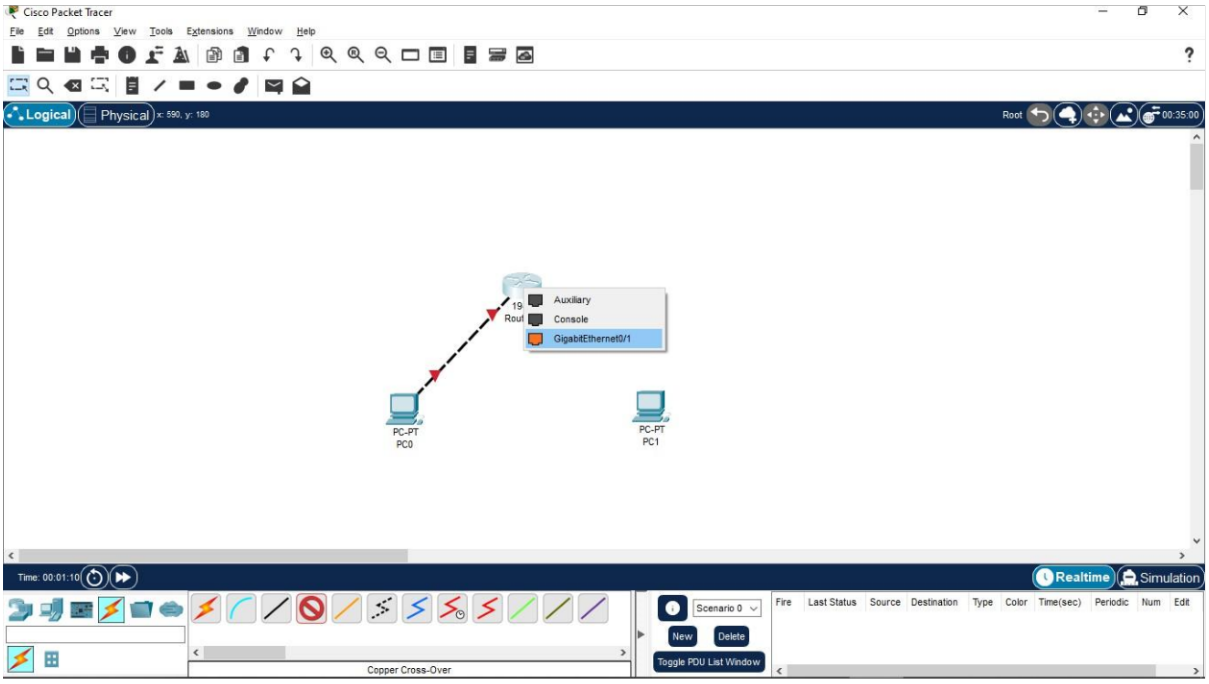
Step 2: Select Connectivity & Copper cross-over cable. Click on PC0 to get the interface options. Select Fa0/0



Step 3: Click on router0 to get the interface options and select GigabitEthernet0/0.



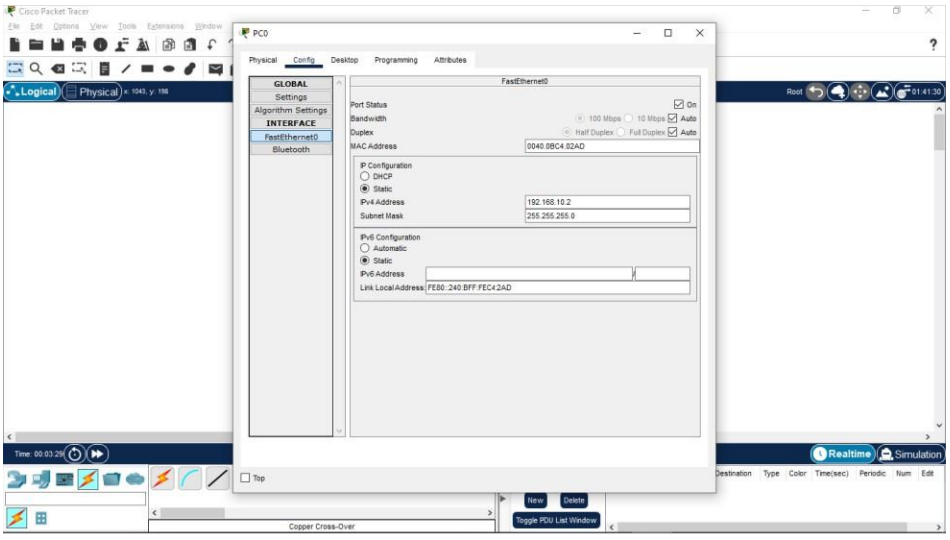
Step 4: Now PC0 and Router0 are physically connected. Again select copper cross-over cable and again click on Router0 to get the interface options and select GigabitEthernet0/1.



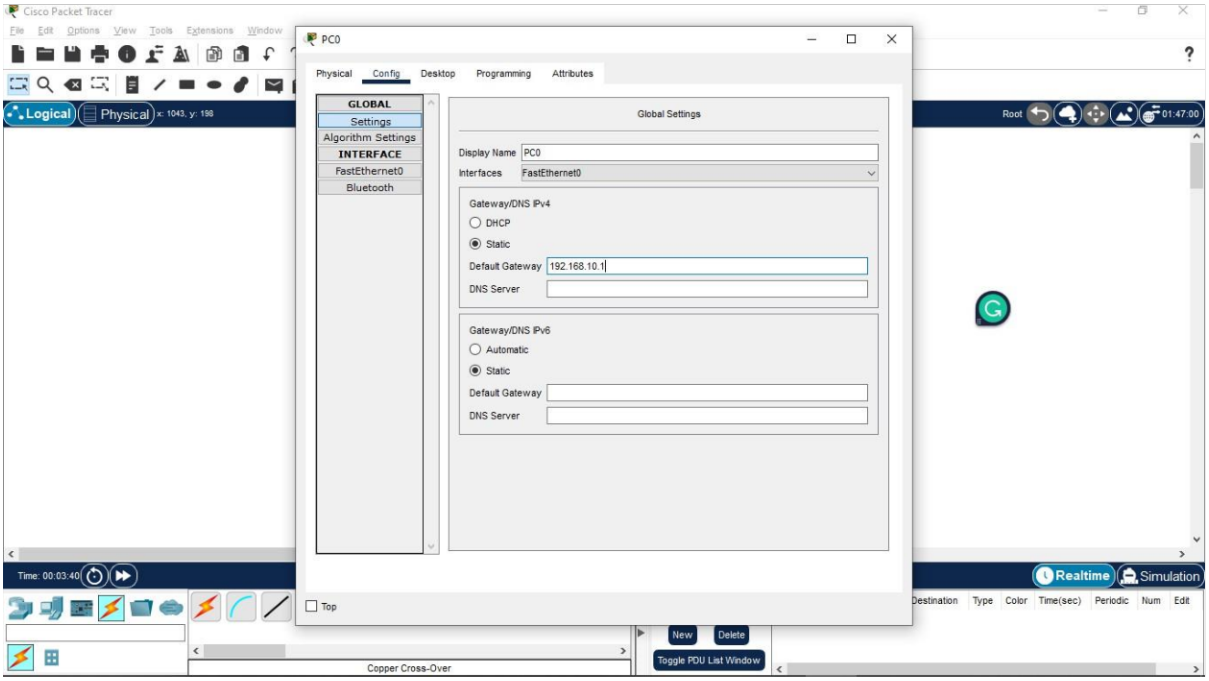
Step 5: Click on PC1 to get the interface options and select Fa0/0.

Step 6: Now the PCs are physically connected through Router. To establish logical connectivity,

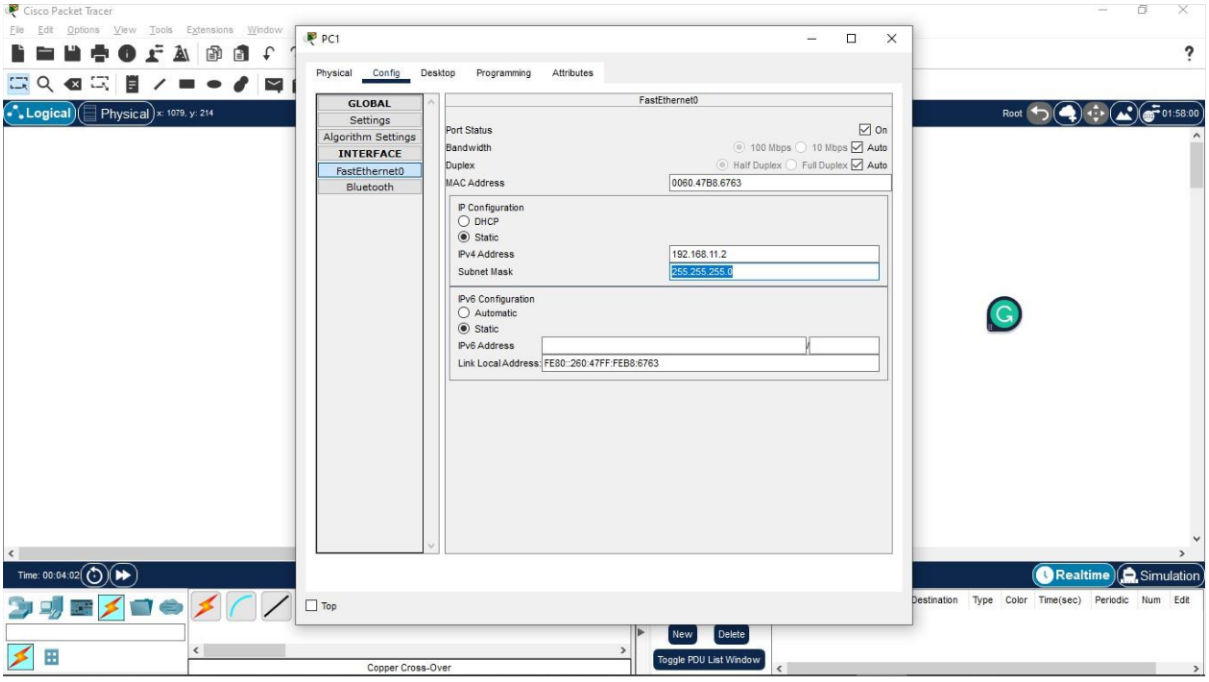
- Click on PC0.
- Select Config tab.
- Click on FastEhternet0/0 in the left pane.
- Configure the ip address 192.168.10.2 and subnet mask 255.255.255.0



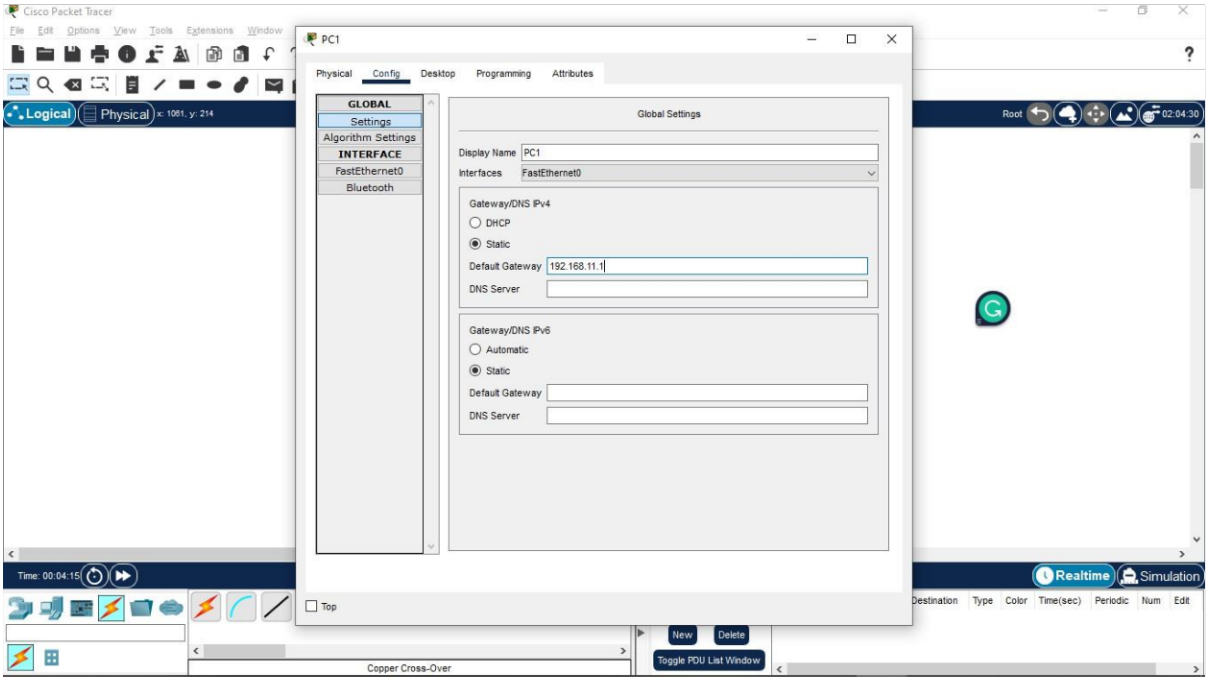
Step 7: Now click on settings and configure the gateway as 192.168.10.1



Step 8: Repeat the same procedure for PC1 and Configure the ip address 192.168.11.2 and subnet mask 255.255.255.0

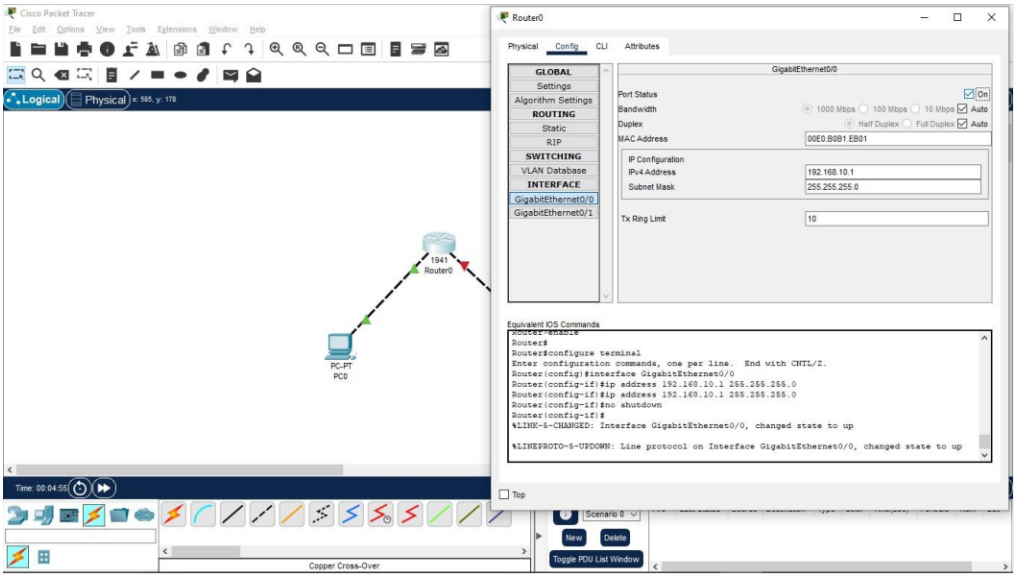


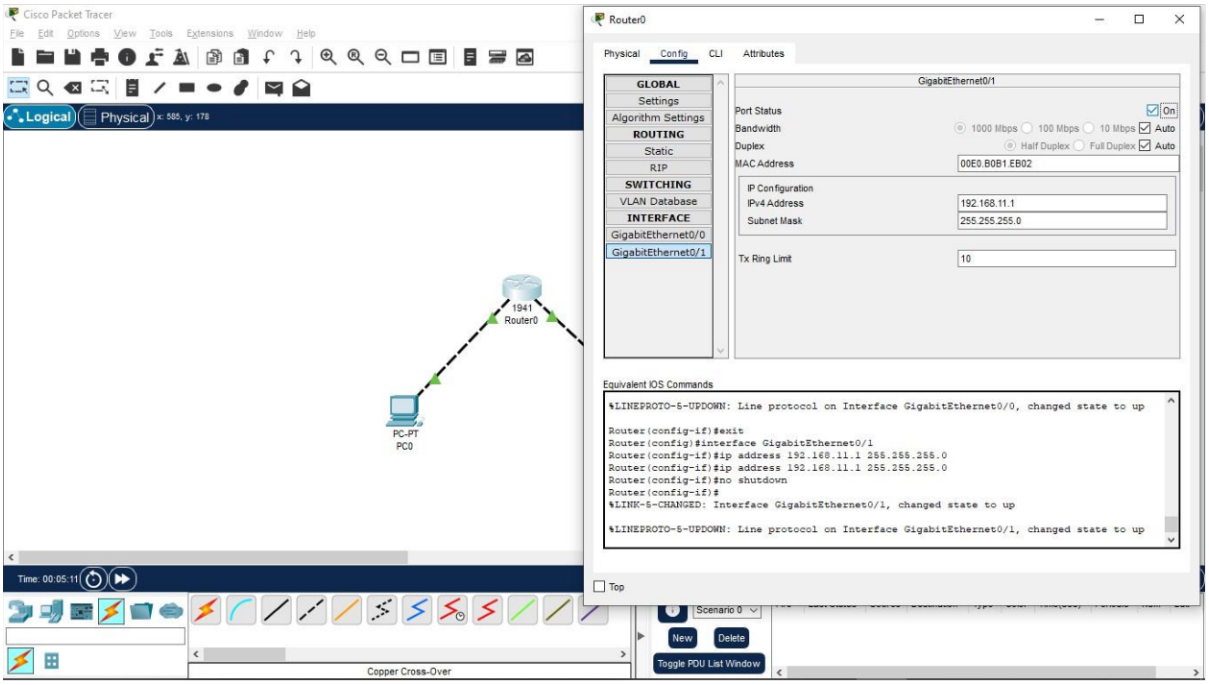
Step 9: Now click on settings and configure the gateway as 192.168.11.1



Step 10: Router configuration

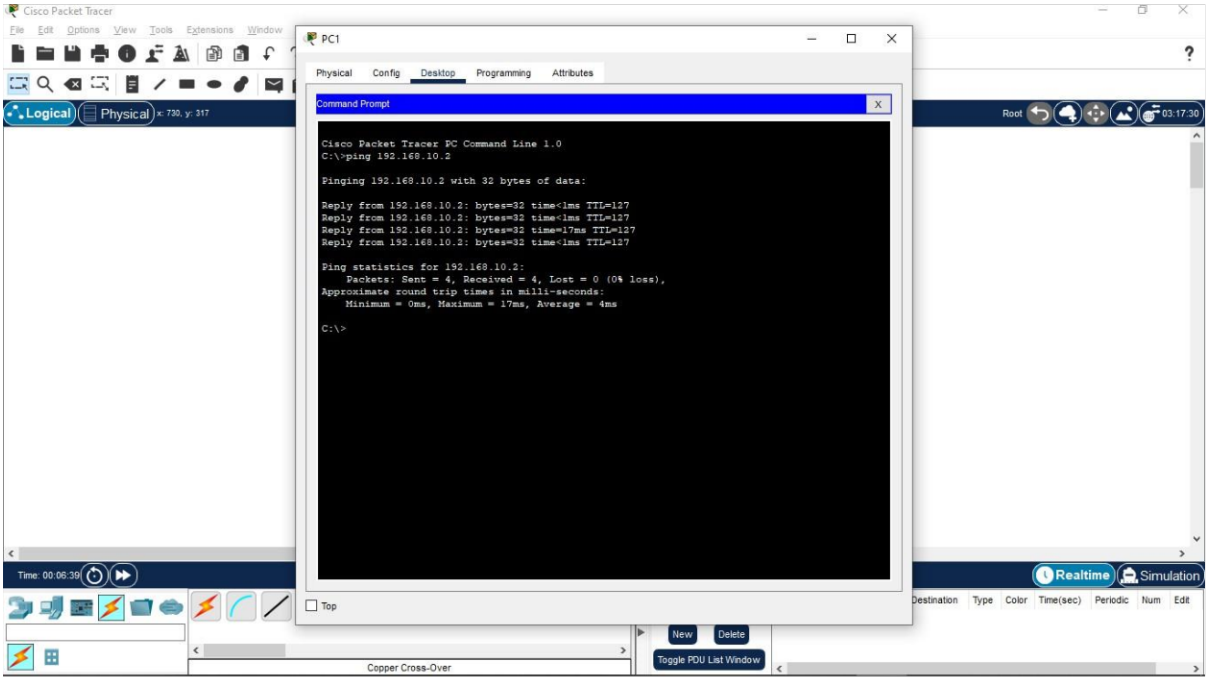
- Click on Router0 and select Config tab.
- Click on GigabitEthernet0/0 in the left pane and configure the ip address 192.168.10.1 and subnet mask 255.255.255.0
- Click on GigabitEthernet0/1 in the left pane and configure the ip address 192.168.11.1 and subnet mask 255.255.255.0





Step 11: Now both the PCs are physically and logically connected. To check the logical connectivity,

- Click on PC1.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.2 to fetch the output as follows



EXERCISE 4 - B: SUBNETTING IN WAN CONFIGURATION (DTE AND DCE)

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

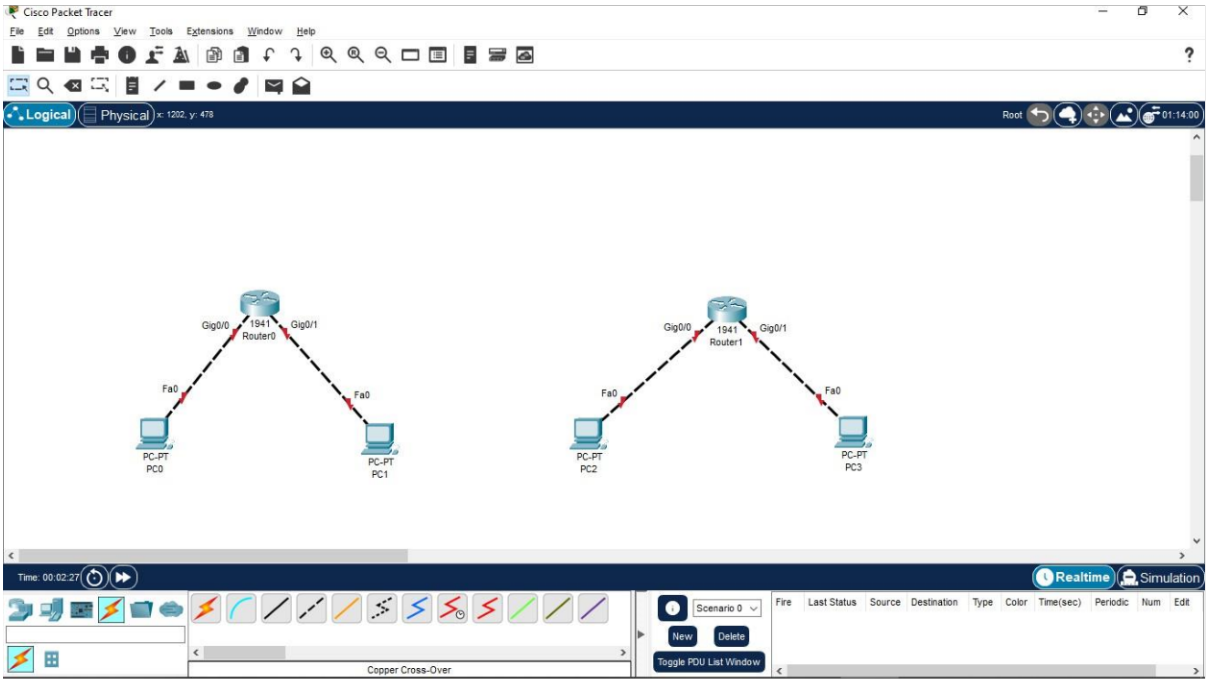
Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

Addressing Table:

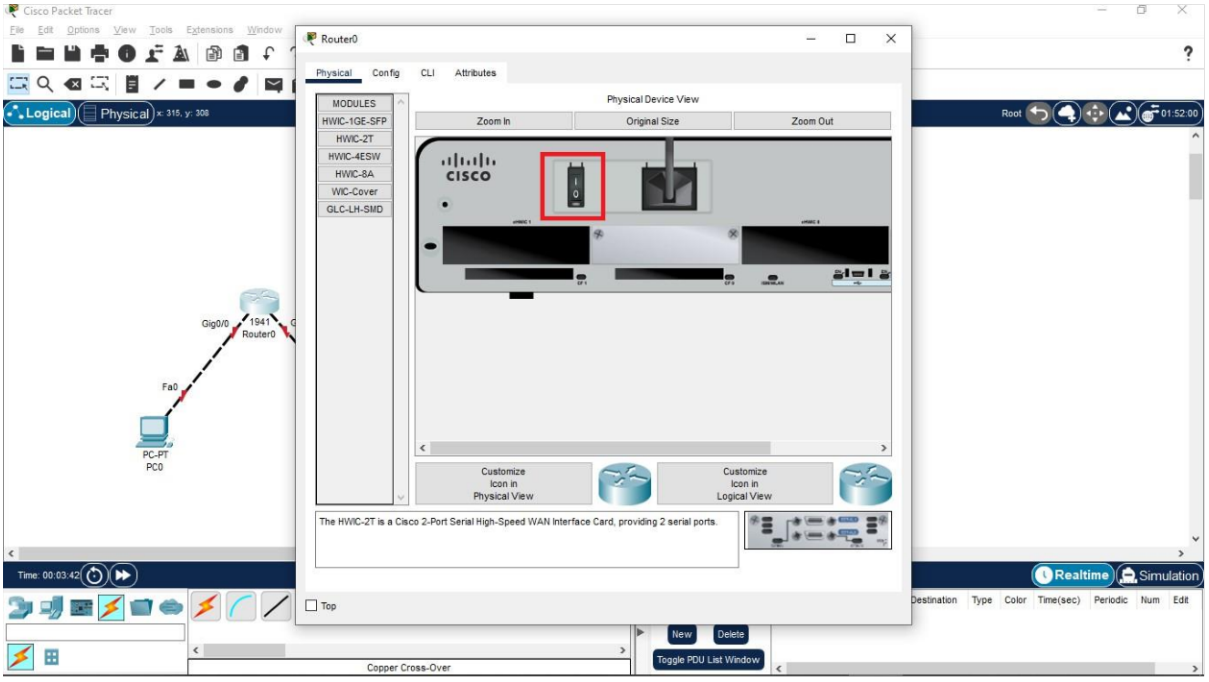
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

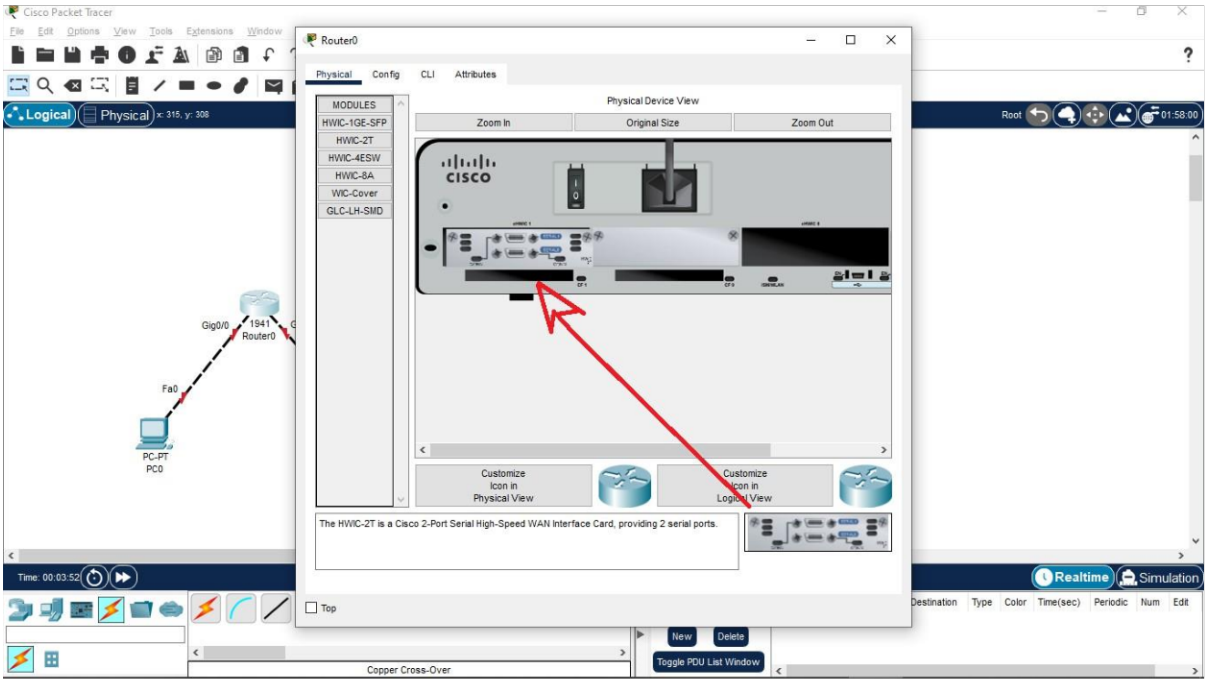
Step 1: Drag 4 PCs and 2 routers in the console area as shown in figure



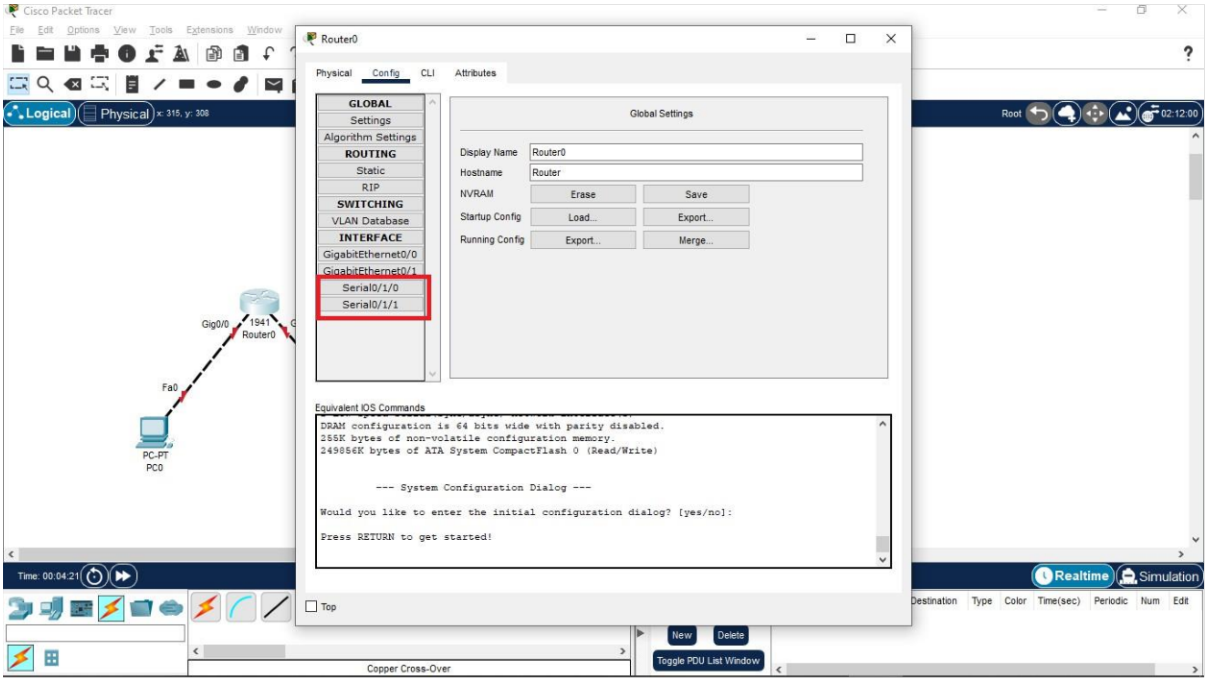
Step 2: Click on Router0 and go to Physical tab. Click HWIC2T in the left pane and Click on zoom in. Switch off the Hardware



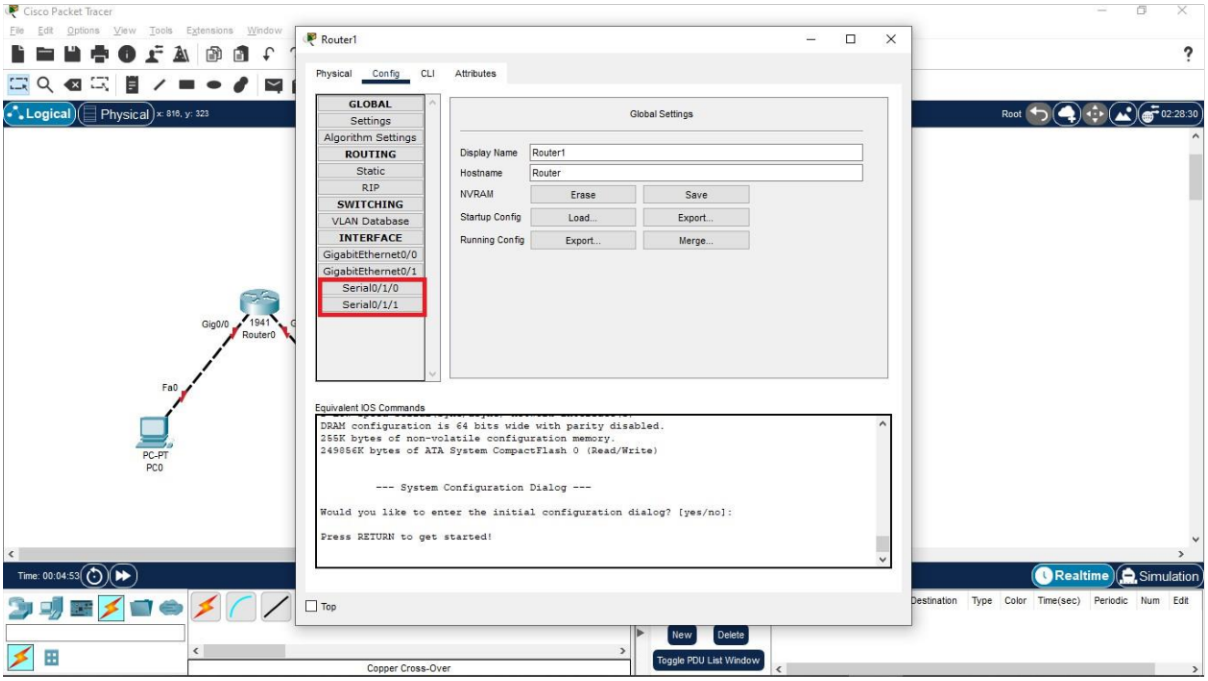
Step 3: Find the console from lower right corner. Drag and drop the console in the empty area as shown in the figure.



Step 4: Now again switch on the hardware and check in config tab for 2 serial ports added.

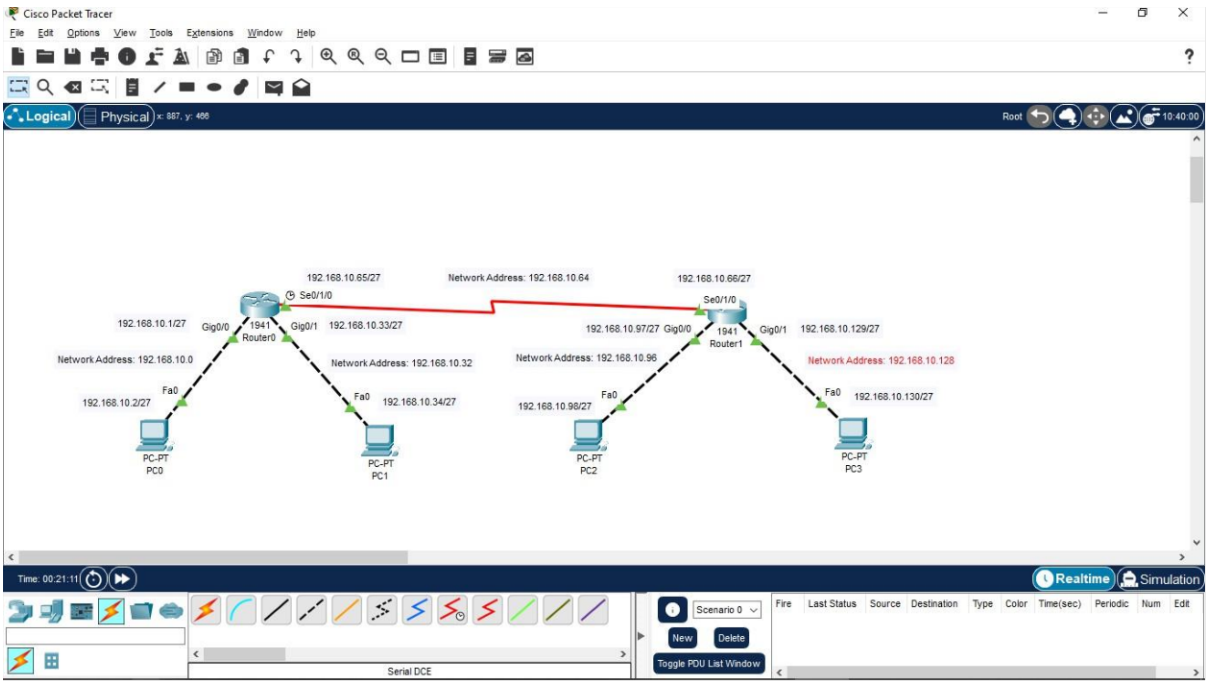


Step 5: Repeat the same procedure (Step 3 and Step 4) for Router1



Step 6: Now the PCs are physically connected through Router. To establish logical connectivity, assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the following table.

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-



Scenario with Network Address for each link

Step 7:

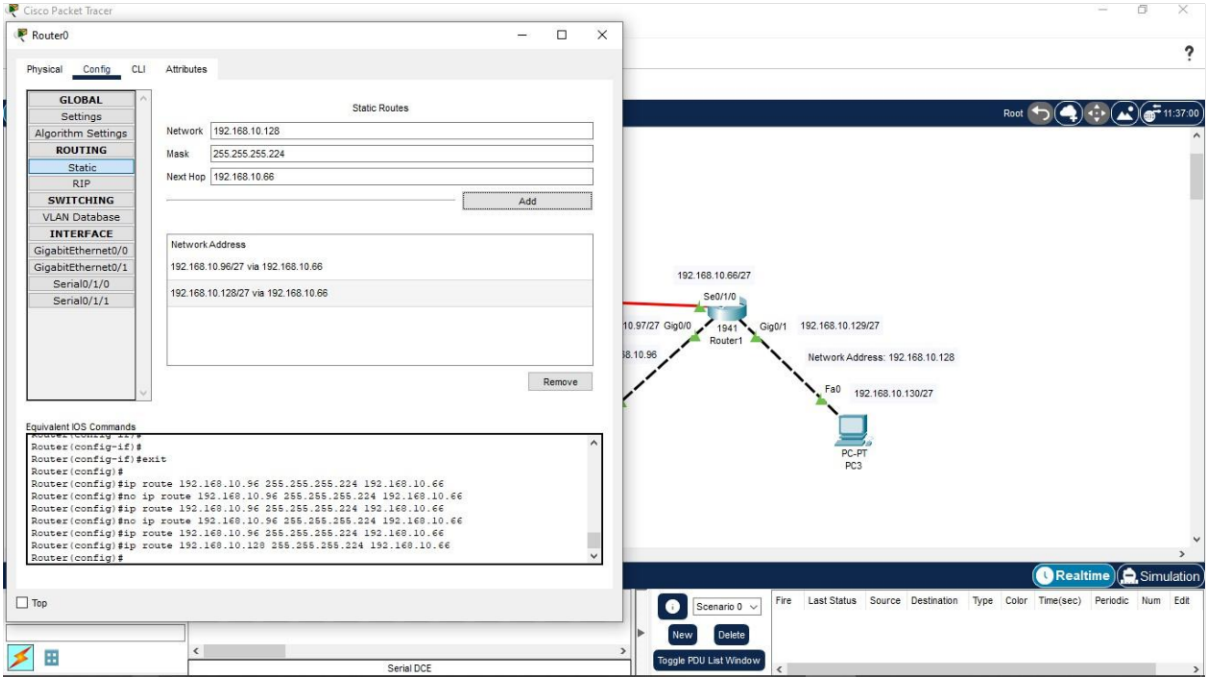
- **To enable packet transmission among the devices in the scenario, Static Routing has to be configured.**
- **To configure static routing, unknown networks and next-hop IP address to reach the unknown network for Router0 and Router1 has to be determined.**
- **Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.**
- **Unknown networks for the routers are derived in the following table**

Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-

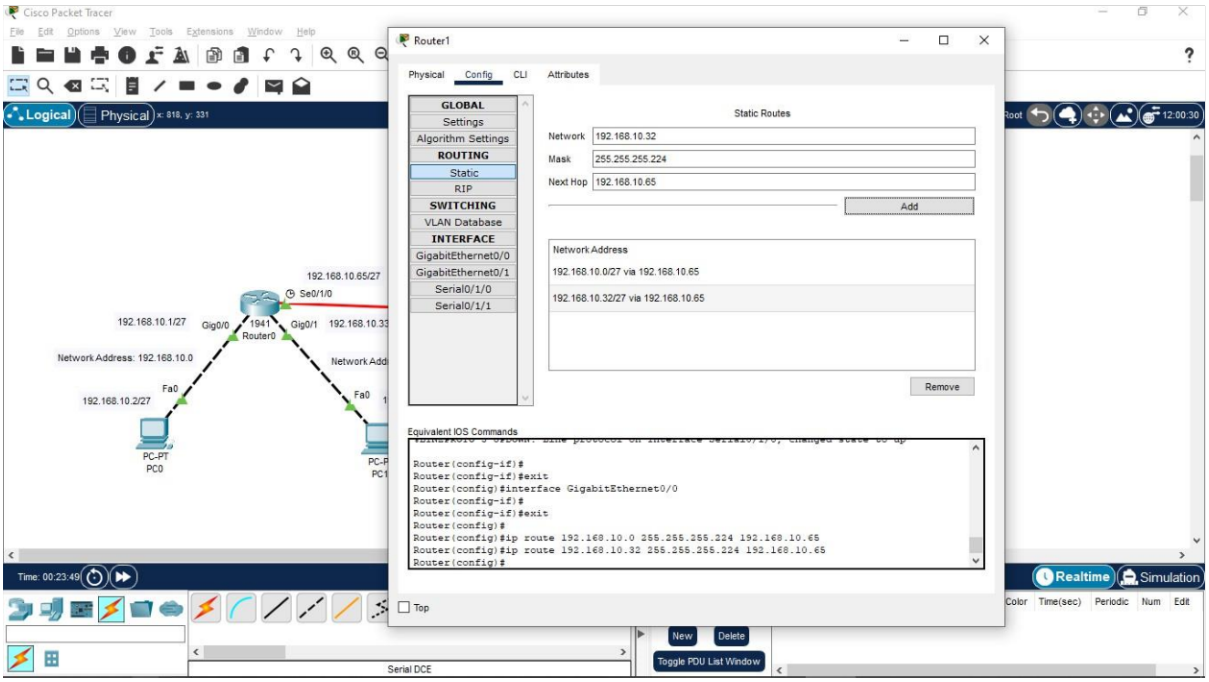
Only unknown networks should be configured for static routing

Step 8:

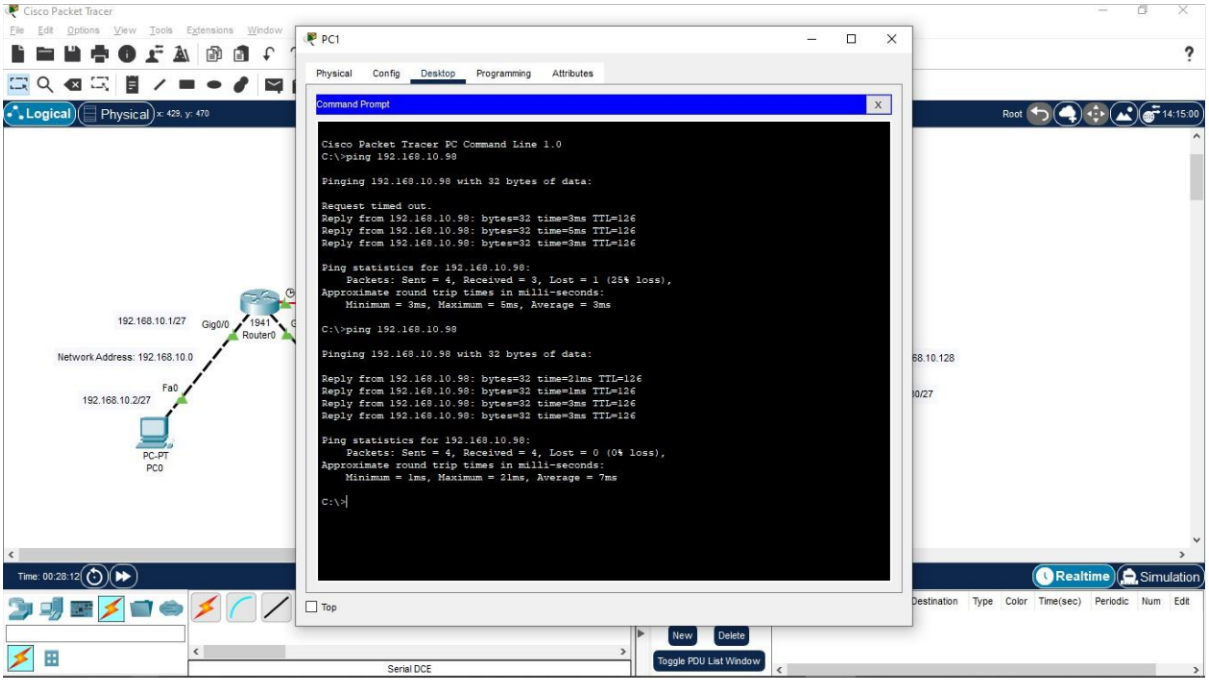
- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (refer unknown network for router0) => Subnet Mask for network address => Next-hop address => Add network
- Repeat the same to add the next network.
- Two networks should be added for the given scenario



Step 9: Repeat the same procedure to configure for Router1



Step 10: After configuration of Static routing in both Routers, Check the connectivity among any two devices using Ping Command



Step 11:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router#

Follow the command “show ip route” to get the routing table of a router

```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 8 subnets, 2 masks
C 192.168.10.0/27 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
C 192.168.10.32/27 is directly connected, GigabitEthernet0/1
L 192.168.10.33/32 is directly connected, GigabitEthernet0/1
C 192.168.10.64/27 is directly connected, Serial0/1/0
L 192.168.10.65/32 is directly connected, Serial0/1/0
S 192.168.10.96/27 [1/0] via 192.168.10.66
S 192.168.10.128/27 [1/0] via 192.168.10.66
```

Result:

Thus, the implementation of IP addressing & subnetting is done and verified in packet tracer.

EXERCISE 5: DEMONSTRATION OF STATIC AND DEFAULT ROUTING

Exercise 5. a: Demonstration of Static Routing

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

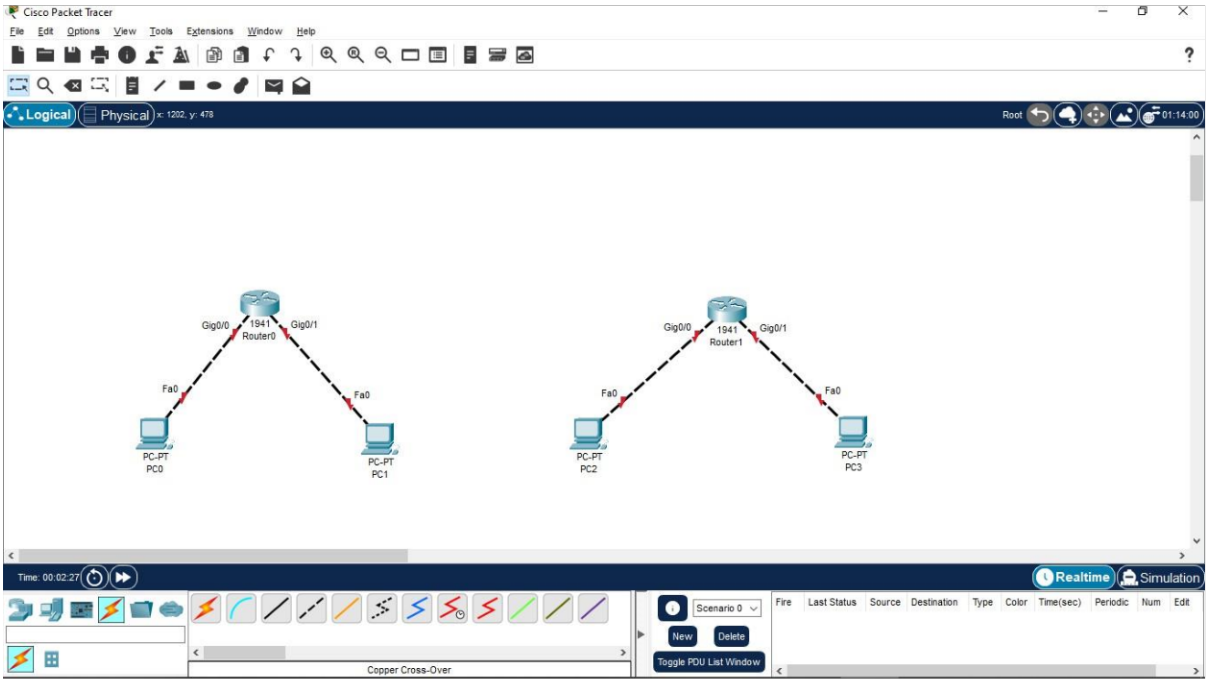
Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

Addressing Table:

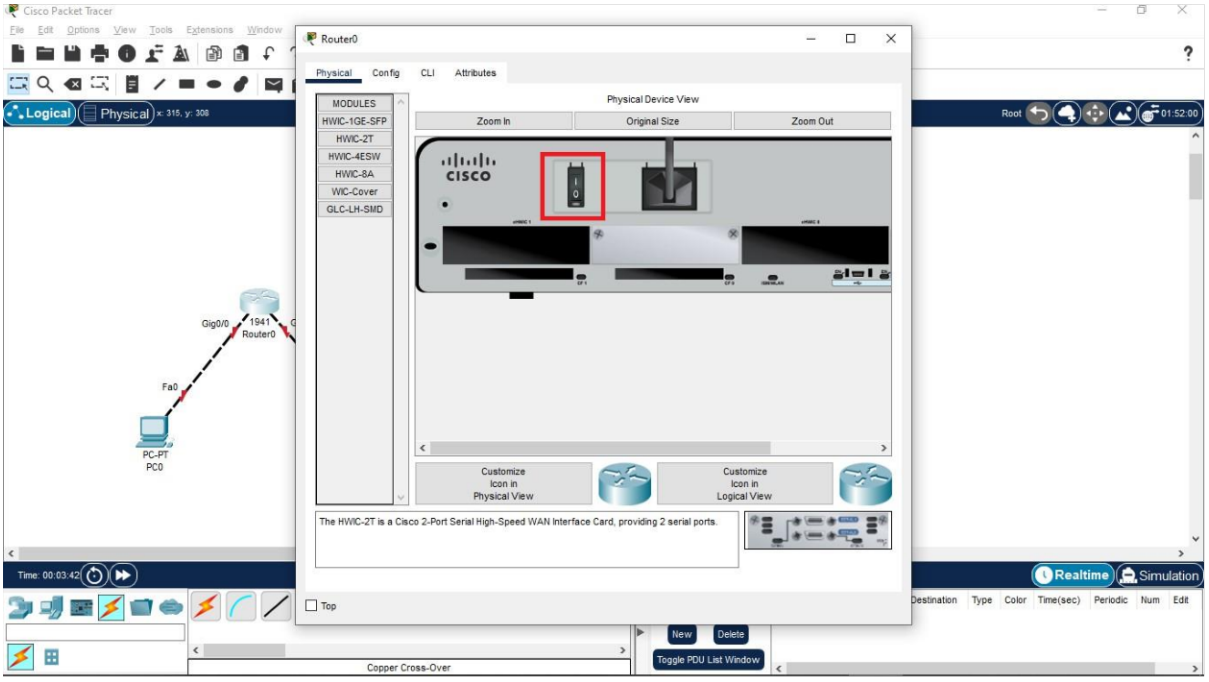
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

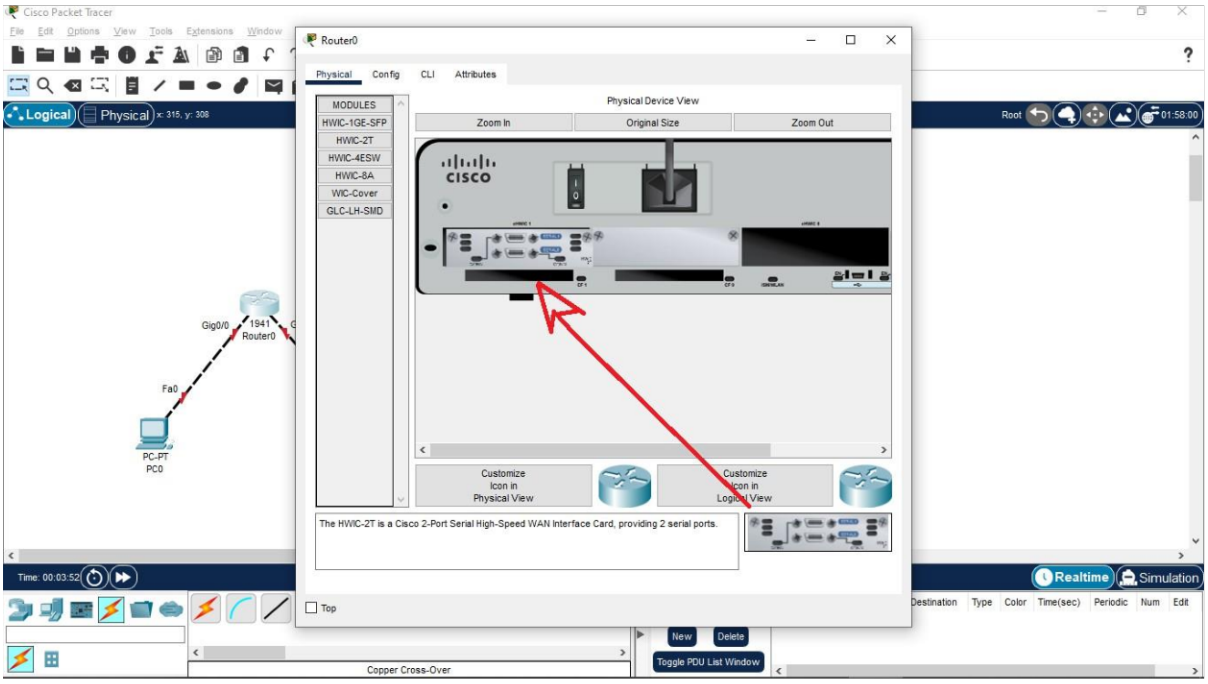
Step 1: Drag 4 PCs and 2 routers in the console area as shown in figure



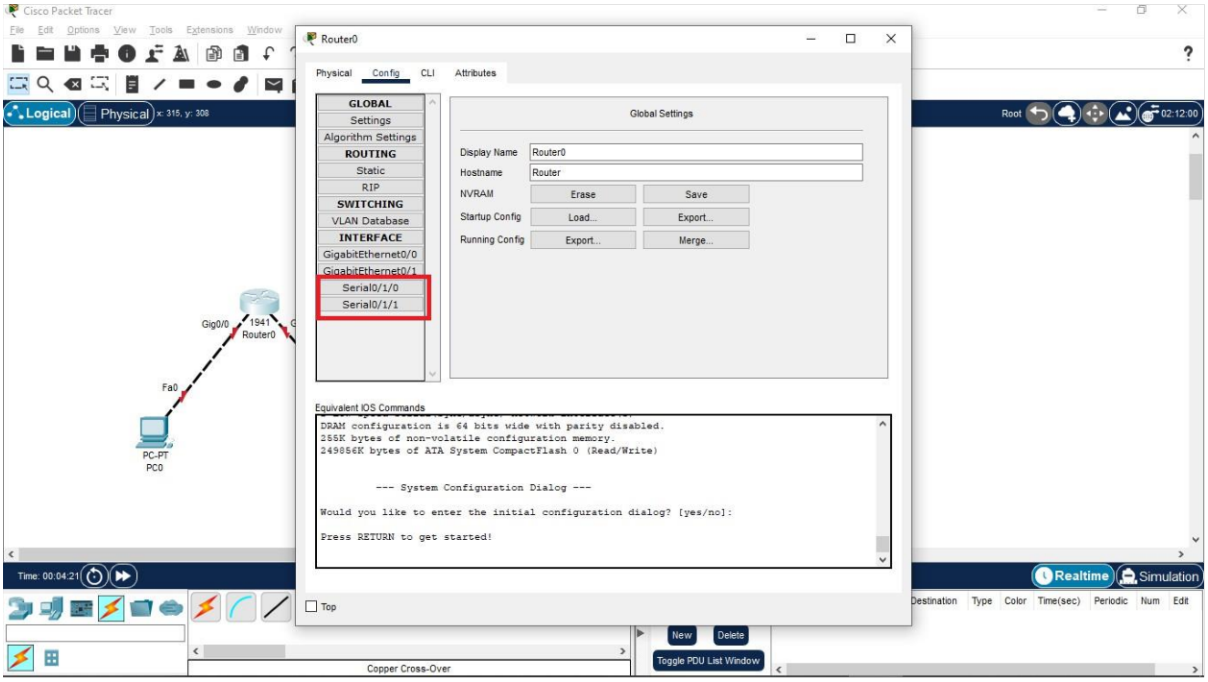
Step 2: Click on Router0 and go to Physical tab. Click HWIC2T in the left pane and Click on zoom in. Switch off the Hardware



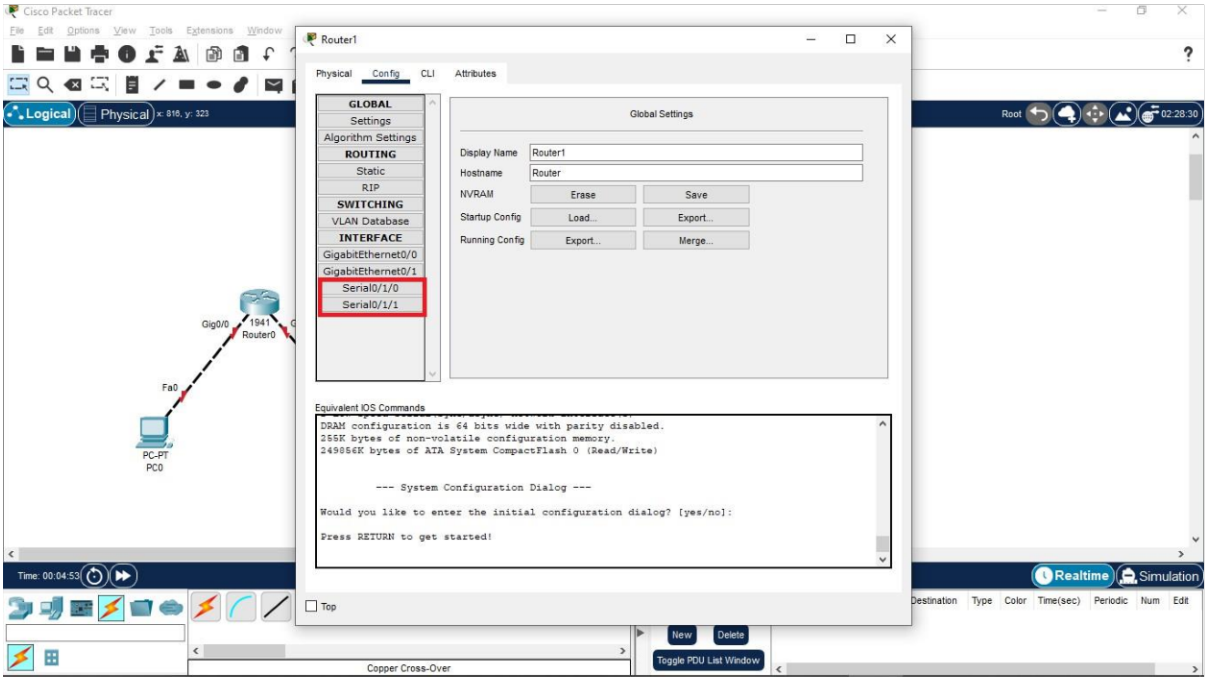
Step 3: Find the console from lower right corner. Drag and drop the console in the empty area as shown in the figure.



Step 4: Now again switch on the hardware and check in config tab for 2 serial ports added.

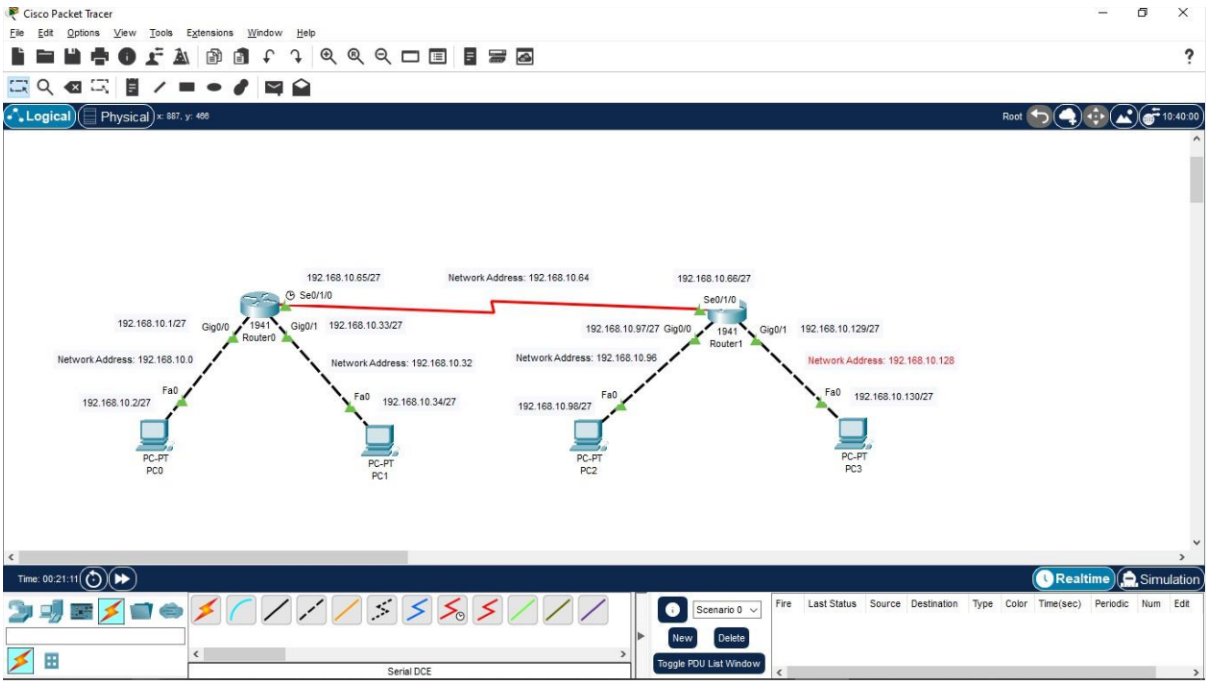


Step 5: Repeat the same procedure (Step 3 and Step 4) for Router1



Step 6: Now the PCs are physically connected through Router. To establish logical connectivity, assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the following table.

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-



Scenario with Network Address for each link

Step 7:

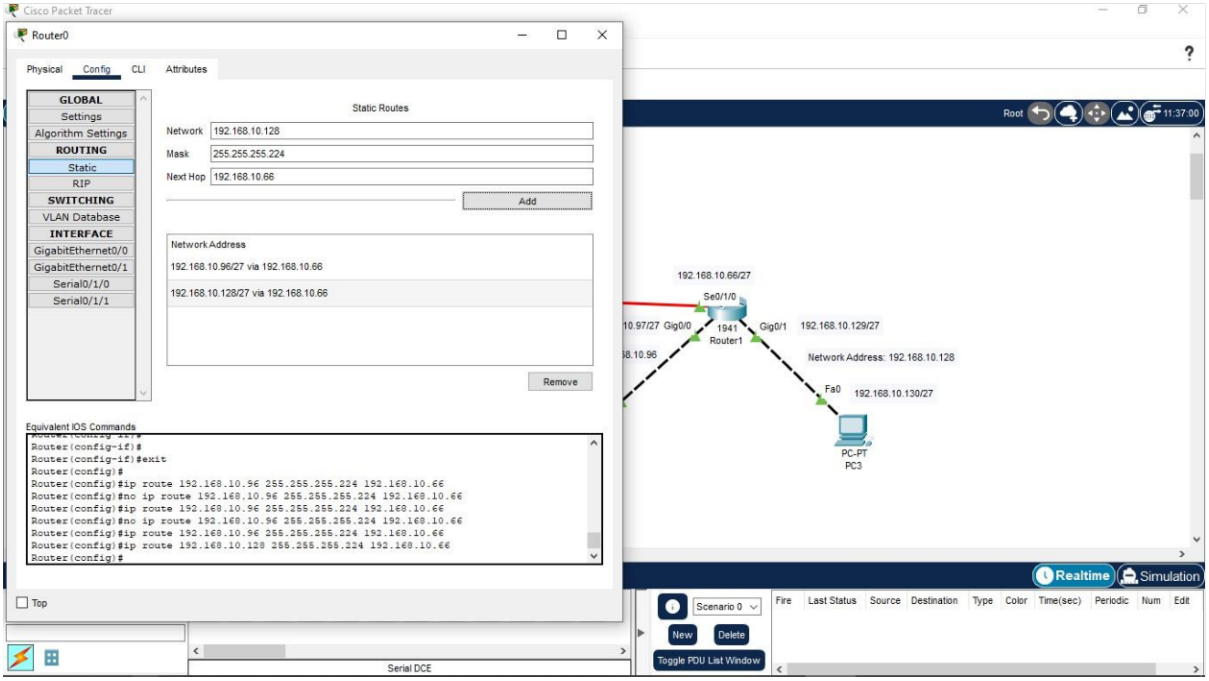
- **To enable packet transmission among the devices in the scenario, Static Routing has to be configured.**
- **To configure static routing, unknown networks and next-hop IP address to reach the unknown network for Router0 and Router1 has to be determined.**
- **Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.**
- **Unknown networks for the routers are derived in the following table**

Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-

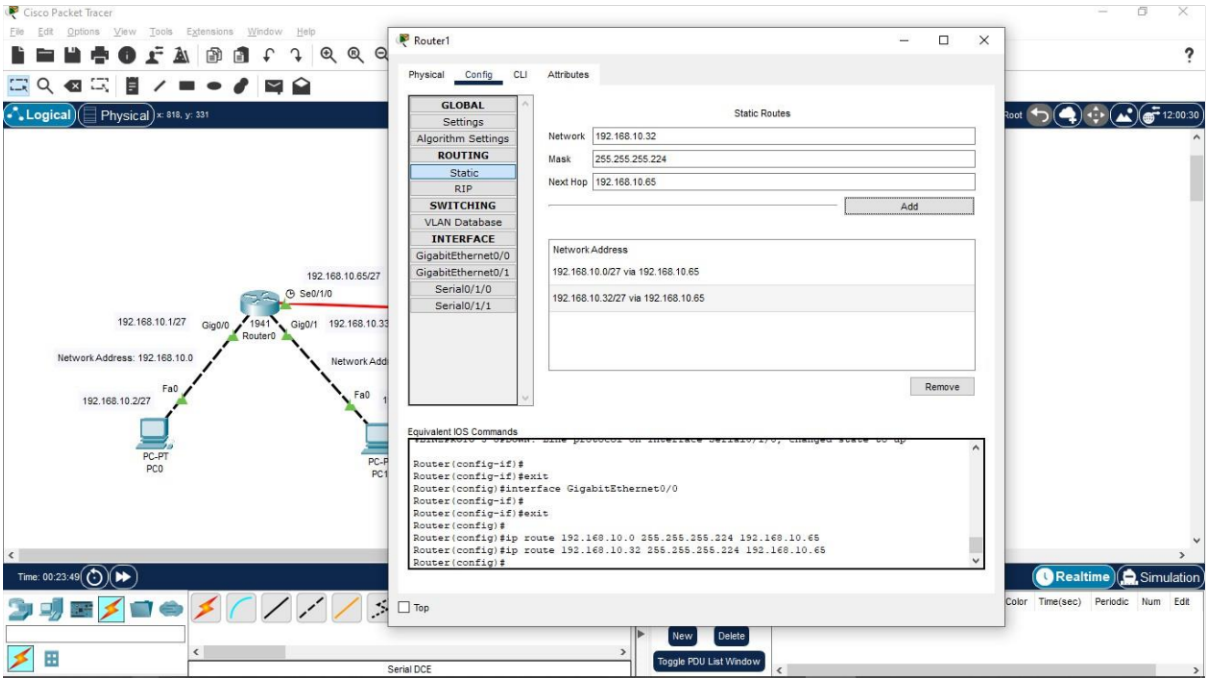
Only unknown networks should be configured for static routing

Step 8:

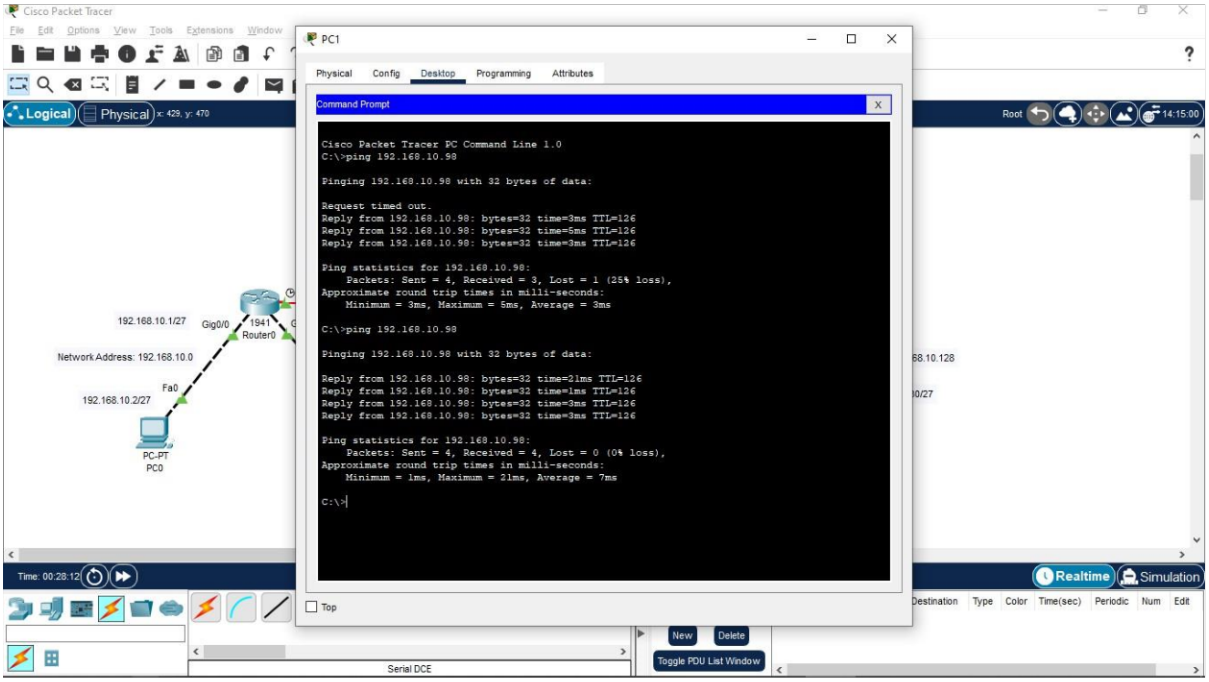
- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (refer unknown network for router0) => Subnet Mask for network address => Next-hop address => Add network
- Repeat the same to add the next network.
- Two networks should be added for the given scenario



Step 9: Repeat the same procedure to configure for Router1



Step 10: After configuration of Static routing in both Routers, Check the connectivity among any two devices using Ping Command



Step 11:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router#

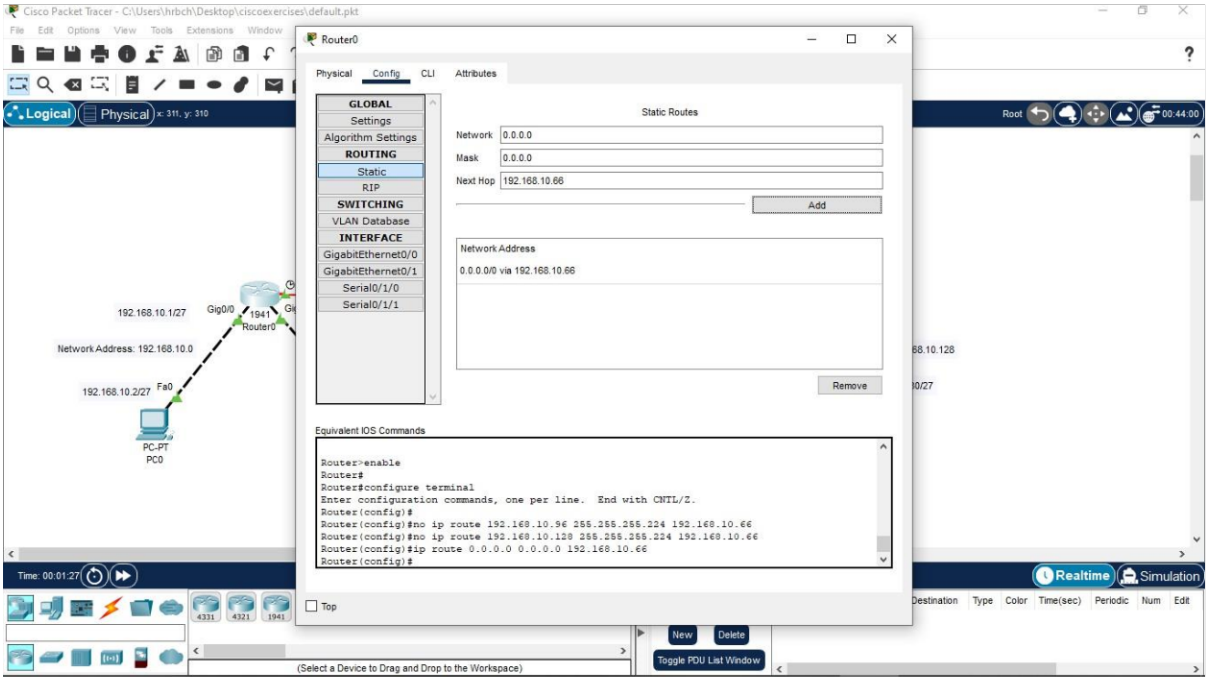
Follow the command “show ip route” to get the routing table of a router

```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 8 subnets, 2 masks
C 192.168.10.0/27 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
C 192.168.10.32/27 is directly connected, GigabitEthernet0/1
L 192.168.10.33/32 is directly connected, GigabitEthernet0/1
C 192.168.10.64/27 is directly connected, Serial0/1/0
L 192.168.10.65/32 is directly connected, Serial0/1/0
S 192.168.10.96/27 [1/0] via 192.168.10.66
S 192.168.10.128/27 [1/0] via 192.168.10.66
```

5 b: For Default Routing replace Step 8 and Step 9 with Step 12 and Step 13

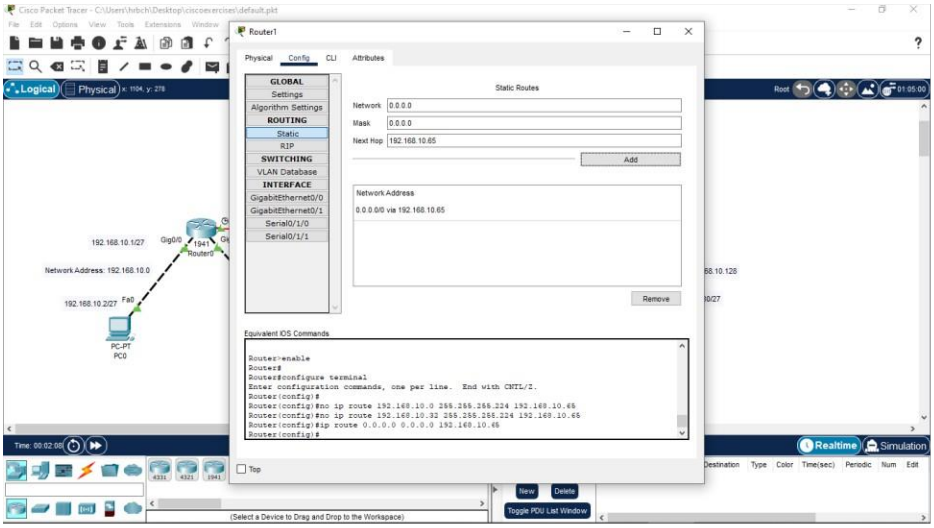
Step 12:

- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (as 0.0.0.0) => Subnet Mask (as 0.0.0.0) => Next-hop address (192.168.10.66) => Add network
- One entry should be added for the given scenario

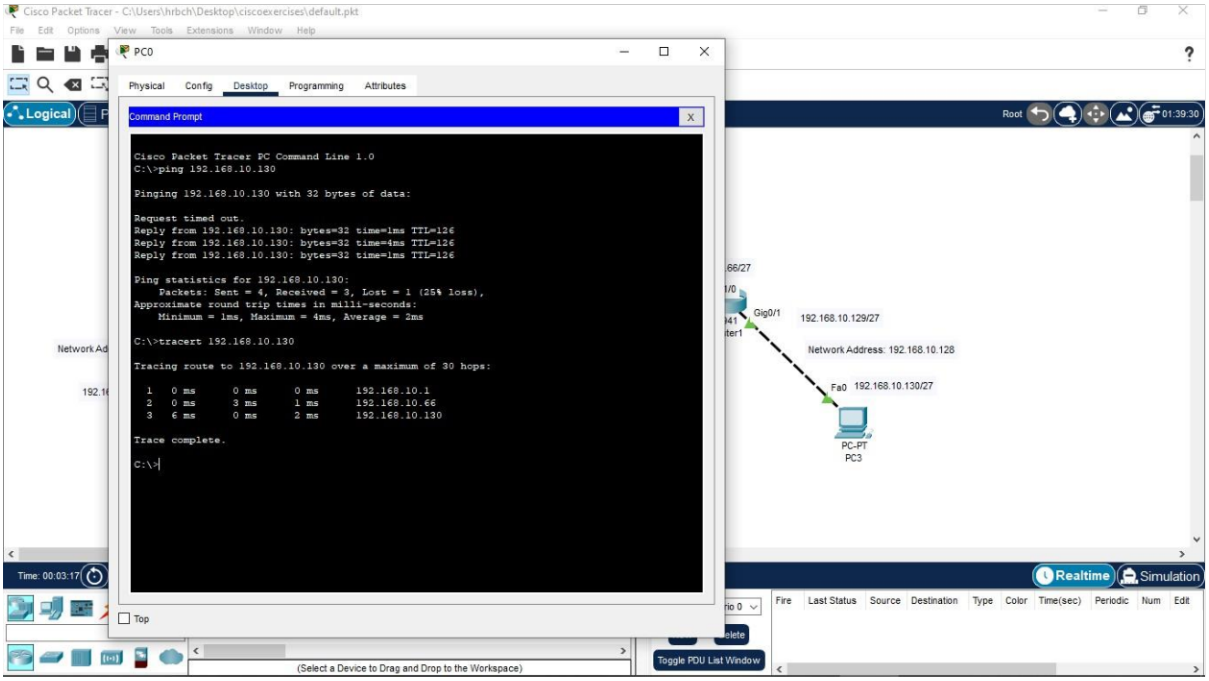


Step 13: Repeat the same procedure to configure for Router1.

Router1 => Config tab => Static =>Enter network address (as 0.0.0.0) => Subnet Mask (as 0.0.0.0) => Next-hop address (192.168.10.65) => Add network



Output for Default Routing: Click on PC0 => Desktop => Command Prompt and check **"tracert 192.168.10.130"** command



EXERCISE 6: NAT CONFIGURATION

AIM:

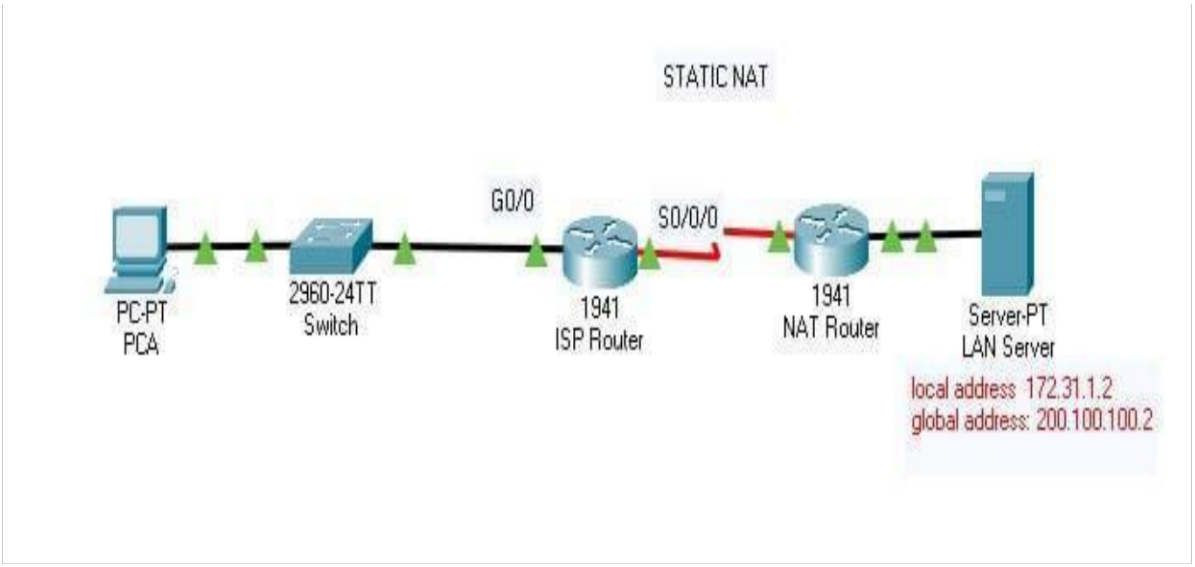
To configure the Network Address Translation (NAT) on a Cisco router.

a. Static NAT configuration

Steps to configure static NAT
Static NAT can be configured using the following two steps.
They are:
1. Creating a mapping between the private internal address and public global address using the `ip nat inside source static [private-address] [public-address]` global configuration command.

2. After the mapping is made, the interfaces taking part in the NAT translation are configured as either `inside` or `outside` with respect to NAT.

The router interface associated within the LAN is assigned the `inside` interface using the `ip nat inside` interface mode command.
Similarly, the router interface associated with the internet is assigned the outside interface using the `ip nat inside` interface mode command.



Static NAT topology

In the figure above, the Gigabit 0/0 `(g0/0)` interface is the `inside` interface because it is connected to the LAN. In contrast, the `S0/0/0` interface is configured as the `outside` interface because it is connected to the internet.

Configuring static NAT

To configure a static NAT between the private address `172.31.1.2` and public address `200.100.100.2`:

- Map the server's private address `172.31.1.2` to the public routable address `200.100.100.2` using the command `ip nat inside source static 172.31.1.2 200.100.100.2`.
- Enter the "interface serial `s0/0/0/`" command and identify the interface as the outside interface using the command `ip nat outside`.
- Enter the "interface gigabitethernet `g0/0`" command and identify it as the inside interface relative to NAT using the `ip nat inside` command.

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 172.31.1.2 200.100.100.2
Router(config)#interface s0/0/0
Router(config-if)#ip nat outside
Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#
```

Static NAT configuration

b. Dynamic NAT configuration

Steps to configure dynamic NAT

Dynamic NAT still requires that both the inside and the outside interfaces be configured.

For allocation, it uses an access control list (ACL) to specify which private addresses are subject to translation and a NAT pool of registered IP addresses.

1. Create an ACL using the `access-list 1 permit address wildcard mask` command.

2. Create a NAT pool using the `ip nat pool [name] [first-address] [last-address] [netmask] [subnet mask]` global configuration command.

This pool will contain the public addresses for the translation. Because, ISP assigns the public addresses contiguous to the organizations.

The `first address` is the least in the given address range. And, the `last address` is the highest address of that range.

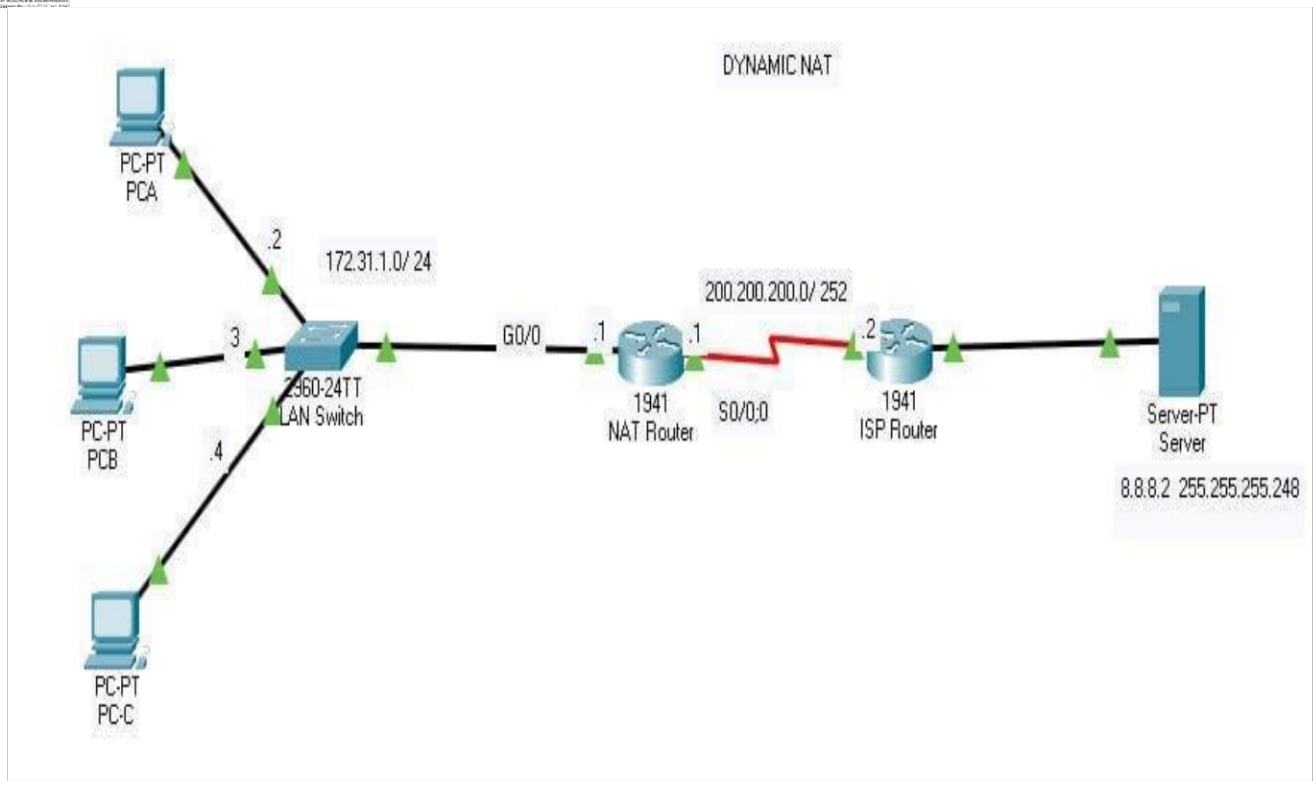
The netmask identifies the network to which of these addresses belong to, using the `ip nat inside source list [access-list] [number] pool [name]` command to bind the ACL and the NAT Pool created.
In this case, the ACL number is `1`, and the NAT POOL is `LAN`.

NOTE: Different ACL numbers and pool names can be created and used, but ACL `1` and pool name `LAN` will be used throughout this tutorial for simplicity.
1. Use the `ip nat inside` interface command to enable the `inside` interface for NAT translation

2. Use the `ip nat outside` interface command to enable the `outside` interface for NAT translation.

Configuring dynamic NAT

An organization is assigned with two public addresses: `200.100.100.1` and `200.100.100.2`. It wants to allow its internal hosts, in the private network `172.31.1.0` and `255.255.255.0` to reach the internet using dynamic NAT.



Dynamic NAT topology

To configure the dynamic NAT for the network topology above:

- Create an access list that will specify the private addresses that are allowed to be translated using the `access-list 1 permit 172.31.1.0 0.0.0.255`.
- Creates a pool that will contain the public addresses to be utilized for translation using the `ip nat pool LAN 200.100.100.1 200.100.100.1 netmask 255.255.255.0`.
- Bind the access list and the pool together using the `ip inside source list 1 pool LAN`. This allows for the dynamic translation of the private addresses and the public addresses in a NAT pool named `LAN`.
- Enter the `interface serial 0/0/0/` command and identify it as an outside interface using the `ip nat outside` command.
- Enter the `interface gigabitethernet g0/0` command and identify it as the `inside` interface using the `ip nat inside` command.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 1 permit 172.31.1.0 0.0.0.255
Router(config)#ip nat pool LAN 200.100.100.1 200.100.100.2 netmask 255.255.255.0
Router(config)#ip nat inside source list 1 pool LAN
Router(config)#interface s0/0/0
Router(config-if)#ip nat inside
Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
```

Result:

Thus, we have configured the Network Address Translation on a cisco router.

EXERCISE 7. A: DEMONSTRATION OF RIP V1

[Note: RIP v1 does not support classless addressing mode and RIP v2 must be used in Subnet enabled Scenario]

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers [1941]	2
Serial DCE	1

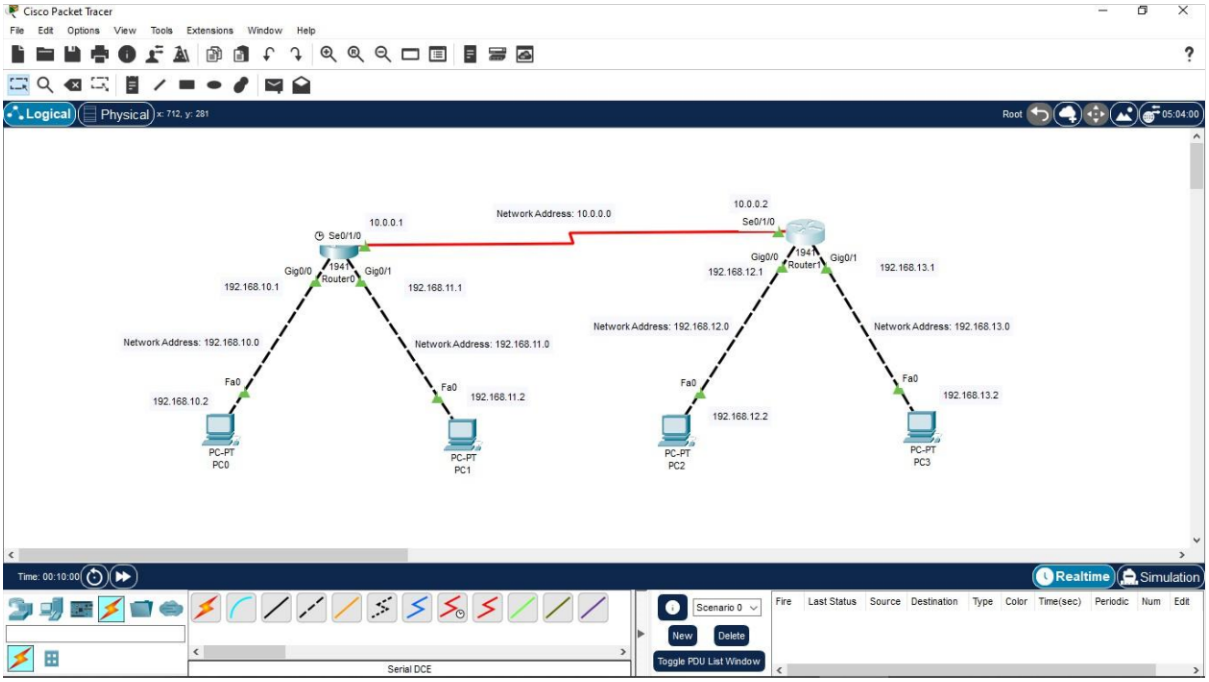
Addressing Table:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/0	192.168.11.2	255.255.255.0	192.168.11.1
PC2	Fa0/0	192.168.12.2	255.255.255.0	192.168.12.1
PC3	Fa0/0	192.168.13.2	255.255.255.0	192.168.13.1
Router0	Gigabit 0/0	192.168.10.1	255.255.255.0	-
Router0	Gigabit 0/1	192.168.11.1	255.255.255.0	-
Router0	Se0/1/0	10.0.0.1	255.0.0.0	-
Router1	Gigabit 0/0	192.168.12.1	255.255.255.0	-
Router1	Gigabit 0/1	192.168.13.1	255.255.255.0	-
Router1	Se0/1/0	10.0.0.2	255.0.0.0	-

Procedure:

Step 1:

- Drag 4 PCs and 2 routers in the console area as shown in the figure.
- Follow the procedure for connecting **Serial DCE cable from Exercise 4.**
- Assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the Addressing Table



Step 2:

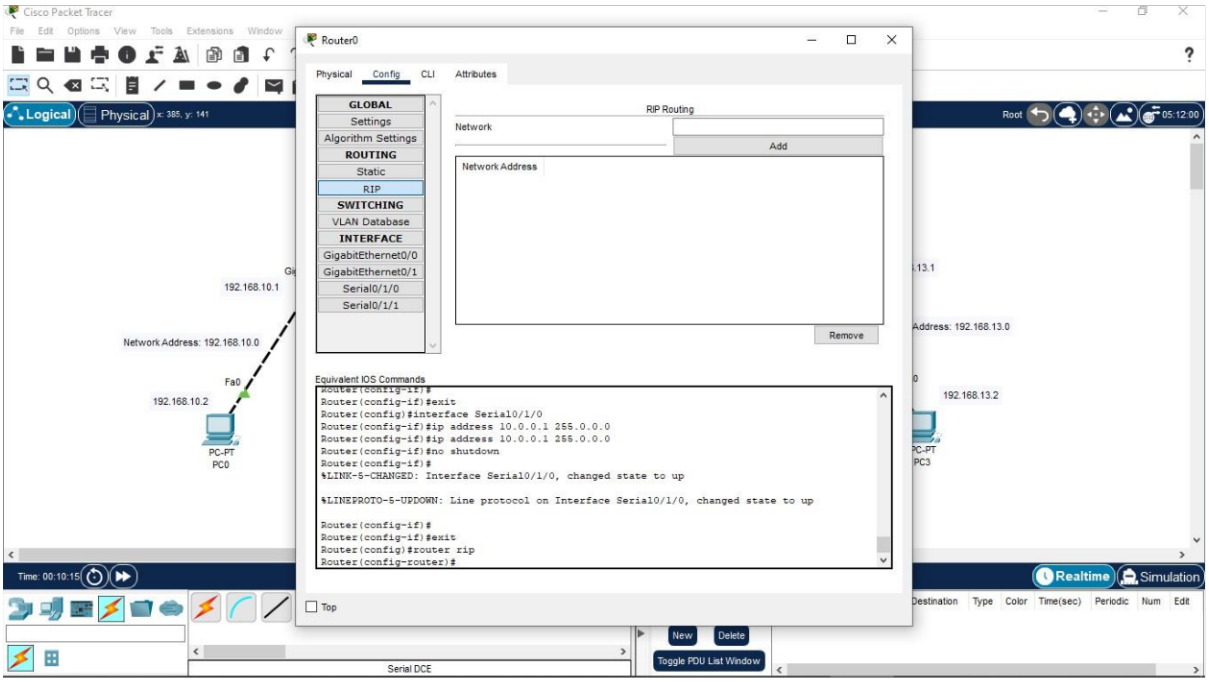
- To enable packet transmission among the devices in the scenario, RIP Routing has to be configured.
- To configure RIP routing, **known networks** for Router0 and Router1 has to be determined.
- Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.
- The **known networks** for the routers are derived in the following table

Device	Known Networks	Subnet Mask
Router0	192.168.10.0	255.255.255.0
Router0	192.168.11.0	255.255.255.0
Router0	10.0.0.0	255.0.0.0
Router1	192.168.12.0	255.255.255.0
Router1	192.168.13.0	255.255.255.0
Router1	10.0.0.0	255.0.0.0

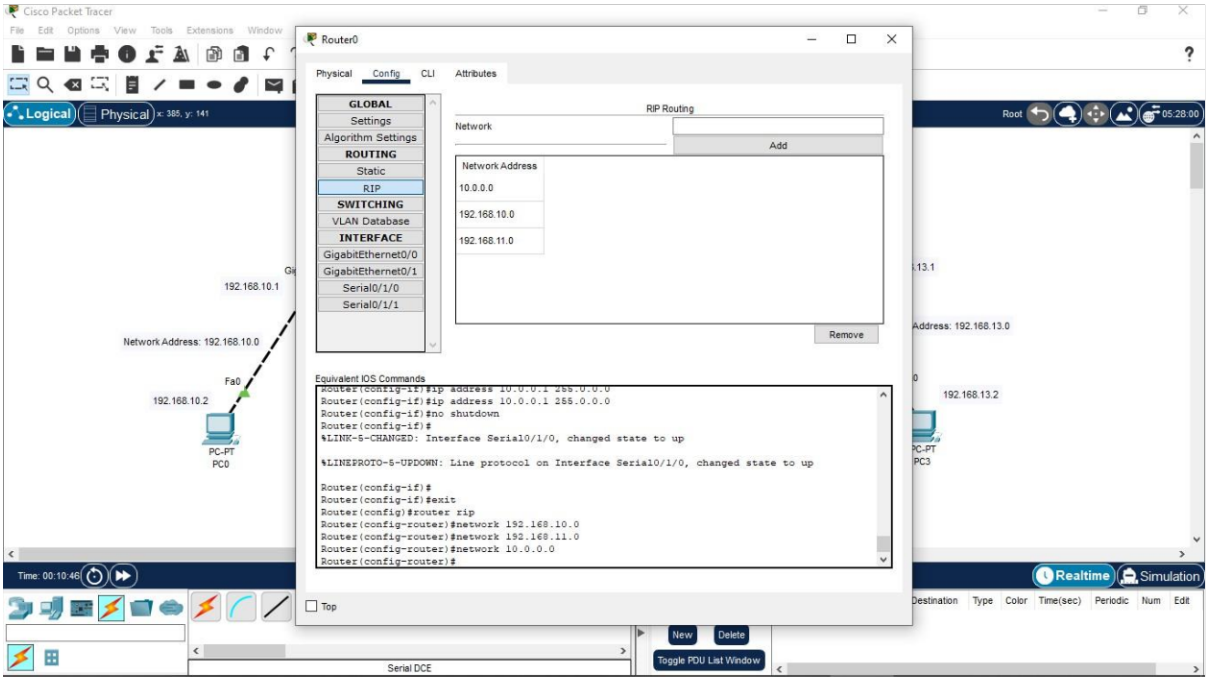
Only known networks should be configured for RIP routing

Step 3:

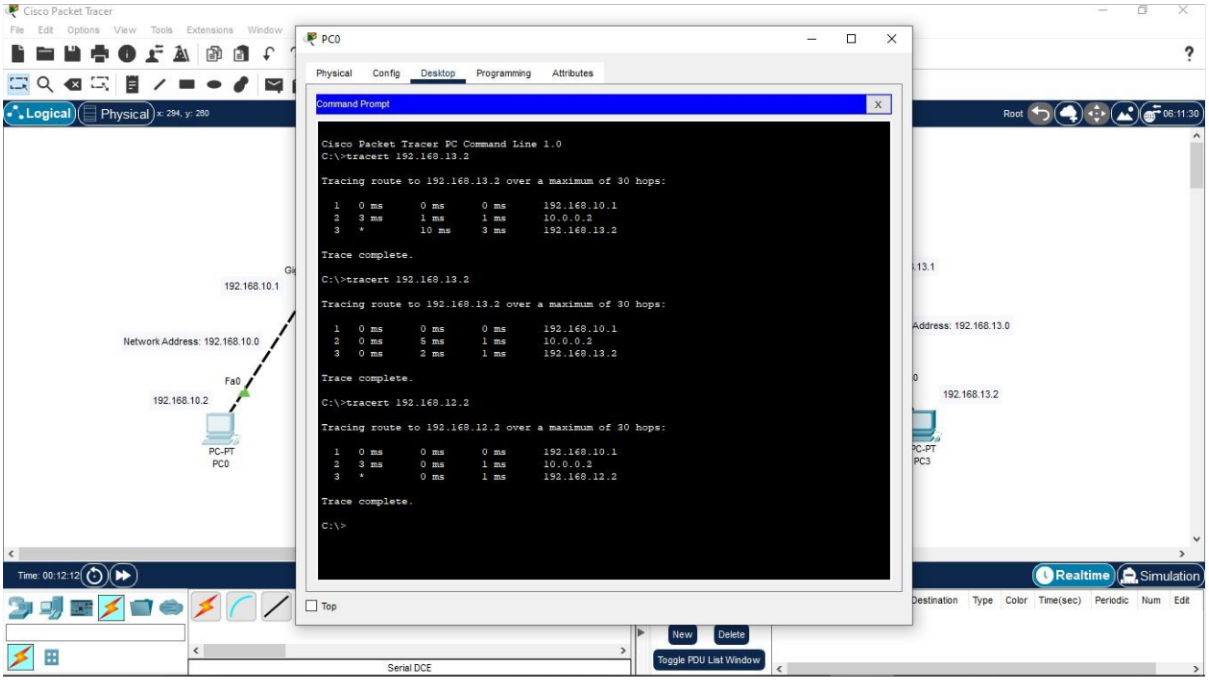
- To configure RIP routing, click on
  - Router0 => Config tab => RIP =>Enter network address (refer table) => Add network
- Repeat the same to add the next network.
- Three networks should be added for the given scenario



Step 4: Repeat the same procedure to configure for Router1

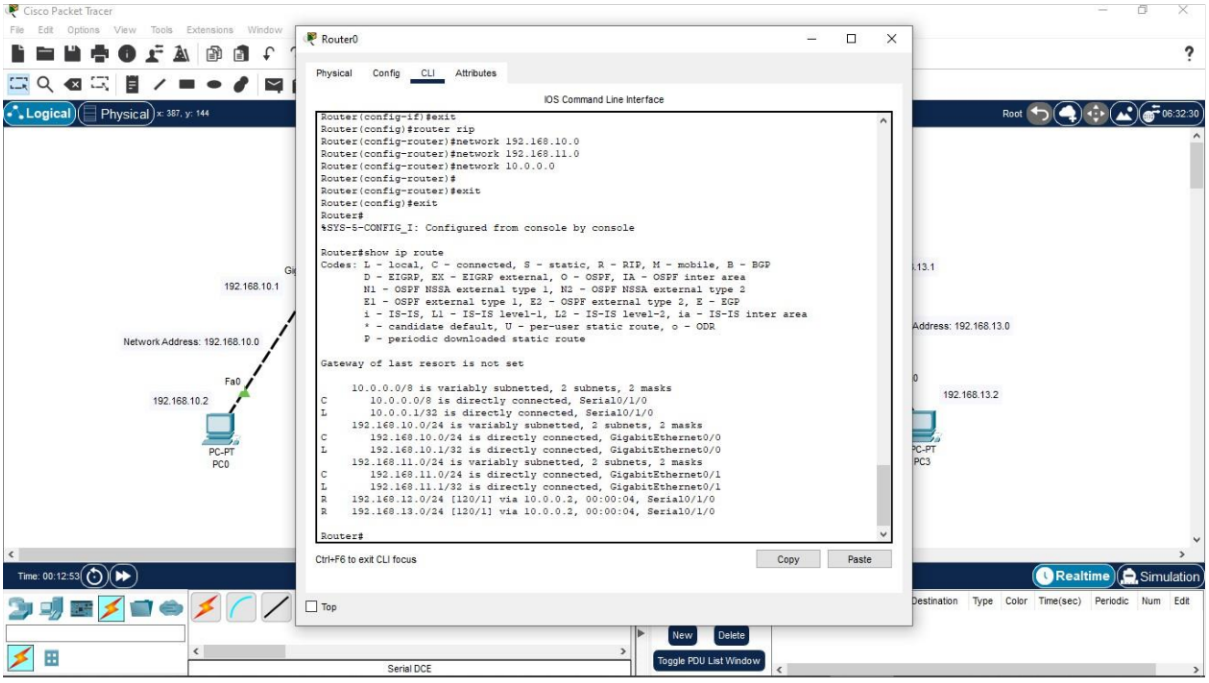


Step 5: After configuration of RIP routing in both Routers, Check the connectivity among any two devices using Ping Command or tracertcommand



Step 6:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router# show ip route



EXERCISE 7. B: DEMONSTRATION OF RIP V2

Objective: To demonstrate the configuration of RIP v2

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

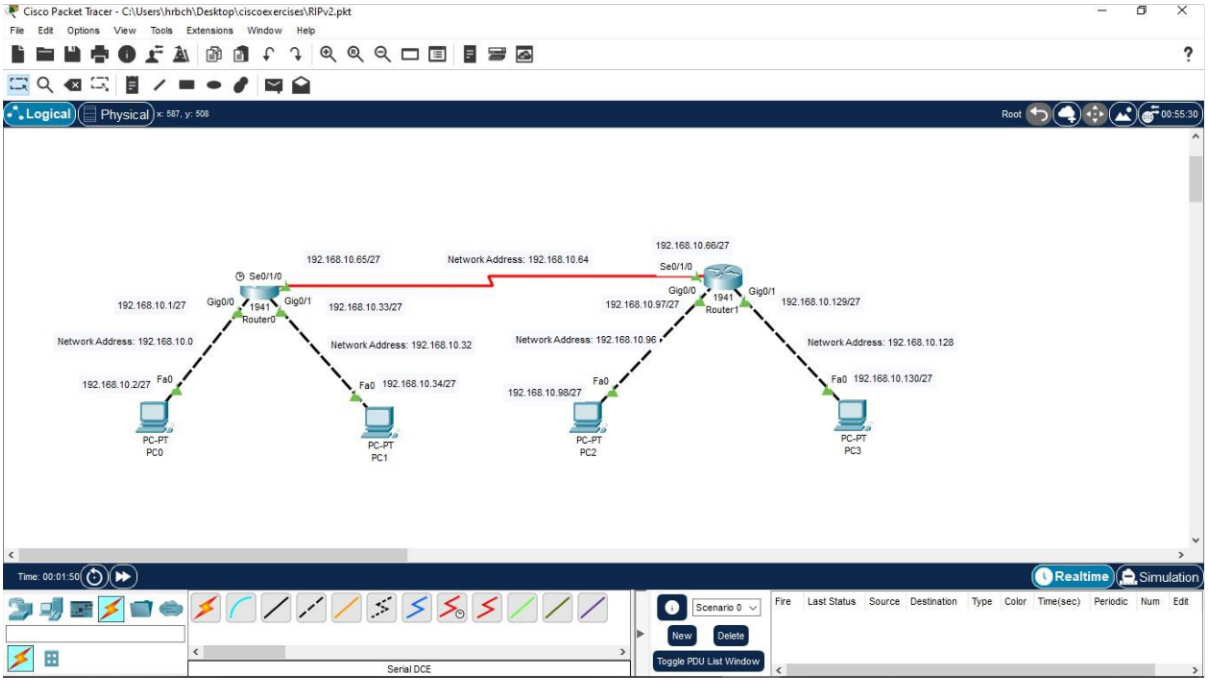
Addressing Table:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

Step 1:

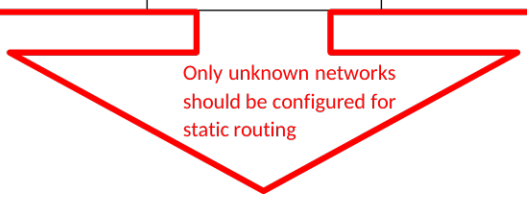
- Drag 4 PCs and 2 routers in the console area as shown in the figure.
- Follow the procedure for connecting **Serial DCE cable from Exercise 4.**
- Assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the Addressing Table



Step 2:

- To enable packet transmission among the devices in the scenario, RIP v2 Routing has to be configured.
- To configure RIP v2 routing, **known networks** for Router0 and Router1 has to be determined.
- Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.
- The **known networks** for the routers are derived in the following table

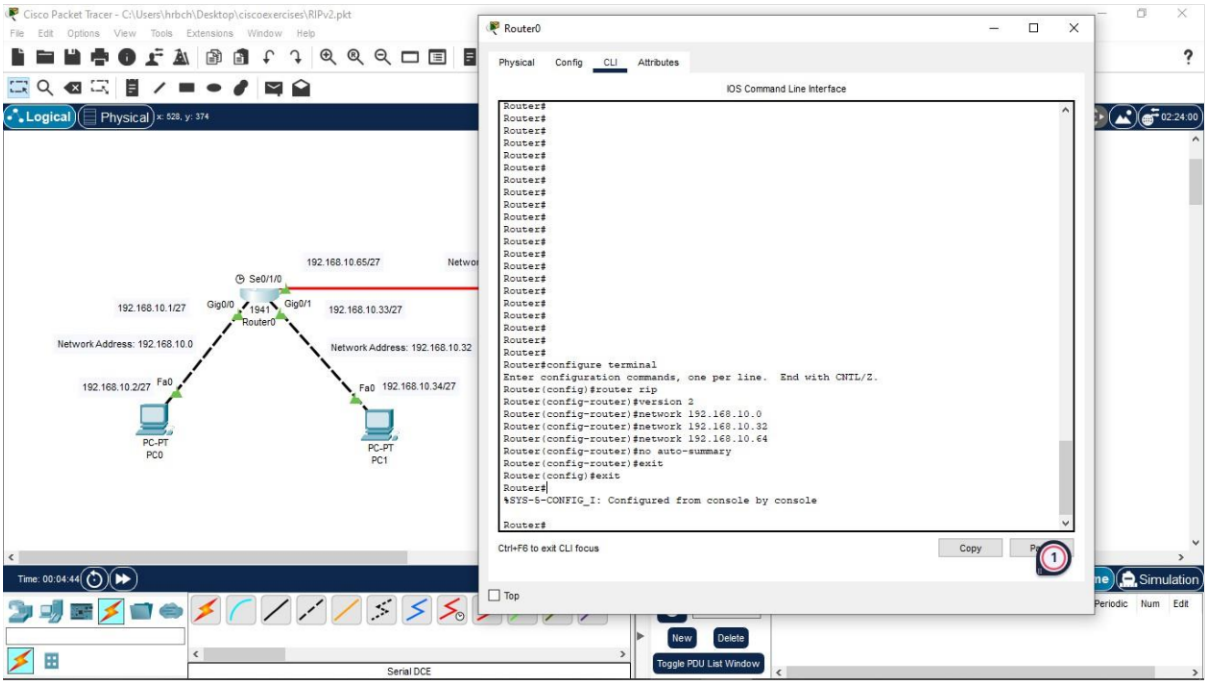
Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-



Step 3:

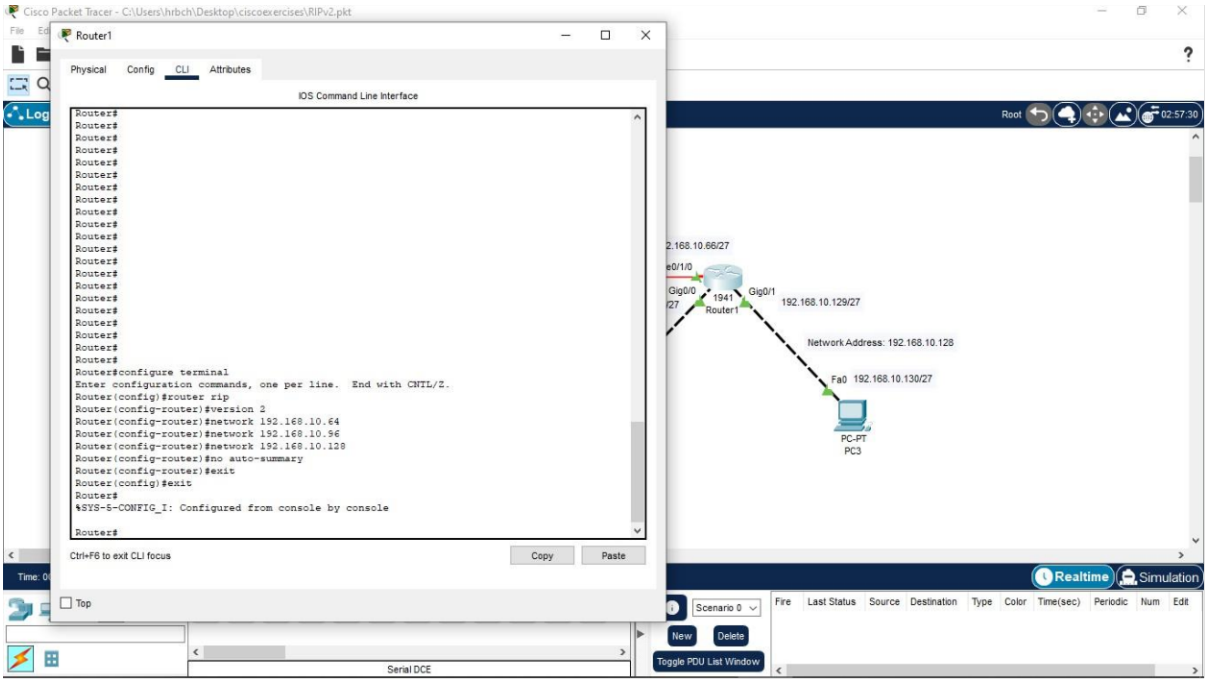
- To configure static routing, click on
  - Router0 => Config tab => CLI
- Type “enable” in “Router>” prompt and type the following commands to configure RIP v2 in Router0

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.0
Router(config-router)#network 192.168.10.32
Router(config-router)#network 192.168.10.64
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console


Router#
```

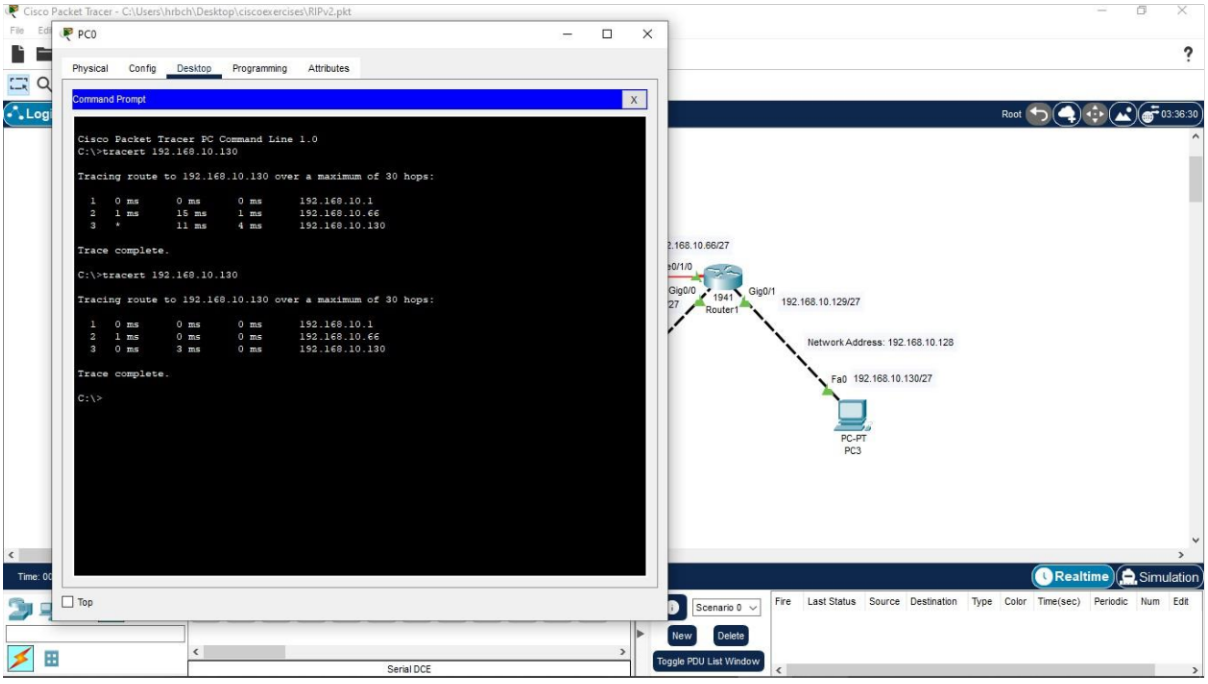


Step 4: Repeat the same procedure to configure for Router1

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.64
Router(config-router)#network 192.168.10.96
Router(config-router)#network 192.168.10.128
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console


Router#
```



Step 5: After configuration of RIP v2 routing in both Routers, Check the connectivity among any two devices using Ping Command/ tracert command



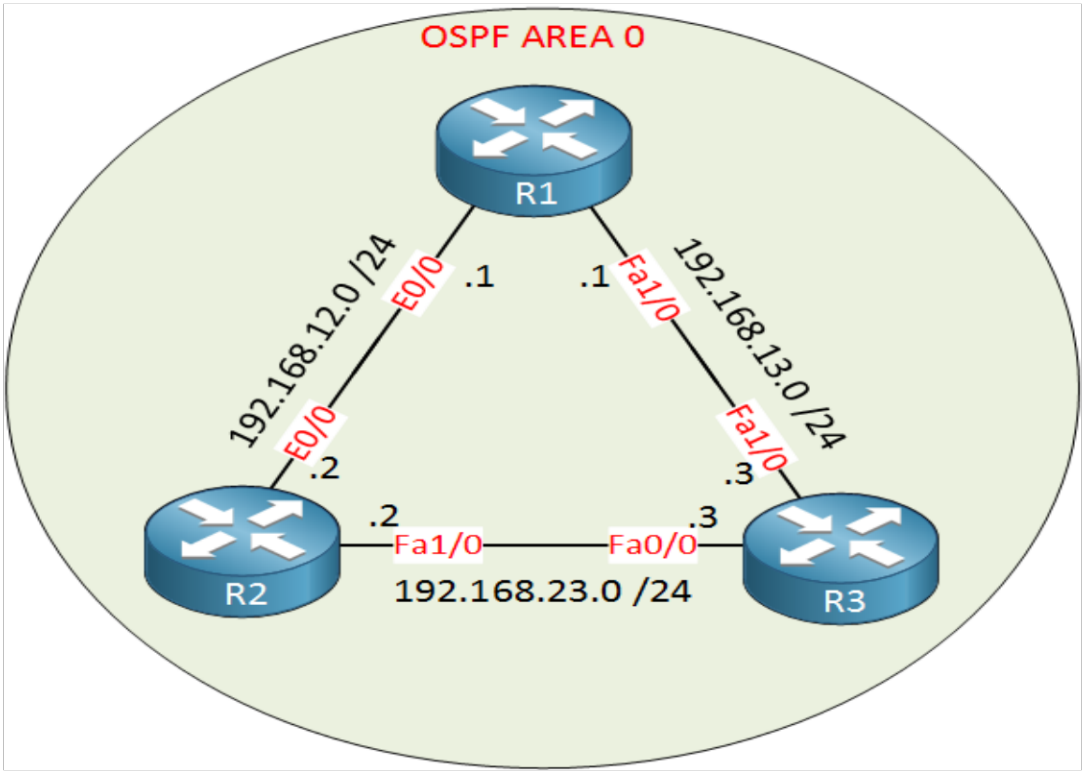
Result:

The demonstration of RIP v1 & RIP v2 is done successfully in the packet tracer.

EXERCISE 8: IMPLEMENTATION OF SINGLE AREA OSPF

Aim:

To configure the basic OSPF methodology using packet tracer.



This is the topology that we'll use. All routers are in OSPF Area 0. Note that the link between R2 and R1 is an Ethernet (10Mbit) link. All other links are FastEthernet (100Mbit) interfaces.

We'll start with the configuration between R2 and R3:

```
R2(config)#router ospf 1

R2(config-router)#network 192.168.23.0 0.0.0.255 area 0

R3(config)#router ospf 1

R3(config-router)#network 192.168.23.0 0.0.0.255 area 0
```

I need to use the router ospf command to get into the OSPF configuration. The number "1" is a process ID and you can choose any number you like.

The second step is to use the network command. It works similar to RIP but it is slightly different, let me break it down for you:

```
network 192.168.23.0 0.0.0.255
```

Just like RIP the network command does two things:

- Advertise the networks that fall within this range in OSPF.
- Activate OSPF on the interface(s) that fall within this range. This means that OSPF will send hello packets on the interface.

Behind 192.168.23.0 you can see it says 0.0.0.255. This is not a subnet mask but a **wildcard mask**. A wildcard mask is a **reverse subnet mask**. Let me give you an example:

Subnetmask	**255**	**255**	**255**	**0**
	11111111	11111111	11111111	00000000
Wildcardmask	**0**	**0**	**0**	**255**
	00000000	00000000	00000000	11111111

When I say reverse subnet mask I mean that the binary 1s and 0s of the wildcard mask are flipped compared to the subnet mask. A subnet mask of 255.255.255.0 is the same as wildcard mask 0.0.0.255. Don't worry about this too much for now as I'll explain wildcard masks to you when we talk about access-lists!

OSPF uses areas so you need to specify the area:

```
area 0
```

In our example we have configured single area OSPF. All routers belong to area 0.

After typing in my network command you'll see this message in the console:

```
R3# %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on FastEthernet0/0 from
LOADING to FULL, Loading Done

R2# %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.3 on FastEthernet1/0 from
LOADING to FULL, Loading Done
```

Great! It seems that R3 and R2 have become neighbors. There's another command we can use to verify that we have become neighbors:

```
R3#show ip ospf neighbor
```

```
Neighbor ID Pri State Dead Time Address Interface

192.168.23.2 1 FULL/BDR 00:00:36 192.168.23.2 FastEthernet0/0

R2#show ip ospf neighbor



Neighbor ID Pri State Dead Time Address Interface

192.168.23.3 1 FULL/DR 00:00:32 192.168.23.3 FastEthernet1/0
```

Show ip ospf neighbor is a great command to see if your router has OSPF neighbors. When the state is full you know that the routers have successfully become neighbors.

Each OSPF router has a router ID and we check it with the show ip protocols command:

```
R2#show ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.2
R3#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.3
```

Above you see the router ID of R2 and R3. They used their highest active IP address as the router ID. Let's create a loopback on R2 to see if the router ID changes…

```
R2(config)#interface loopback 0

R2(config-if)#ip address 2.2.2.2 255.255.255.0
```

This is how you create a loopback interface. You can pick any number that you like it really doesn't matter.

```
R2#show ip protocols
Routing Protocol is "ospf 1"
```

```
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.2
```

The router ID still the same. We need to reset the OSPF process before the change will take effect, this is how you do it:

```
R2#clear ip ospf process

Reset ALL OSPF processes? [no]: yes
```

Use clear ip ospf process to reset OSPF. Let's see if there is a difference:

```
R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 2.2.2.2
```

We can also change the router ID manually. Let me demonstrate this on R3:

```
R3#show ip protocols

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.3
```

Right now it's 192.168.23.3…

```
R3(config-router)#router-id 3.3.3.3
Reload or use "clear ip ospf process" command, for this to take effect
R3#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```

The router is friendly enough to warn me to reload or clear the OSPF process. Let's verify our configuration:

```
R3#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 3.3.3.3
```

As you can see above the router ID is now 3.3.3.3.

Changing the router ID isn't something you would normally do. IP addresses on your router have to be unique so your OSPF router ID will also be unique. Understanding how OSPF selects a router ID is something you have to understand for the exam however.
Right now we have an OSPF neighbor adjacency between R2 and R3. Let's configure our routers so that R2/R1 and R1/R3 also become OSPF neighbors:

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0

R1(config-router)#network 192.168.13.0 0.0.0.255 area 0

R3(config)#router ospf 1

R3(config-router)#network 192.168.13.0 0.0.0.255 area 0
```

I'll advertise all networks in OSPF. Before we check the routing table it's a good idea to see if our routers have become OSPF neighbors:

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface

192.168.13.1      1   FULL/BDR        00:00:31    192.168.12.1    Ethernet0/0

3.3.3.3           1   FULL/DR         00:00:38    192.168.23.3    FastEthernet1/0

R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface

3.3.3.3           1   FULL/BDR        00:00:33    192.168.13.3    FastEthernet1/0

2.2.2.2           1   FULL/DR         00:00:30    192.168.12.2    Ethernet0/0
```

```
R3#show ip ospf neighbor

Neighbor ID    Pri   State      Dead Time   Address         Interface

192.168.13.1     1   FULL/DR    00:00:37    192.168.13.1    FastEthernet1/0

2.2.2.2          1   FULL/BDR   00:00:30    192.168.23.2    FastEthernet0/0
```

Excellent our routers have become OSPF neighbors and the state is full which means they are done exchanging information. Let's check the routing tables:

```
R2#show ip route ospf

O    192.168.13.0/24 [110/2] via 192.168.23.3, 00:09:45, FastEthernet1/0
```

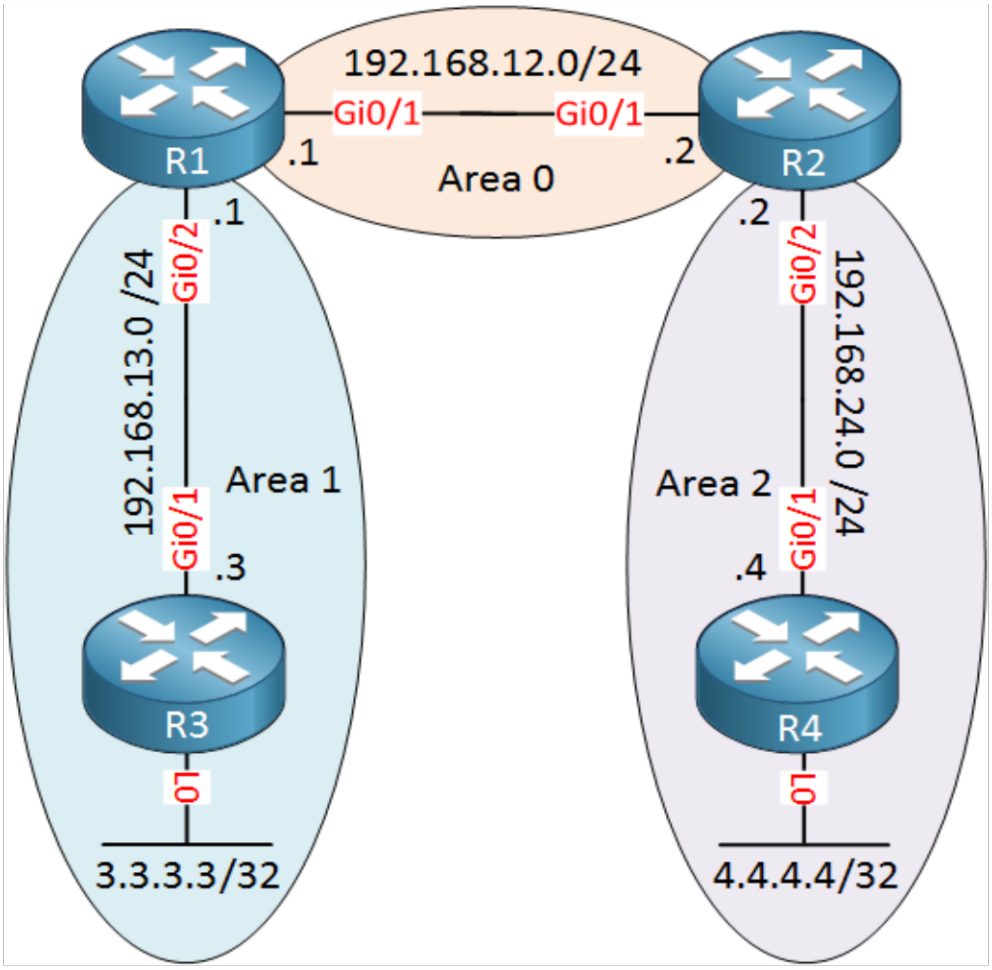
Result:

The implementation and verification of single area OSPF is done in packet tracer.

EXERCISE 9: IMPLEMENTATION OF MULTI AREA OSPF

Aim:

To configure the multi area OSPF topology using packet tracer.



Let's start with all network commands to get OSPF up and running. The network command defines to which area each interface will belong.First, we will configure R1 and R2 for the backbone area:

```
R1(config)#router ospf 1

R1(config-router)#network 192.168.12.0 0.0.0.255 area 0

R2(config)#router ospf 1

R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
```

Let's configure R1 and R3 for area 1:

```
R1(config)#router ospf 1

R1(config-router)#network 192.168.13.0 0.0.0.255 area 1

R3(config)#router ospf 1
```

```
R3(config-router)#network 192.168.13.0 0.0.0.255 area 1

R3(config-router)#network 3.3.3.3 0.0.0.0 area 1
```

And last but not least, R2 and R4 for area 2:

```
R2(config)#router ospf 1

R2(config-router)#network 192.168.24.0 0.0.0.255 area 2

R4(config)#router ospf 1

R4(config-router)#network 192.168.24.0 0.0.0.255 area 2

R4(config-router)#network 4.4.4.4 0.0.0.0 area 2
```

Those are all the network commands we need.

Verification

Let's verify our work. First, let's make sure we have OSPF neighbors:

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface

192.168.24.2      1   FULL/DR         00:00:36    192.168.12.2
GigabitEthernet0/1

3.3.3.3           1   FULL/BDR        00:00:34    192.168.13.3
GigabitEthernet0/2
```

R1 has formed a neighbor adjacency with R2 and R3. Let's check R2:

```
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
```

```
192.168.13.1      1   FULL/BDR        00:00:34    192.168.13.1
GigabitEthernet0/1

4.4.4.4           1   FULL/BDR        00:00:30    192.168.24.4
GigabitEthernet0/2
```

R2 has formed neighbor adjacencies with R1 and R4. The s**how ip** ospf **neighbor** command, however, doesn't tell me anything about the areas that are used. If you want to see this, you could add the **detail** parameter like this:

```
R2#show ip ospf neighbor detail

 Neighbor 192.168.13.1, interface address 192.168.12.1

    In the area 0 via interface GigabitEthernet0/1

    Neighbor priority is 1, State is FULL, 6 state changes

    DR is 192.168.12.2 BDR is 192.168.12.1

    Options is 0x12 in Hello (E-bit, L-bit)

    Options is 0x52 in DBD (E-bit, L-bit, O-bit)

    LLS Options is 0x1 (LR)

    Dead timer due in 00:00:33

    Neighbor is up for 00:17:30

    Index 1/1/1, retransmission queue length 0, number of retransmission 0

    First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

    Last retransmission scan length is 0, maximum is 0

    Last retransmission scan time is 0 msec, maximum is 0 msec

 Neighbor 4.4.4.4, interface address 192.168.24.4

    In the area 2 via interface GigabitEthernet0/2

    Neighbor priority is 1, State is FULL, 6 state changes

    DR is 192.168.24.2 BDR is 192.168.24.4
```

```
    Options is 0x12 in Hello (E-bit, L-bit)

    Options is 0x52 in DBD (E-bit, L-bit, O-bit)

    LLS Options is 0x1 (LR)

    Dead timer due in 00:00:31

    Neighbor is up for 00:15:57

    Index 1/1/2, retransmission queue length 0, number of retransmission 0

    First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

    Last retransmission scan length is 0, maximum is 0

    Last retransmission scan time is 0 msec, maximum is 0 msec
```

Above you can see that interface GigabitEthernet0/1 is in area 0 and interface GigabitEthernet0/2 is in area 2. Another good command to find area information is **show ip protocols**:

```
R2#show ip protocols

*** IP Routing is NSF aware ***


Routing Protocol is "application"

  Sending updates every 0 seconds

  Invalid after 0 seconds, hold down 0, flushed after 0

  Outgoing update filter list for all interfaces is not set

  Incoming update filter list for all interfaces is not set

  Maximum path: 32

  Routing for Networks:

  Routing Information Sources:

    Gateway         Distance      Last Update
```

```
  Distance: (default is 4)

Routing Protocol is "ospf 1"

  Outgoing update filter list for all interfaces is not set

  Incoming update filter list for all interfaces is not set

  Router ID 192.168.24.2

  It is an area border router

  Number of areas in this router is 2. 2 normal 0 stub 0 nssa

  Maximum path: 4

  Routing for Networks:

    192.168.12.0 0.0.0.255 area 0

    192.168.24.0 0.0.0.255 area 2

  Routing Information Sources:

    Gateway         Distance      Last Update

    4.4.4.4              110      00:16:04

    192.168.13.1         110      00:16:53

  Distance: (default is 110)
```

Above you can see which networks belong to which area:

- Network 192.168.12.0 in area 0.
- Network 192.168.24.0 in area 2.

Let's check our routing tables. Let's start with R1:

```
R1#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
       E1 - OSPF external type 1, E2 - OSPF external type 2

       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

       ia - IS-IS inter area, * - candidate default, U - per-user static
route

       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

       a - application route

       + - replicated route, % - next hop override, p - overrides from PfR



Gateway of last resort is not set

      3.0.0.0/32 is subnetted, 1 subnets

O        3.3.3.3 [110/2] via 192.168.13.3, 00:01:47, GigabitEthernet0/2

      4.0.0.0/32 is subnetted, 1 subnets

O IA     4.4.4.4 [110/3] via 192.168.12.2, 00:00:54, GigabitEthernet0/1

O IA  192.168.24.0/24 [110/2] via 192.168.12.2, 00:01:44, GigabitEthernet0/1
```

Above we see three OSPF entries. The first one is for 3.3.3.3/32, the loopback interface of R3. It shows up with an O since this is an intra-area route. R1 has also learned about 4.4.4.4/32 and 192.168.24.0/24. These two entries show up as O IA since they are inter-area routes.

R2 has a similar output:

```
R2#show ip route ospf



      3.0.0.0/32 is subnetted, 1 subnets

O IA     3.3.3.3 [110/3] via 192.168.12.1, 00:02:19, GigabitEthernet0/1

      4.0.0.0/32 is subnetted, 1 subnets

O        4.4.4.4 [110/2] via 192.168.24.4, 00:01:29, GigabitEthernet0/2
```

```
O IA   192.168.13.0/24 [110/2] via 192.168.12.1, 00:02:24, GigabitEthernet0/1
```

Above we see that R2 has learned about 3.3.3.3/32 and 192.168.13.0/24 which area inter-area routes. 4.4.4.4/32 is an intra-area route.

Let's check R3:

```
R3#show ip route ospf

      4.0.0.0/32 is subnetted, 1 subnets

O IA     4.4.4.4 [110/4] via 192.168.13.1, 00:01:57, GigabitEthernet0/1

O IA   192.168.12.0/24 [110/2] via 192.168.13.1, 00:02:50, GigabitEthernet0/1

O IA   192.168.24.0/24 [110/3] via 192.168.13.1, 00:02:47, GigabitEthernet0/1
```

Everything that R3 has learned is from another area, that's why we only see inter-area routes here. The same thing applies to R4:

```
R4#show ip route ospf

      3.0.0.0/32 is subnetted, 1 subnets

O IA     3.3.3.3 [110/4] via 192.168.24.2, 00:02:13, GigabitEthernet0/1

O IA   192.168.12.0/24 [110/2] via 192.168.24.2, 00:02:13, GigabitEthernet0/1

O IA   192.168.13.0/24 [110/3] via 192.168.24.2, 00:02:13, GigabitEthernet0/1
```

Just to be sure, let's try a quick ping between R3 and R4 to prove that our multi-area OSPF configuration is working:

```
R3#ping 4.4.4.4 source 3.3.3.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:

Packet sent with a source address of 3.3.3.3

!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/11/13 ms
```

Our ping is successful. That will be all for now.

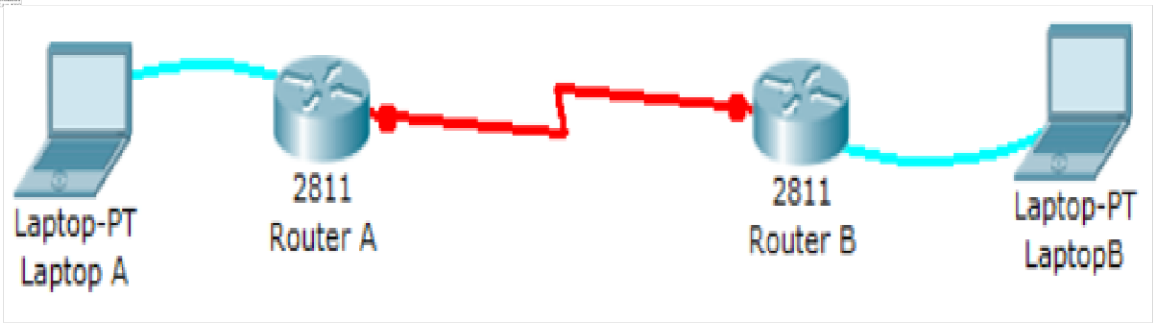
Result:

Thus, we have configured multiple OSPF areas and verified OSPF routes in the routing table that are from different areas.

EXERCISE 10-A: PPP CONFIGURATION

Objective:

To configure PPP on a serial link in Packet Tracer.



1. Use the connected laptops to find the DCE and DTE routers. You can connect to the routers using CLI.

2. Configure the routers with the following parameters :

- Clock : 250000
- PPP link between the routers
- DCE IP : 192.168.10.5/30

- DTE IP : 192.168.10.6/30

3. Check IP connectivity between the two routers using the ping command.

1. Use the connected laptops to find the DCE and DTE routers

The **show controllers <serial interface>** command is used to determine which side of the cable is the DCE side.

In this example, Router-A is the DTE side, and Router-B the DCE side (DCE V.35, clock rate set).

```
Router-A#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DTE V.35 TX and RX clocks detected

Router-B#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 2000000
```

2. Configure the routers with the following parameters

Router-B beeing the DCE, clock rate has to be configured on Router-B serial 0/0/0 interface

```
Router-B(config)#interface serial 0/0/0
Router-B(config-if)#clock rate 250000
```

Then, configure PPP encapsulation and IP address on Router-B serial 0/0/0 interface.
The **encapsulation ppp** configures PPP protocol on the serial interface. PPP authenication can be oprtionnally configured using the following IOS commands which are not used in this lab :

- ppp authentication : Set PPP link authentication method
- ppp pap: Set PAP authentication parameters

Router-B beeing the DCE side of the serial link, the 192.168.10.5/30 IP address is configured on Router-B serial 0/0/0 interface. Don't forget to enable the interface with a **no shutdown** command.

```
Router-B(config)#interface serial 0/0/0
Router-B(config-if)#encapsulation ppp
Router-B(config-if)#ip address 192.168.10.5 255.255.255.252
Router-B(config-if)#no shutdown
```

The show interfaces serial 0/0/0 confirms that PPP encapsulation is enabled on the interface : *Encapsulation PPP, loopback not set, keepalive set (10 sec)*

```
Router-B#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 192.168.10.5/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  [...]
```

Finally, configure PPP encapsulation and IP address on Router-A serial 0/0/0 interface. The link becomes up as both routers are correctly configured.

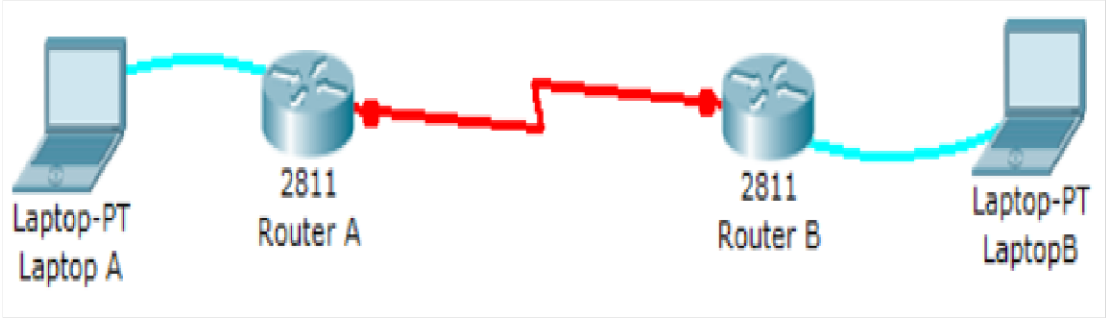
```
Router-A(config)#interface serial 0/0/0
Router-A(config-if)#encapsulation ppp
Router-A(config-if)#ip address 192.168.10.6 255.255.255.252
Router-A(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

3. Check IP connectivity between the two routers using the ping command.

Issue a ping from Router-A to Router-B to test network connectivity between the two routers.

```
Router-A# ping 192.168.10.5
```

EXERCISE 10-B: HDLC CONFIGURATION



1. Use the connected laptops to find the DCE and DTE routers

The show controllers <serial interface> command is used to determine which side of the cable is the DCE side.

In this example, Router-A is the DTE side, and Router-B the DCE side (DCE V.35, clock rate set).

```
Router-A#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DTE V.35 TX and RX clocks detected

Router-B#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 2000000
```

2. Configure the routers with the following parameters

Router-B beeing the DCE, clock rate has to be configured on Router-B serial 0/0/0 interface

```
Router-B(config)#interface serial 0/0/0
Router-B(config-if)#clock rate 250000
```

Then, configure HDLC encapsulation and IP address on Router-B serial 0/0/0 interface.
The **encapsulation hdlc** configures HDLC protocol on the serial interface.

Router-B beeing the DCE side of the serial link, the 192.168.1.5/30 IP address is configured on Router-B serial 0/0/0 interface. Don't forget to enable the interface with a **no shutdown** command.

```
Router-B(config)#interface serial 0/0/0
Router-B(config-if)#encapsulation hdlc
Router-B(config-if)#ip address 192.168.10.5 255.255.255.252
Router-B(config-if)#no shutdown
```

The show interfaces serial 0/0/0 confirms that HDLC encapsulation is enabled on the interface
: *Encapsulation HDLC, loopback not set, keepalive set (10 sec)*

```
Router-B#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
```

```
  Hardware is HD64570
  Internet address is 192.168.10.5/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  [...]
```

Finally, configure HDLC encapsulation and IP address on Router-A serial 0/0/0 interface. The link becomes up as both routers are correctly configured.

```
Router-A(config)#interface serial 0/0/0
Router-A(config-if)#encapsulation hdlc
Router-A(config-if)#ip address 192.168.10.6 255.255.255.252
Router-A(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

3. Check IP connectivity between the two routers using the ping command.

Issue a ping from Router-A to Router-B to test network connectivity between the two routers.

```
Router-A#ping 192.168.10.5
```

Result:

Thus, we have configured PPP and HDLC protocols and verified in the packet tracer.

EXERCISE 11-A: BASIC BGP CONFIGURATION

Aim:

To configure BGP routers with different Autonomous System numbers configuration between two routers.

First we need to configure some interfaces on two routers as follows:

```
R1(config)#interface fastethernet0/0
R1(config-if)#ip address 11.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#interface loopback 0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
```

```
R2(config)#interface fastethernet0/0
R2(config-if)#ip address 11.0.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#interface loopback 0
R2(config-if)#ip address 2.2.2.2 255.255.255.0
```

So we have just configured interface fa0/0 and loopback0 on both routers. Next we will configure the BGP configuration part on R1:

```
R1(config)#router bgp 1
R1(config-router)#neighbor 11.0.0.2 remote-as 2
```

The configuration is very simple with only two lines on R1. In the first line, BGP configuration begins with a familiar type of command: the **router bgp** command, where **AS number** is the BGP AS number used by that router (same as EIGRP, OSPF configuration).

The next command defines the IP address of the neighbor. Unlike OSPF or EIGRP, BGP cannot discover its neighbors automatically so we have to explicitly declare them. We also have to know and declare the neighbor's BGP AS number as well. In this case R1 wants to establish BGP neighbor relationship with R2 (in BGP AS 2) so it choose an interface on R2 (Fa0/0: **11.0.0.2**) and specify R2 is in **BGP AS 2** via the command "neighbor **11.0.0.2** remote-as **2**". At the other end R2 will do the same thing for R1 to set up BGP neighbor relationship.

```
R2(config)#router bgp 2
R2(config-router)#neighbor 11.0.0.1 remote-as 1
```

After a moment we should see a message (on each router) similar to the following, letting us know that an adjacency has been formed:

On R1:

```
*Aug 17 00:09:38.453: %BGP-5-ADJCHANGE: neighbor 11.0.0.2 Up
```

On R2:

```
*Aug 17 00:09:38.453: %BGP-5-ADJCHANGE: neighbor 11.0.0.1 Up
```

So after forming BGP neighbor relationship we can verify by using the "show ip bgp summary" command on both routers:

```
R1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 1
BGP table version is 1, main routing table version 1

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
11.0.0.2        4     2      19      19        1    0    0 00:16:21        0
R2#show ip bgp summary
BGP router identifier 2.2.2.2, local AS number 2
BGP table version is 1, main routing table version 1

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
11.0.0.1        4     1      20      20        1    0    0 00:17:13        0
```

Let's try advertising the loopback 0 interface on R1 to R2:

```
R1(config-router)#network 1.1.1.0 mask 255.255.255.0
```

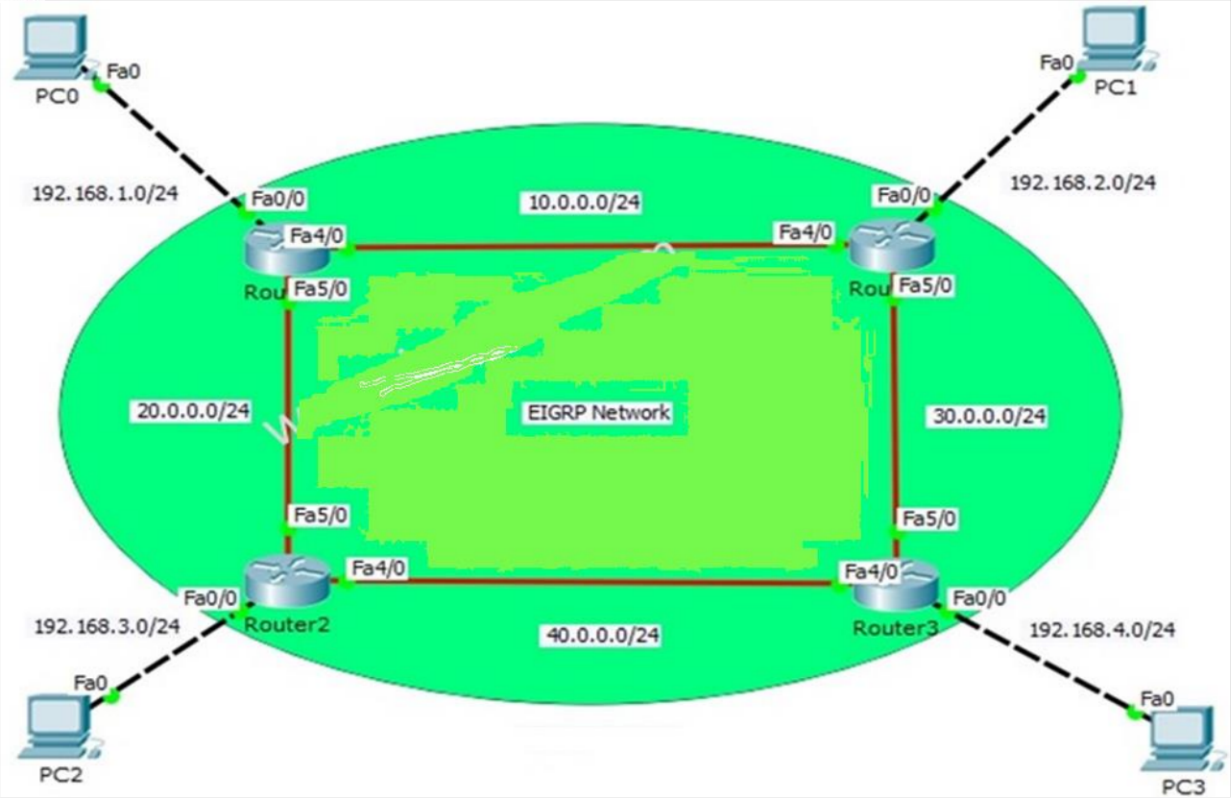
Now the BGP routing tables on these two routers contain this route:

```
R1#sh ip bgp
BGP table version is 4, local router ID is 11.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 1.1.1.0/24       0.0.0.0                  0         32768 i
```

EXERCISE 12: EIGRP CONFIGURATION

Objective:

To configure and verify the EIGRP routing in packet tracer.



First, Assign the IP addresses for PC's as given below.

PC0: 192.168.1.1 255.255.255.0 GW:192.168.1.2

PC1: 192.168.2.1 255.255.255.0 GW:192.168.2.2

PC2: 192.168.3.1 255.255.255.0 GW:192.168.3.2

PC3: 192.168.4.1 255.255.255.0 GW:192.168.4.2

EIGRP Configuration

```
Router0(config)# interface FastEthernet0/0

Router0(config-if)# ip address 192.168.1.2 255.255.255.0

Router0(config-if)# no shutdown

Router0(config-if)# exit

Router0(config)# interface FastEthernet4/0

Router0(config-if)# ip address 10.0.0.1 255.255.255.0

Router0(config-if)# no shutdown

Router0(config-if)# exit

Router0(config)# interface FastEthernet5/0
```

```
Router0(config-if)# ip address 20.0.0.1 255.255.255.0

Router0(config-if)# no shutdown

Router0(config-if)# end

Router0# copy running-config startup-config
```

```
Router1(config)# interface FastEthernet0/0

Router1(config-if)# ip address 192.168.2.2 255.255.255.0

Router1(config-if)# no shutdown

Router1(config-if)# exit

Router1(config)# interface FastEthernet4/0

Router1(config-if)# ip address 10.0.0.2 255.255.255.0

Router1(config-if)# no shutdown

Router1(config-if)# exit

Router1(config)# interface FastEthernet5/0

Router1(config-if)# ip address 30.0.0.1 255.255.255.0

Router1(config-if)# no shutdown

Router1(config-if)# end

Router1# copy running-config startup-config
```

```
Router2(config)# interface FastEthernet0/0

Router2(config-if)# ip address 192.168.3.2 255.255.255.0

Router2(config-if)# no shutdown

Router2(config-if)# exit

Router2(config)# interface FastEthernet4/0

Router2(config-if)# ip address 40.0.0.1 255.255.255.0

Router2(config-if)# no shutdown
```

```
Router2(config-if)# exit

Router2(config)# interface FastEthernet5/0

Router2(config-if)# ip address 20.0.0.2 255.255.255.0

Router2(config-if)# no shutdown

Router2(config-if)# end

Router2# copy running-config startup-config
```

```
Router3(config)# interface FastEthernet0/0

Router3(config-if)# ip address 192.168.4.2 255.255.255.0

Router3(config-if)# no shutdown

Router3(config-if)# exit

Router3(config)# interface FastEthernet4/0

Router3(config-if)# ip address 40.0.0.2 255.255.255.0

Router3(config-if)# no shutdown

Router3(config-if)# exit

Router3(config)# interface FastEthernet5/0

Router3(config-if)# ip address 30.0.0.2 255.255.255.0

Router3(config-if)# no shutdown

Router3(config-if)# end

Router3# copy running-config startup-config
```

Here, our EIGRP Autonomous number will be **100**. And for Router0, **192.168.1.0, 10.0.0.0, 20.0.0.0** network will be added under this EIGRP process. These are the directly connected networks to Router0.

```
Router0(config)# router eigrp 100

Router0(config-router)# network 192.168.1.0

Router0(config-router)# network 10.0.0.0

Router0(config-router)# network 20.0.0.0
```

```
Router0(config-router)# no auto-summary

Router0(config-router)# end

Router0# copy running-config startup-config
```

The similar configuration will be done on Router1 also. Here, only the networks will be changed. The direclty connected networks to Router1 will be added.

```
Router1(config)# router eigrp 100

Router1(config-router)# network 192.168.2.0

Router1(config-router)# network 10.0.0.0

Router1(config-router)# network 30.0.0.0

Router1(config-router)# no auto-summary

Router1(config-router)# end

Router1# copy running-config startup-config
```

For Router2 and Router3 the similar **Cisco EIGRP Configuration** will be done. Only one added networks will be changed. Because each router has different directly connected networks.

```
Router2(config)# router eigrp 100

Router2(config-router)# network 192.168.3.0

Router2(config-router)# network 20.0.0.0

Router2(config-router)# network 40.0.0.0

Router2(config-router)# no auto-summary

Router2(config-router)# end

Router2# copy running-config startup-config
```

```
Router3(config)# router eigrp 100

Router3(config-router)# network 192.168.4.0

Router3(config-router)# network 30.0.0.0
```

```
Router3(config-router)# network 40.0.0.0

Router3(config-router)# no auto-summary

Router3(config-router)# end

Router3# copy running-config startup-config
```

Configuration Verification

```
Router0# show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H Address Interface Hold Uptime SRTT RTO Q Seq
(sec) (ms) Cnt Num
0 10.0.0.2 Fa4/0 14 00:14:53 40 1000 0 12
1 20.0.0.2 Fa5/0 12 00:14:53 40 1000 0 11
```

```
Router1# show ip eigrp interfaces
IP-EIGRP interfaces for process 100
Xmit Queue Mean Pacing Time Multicast Pending
Interface Peers Un/Reliable SRTT Un/Reliable Flow Timer Routes
Fa4/0 1 0/0 1236 0/10 0 0
Fa5/0 1 0/0 1236 0/10 0 0
Fa0/0 0 0/0 1236 0/10 0 0
```

```
Router1# show ip eigrp topology

IP-EIGRP Topology Table for AS 100/ID(192.168.2.2)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - Reply status
```

Result:

Thus, the configuration and verification of EIGRP is successfully done.

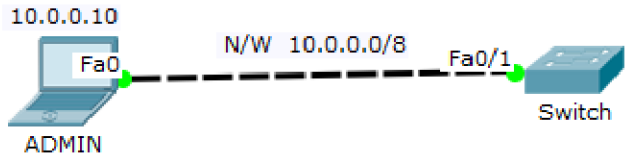
Exercise 13: Configuring Telnet on a switch and a router in Packet Tracer

Objective:

To configure Telnet on a switch and a router in Packet Tracer.

A. Telnet configuration on a switch

1. Create the network topology below in Packet Tracer.



Assign the laptop a static IP address of 10.0.0.10.

The topology above consists of an **ADMIN** laptop and a remote switch. We'll configure Telnet on the switch so that as the admin, you'll be able to access and manage the switch remotely.

2. Configure **enable password** or **enable secret password** on the switch. If you fail to do this,you won't get past the executive mode of the switch even after you establish a telnet connection to the switch.

```
Switch>enable

Switch#config terminal

Switch(config)#enable password admin
```

3.**Configure a VLAN interface on the switch**

We assign an IP address to the **VLAN** interface of the switch so that we can Telnet the switch from the laptop using this address.

```
Switch(config)#int VLAN 1

Switch(config-if)#ip address 10.0.0.20   255.0.0.0

Switch(config-if)#no shut

Switch(config-if)#exit
```

4.Configure a **Telnet password** for remote access.

This password is configured on **VTY** lines. VTY means Virtual Terminal. Before you can manage the switch remotely via Telnet, you'll have to provide this password.

```
Switch(config)#line vty 0 15

Switch(config-line)#password cisco

Switch(config-line)#login
```

Telnet access to the switch is allowed through VTY lines. We can establish up to 16 telnet connections to to the switch at the same time. Tha'ts what '0 15' means.

Next,

5. Test Telnet connectivity.

Go to command prompt of the laptop and type *telnet 10.0.0.20*

Hope you remember that 10.0.0.20 is the VLAN address of the switch through which we can access it remotely.

6. Now provide the Telnet password that you set in step 3. Mine is **cisco.** Notice that password characters **won't show up(no echo)** on the screen as you type them, but just type, then hit ENTER.

After you're authenticated, you will see the the CLI of the remote switch appear.

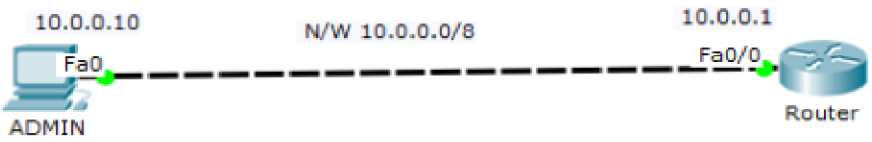
Now provide the enable password **admin** (or yours which you set in step 1) to enter the **privileged executive mode** of the switch. You can then continue to configure your switch the way you desire(but now, **remotely**)

Note that Telnet and enable passwords are different. **Enable password** authenticates you into privileged executive mode of the terminal device(switch, for example), but you're using **Telnet Password** to allow you access the interface of the remote device after connecting to it. You can see that we used telnet password to access the CLI of the remote switch.

B. Telnet Configuration on Router

For a router, Telnet configuration is almost same as that of the switch.

1. Build the network topology below



2. Configure **enable password** or **enable secret password** on the router

```
Router>en

Router#config term

Router(config)#enable password admin
```

3. Configure IP addresses on the admin PC and interface fa0/0 of the router

Router

```
Router(config)#int fa0/0

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shut
```

Admin PC

IP address 10.0.0.10 **Subnet mask** 255.0.0.0 **Default gateway** 10.0.0.1

4. Configure VLAN interface on the router. This interface allows for remote access on a switch or router via protocols such as Telnet or Secure Shell(SSH)

```
Router(config)#int VLAN 1

Router(config-if)#no shutdown
```

As you can see, we've not configured the VLAN interface with an IP address. We could do this but it is unnecessary. We already have an interface fa0/0 of the router with an IP address through which we can Telnet the router from the PC.

4. Configure Telnet password on VTY lines and configure remote login.

```
Router(config)#

Router(config)#line vty 0 15

Router(config-line)#password cisco

Router(config-line)#login
```

5. We can now telnet the router using the IP address of fa0/0 interface. So, in the command prompt of the admin PC type ***telnet 10.0.0.1*** then hit enter key.

6. Provide Telnet Password (that you set in step 4), then hit enter. Correct password allows you access the CLI of the router .

7. Now provide the enable password (that you set in step 2) to be allowed into **privileged executive mode** of the router.

You can now do configurations on the router from the PC remotely.

Result:

Thus, the configuration and verification of Telnet protocol is done successfully.